Sound: a longitudinal mechanical wave in a medium

What sound is: longitudinal wave, s(x,t), λ = v/f, wave vs particle speed.

Sound: a longitudinal mechanical wave in a medium

Sound needs a medium: it is a disturbance that travels through a material by pushing and pulling along the travel direction. Gas molecules oscillate about fixed equilibrium positions while the pattern — the wave — moves at the speed of sound; no bulk gas flows along the pipe. Sound in a fluid is purely longitudinal (transverse waves need shear); in a solid rod, longitudinal and transverse modes can coexist (Oly 10).

The wave pattern of a pure tone is sinusoidal, so everything follows from the displacement field $s(x,t) = s_0 \sin(kx - \omega t)$:

Sound: a longitudinal mechanical wave in a medium (cont.)

s is the small displacement of a fluid element from its equilibrium position, amplitude s_0 is microscopic (audible sounds give $s_0 \sim 10^{-8}$ to 10^{-4} m), $k = 2\pi/\lambda$, $\omega = 2\pi f$, and the wave speed is $v = \omega/k = f\lambda$ in the medium — only the pattern moves at v.

Compressions (pressure maxima) and rarefactions (minima) sit half a wavelength apart. The distance between two consecutive compressions is exactly λ. If a source vibrates with frequency f, it launches f wavefronts per second, and the wavelength in the medium is $\lambda = v/f$ — the medium fixes v, the source fixes f, the medium and source together give λ.

Sound: a longitudinal mechanical wave in a medium (cont.)

A wave with frequency 340 Hz in air at 20 C has $\lambda = 343/340 = 1.009$ m. Direction convention: $s = s_0 \sin(kx - \omega t)$ travels toward $+x$ because a phase point $kx - \omega t =$ const has $dx/dt = \omega/k = v > 0$.

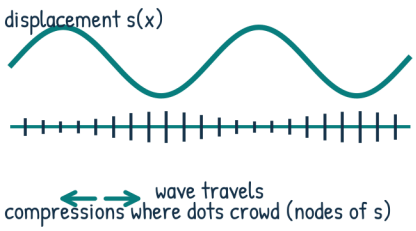
Speed is a bulk property: $v = \sqrt{B/\rho}$ for a fluid (Section JEE-04), so it does not depend on frequency — a bass drum and a piccolo travel together (non-dispersive medium), and the tone reaching a listener is undistorted. Loudness relates to amplitude s_0 or pressure amplitude ΔP_0 (JEE-02/12); pitch relates to f.

Sound: a longitudinal mechanical wave in a medium (cont.)

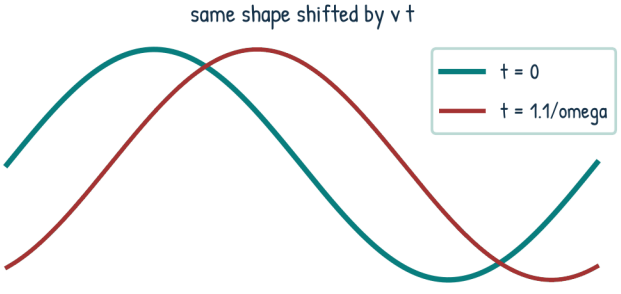
In vacuum $B = 0$, there is no sound, which is itself proof that sound is a pressure-elastic wave. For exam problems the working chain is: given f and v, get λ; given $s(x,t)$, read $s_0, k, \omega, v = \omega/k$ and the travel direction from the sign of the $kx - \omega t$ combination.

WHY / longitudinal means the oscillation of each gas element is along the wave travel direction: layers alternately pile up (compression, higher pressure) and spread out (rarefaction, lower pressure), and this pattern advances at $v = \sqrt{B/\rho}$ — the elements themselves only jiggle about their mean positions.

longitudinal wave: gas parcels oscillate in place



the same wave at two times: shape moves at v



REMEMBER / Sound is a longitudinal wave: s(x,t), lambda = v/f; the wave moves at v, the gas jiggles in place.

Sound: a longitudinal mechanical wave in a medium (cont.)

What sound is: longitudinal wave, s(x,t), λ = v/f, wave vs particle speed.

Sound: a longitudinal mechanical wave in a medium (cont.)

CHECK / given $s = 2 \times 10^{-6} \sin(1000x - 3.43 \times 10^5 t)$: $k = 1000$ rad/m so $\lambda = 2\pi/1000 = 6.28$ mm; $\omega = 3.43 \times 10^5$ rad/s so $f = 54.6$ kHz; $v = \omega/k = 343$ m/s — air at room temperature; the $-\omega t$ sign means travel toward $+x$. $v_{\text{wave}} = \omega/k$ is the speed of the pattern; the gas element itself oscillates with velocity $u = ds/dt = -s_0\omega\cos(kx - \omega t)$, whose amplitude $s_0\omega$ is tiny (for $s_0 = 10^{-6}$ m at 340 Hz: $u_{\max} = 2.1 \times 10^{-3}$ m/s against $v = 343$ m/s). The wave transports energy and momentum; the medium transports neither over long distances.

Particle velocity vs wave speed — the classic confusion (cont.)

Two waves can pass through the same region and superpose (JEE-13) because the governing equation is linear for small s.

One full oscillation of the source emits exactly one wavelength of sound. The period $T = 1/f$ is the time in which the wave advances by λ: $v = \lambda/T = f\lambda$. When a wave crosses into another medium (sound from air into water), f is fixed by the source, v changes with the medium, so λ changes: $\lambda_{\text{water}} = 1480/340 = 4.35$ times longer for the same tone.

WHY / the wave equation for s is $\partial^2 s/\partial t^2 = v^2 \partial^2 s/\partial x^2$ (JEE-04) — linear and second order, so sinusoidal solutions of any f travel at the same v (no dispersion), and the superposition principle holds.

Particle velocity vs wave speed — the classic confusion (cont.)

CHECK / a 200 Hz tuning fork is held in air at 20 C ($v = 343$ m/s): $\lambda = 1.72$ m; the same fork under water ($v = 1480$ m/s): $\lambda = 7.4$ m — same f, same pitch, different wavelength and different speed.

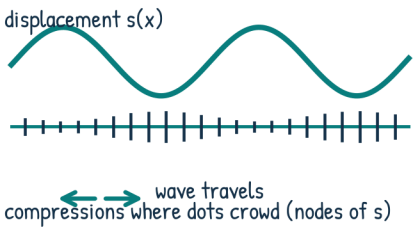
Sources are all vibrating bodies: strings, air columns, membranes, vocal cords. The vibrating surface alternately compresses and rarefies the adjacent air, launching the wave. Receivers are resonant systems too: the ear canal (~2.5 cm closed pipe, fundamental ~3.4 kHz — the ear is most sensitive near 3 kHz), microphone diaphragms, plates.

How sound is produced and detected (practical frame) (cont.)

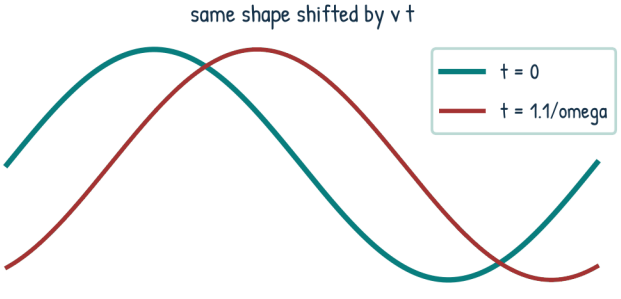
Resonance (JEE-16/17) amplifies sound both at the source and at the receiver, which is why Helmholtz resonators sing at one pitch and why a room has preferred frequencies (room modes = open/closed pipe problem in 3D).

A pure tone (single f) is a sine wave; real sounds are sums of harmonics + noise. The phase velocity of the fundamental defines v; timbre lives in the harmonic content. Ultrasound ($f > 20$ kHz) and infrasound ($f < 20$ Hz) obey exactly the same equations — the audible band is a biological, not physical, boundary. In air at 20 C the audible range spans $\lambda = 17$ m (20 Hz) to 17 mm (20 kHz).

longitudinal wave: gas parcels oscillate in place



the same wave at two times: shape moves at v



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Sound: a longitudinal mechanical wave in a medium (cont.)

What sound is: longitudinal wave, s(x,t), λ = v/f, wave vs particle speed.

How sound is produced and detected (practical frame) (cont.)

in a duct filled with flowing gas of speed w the observed frequency is Doppler-shifted (JEE-25): stationary listener in moving medium hears f unchanged if both source and listener are carried by the same wind; wind matters only for relative motion along the line of sight.

(a) Find v for a wave $s = s_0\sin(2.4x - 1.1\times10^3 t)$ in a metal: $v = \omega/k = 1100/2.4 = 458$ m/s, $\lambda = 2.62$ m, $f = 175$ Hz. (b) A sound wave has $\lambda = 1.62$ m in air at 20 C: $f = 343/1.62 = 212$ Hz. (c) Two speakers 1 m apart vibrate in phase at 340 Hz — the wavelength 1 m sets the fringe spacing (JEE-13). (d) A stone dropped in a well: sound travels down, water splash returns — combines wave travel time with kinematics (paper Section D style).

Worked micro-examples (cont.)

Linearity check: doubling s_0 doubles both ΔP_0 and intensity (JEE-10), and raises the level by 6.02 dB in pressure terms (3.01 dB in intensity terms for doubling power). The wave speed stays exactly the same — amplitude never changes v in linear acoustics (nonlinear acoustics, shocks, are outside the syllabus).

WHY / if v depended on amplitude, a loud note would arrive later than a soft one and music would smear — linear acoustics ($s_0 \ll \lambda$) rules this out for everyday sound.

CHECK / for $f = 1$ kHz, $s_0 = 10^{-5}$ m: $s_0/\lambda = 3\times10^{-8}$ — infinitesimal strain, so the linear wave equation is superbly accurate for ordinary sound.

Strain and acoustic pressure: dP = -B ds/dx

Sound is a small-amplitude compression wave: when a gas element is squeezed (negative $\partial s/\partial x$) its pressure rises. The extra pressure above ambient, $\Delta P = P - P_0$, is the acoustic (excess) pressure. Bulk modulus of the fluid is $B = -V dP/dV$ (positive), and for a displaced column the fractional volume change is exactly the strain $\partial s/\partial x$ (displacement gradient).

Linearizing gives the fundamental relation $\Delta P = -B\,\dfrac{\partial s}{\partial x}$

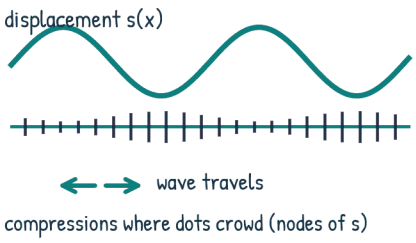
— the minus sign is essential: where displacement decreases with x (elements pile up), pressure rises.

Strain and acoustic pressure: dP = -B ds/dx (cont.)

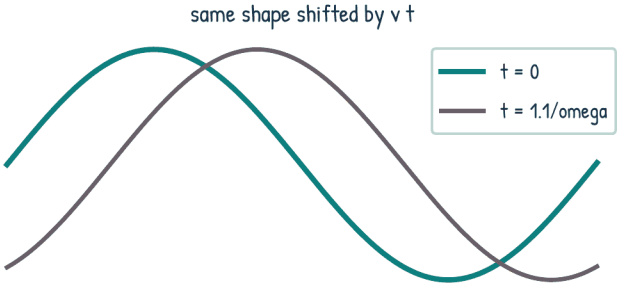
Insert $s = s_0\sin(kx - \omega t)$: $\partial s/\partial x = k s_0\cos(kx - \omega t)$, so $\Delta P = -Bks_0\cos(kx - \omega t)$, i.e. pressure amplitude $\Delta P_0 = Bks_0$. Using $v = \sqrt{B/\rho}$, $B = \rho v^2$ and $k = \omega/v$, this becomes the exam workhorse $\Delta P_0 = Bks_0 = \rho v\omega s_0$. In air at 20 C: $\rho = 1.2$ kg/m^3 , $v = 343$ m/s, so $\Delta P_0 \approx 412\,\omega s_0$ (SI).

Example: $s_0 = 2\times10^{-6}$ m at $f = 340$ Hz gives $\Delta P_0 = 1.2\times343\times2136\times2\times10^{-6} = 1.76$ Pa.

longitudinal wave: gas parcels oscillate in place



the same wave at two times: shape moves at v



REMEMBER / Sound is a longitudinal wave: s(x,t), lambda = v/f; the wave moves at v, the gas jiggles in place.

Strain and the acoustic pressure Delta P = -B,partial s/partial x

Strain, acoustic pressure, Δ P = -B ∂ s/ ∂ x, Δ P 0 = ρ v ω s 0 , bulk modulus frame.

Strain and acoustic pressure: dP = -B ds/dx (cont.)

Physical sizes: atmospheric pressure $P_0 = 1.013 \times 10^5$ Pa, while audible acoustic pressures run 2×10^{-5} Pa (threshold) to ~20 Pa (threshold of pain) — $\Delta P \ll P_0$ always, which is why the linear treatment works and why air density barely changes. Threshold of hearing: $s_0 = \Delta P_0/(\rho v \omega) \approx 2.87 \times 10^{-5}/(412 \cdot 2\pi \cdot 1000) \approx 1.1 \times 10^{-11}$ m at 1 kHz — a nuclear-scale displacement, yet the ear detects it.

WHY / the sign in $\Delta P = -B\partial s/\partial x$ is Hooke-type restoring physics: stretch the element (positive strain) and pressure drops below ambient — the fluid pulls back; squeeze and pressure rises — exactly like a spring $F = -kx$ with B playing the stiffness role.

Strain and acoustic pressure: dP = -B ds/dx (cont.)

CHECK / where s is at a maximum ($s = s_0$ at $kx - \omega t = \pi/2$), $\partial s/\partial x = 0$, so $\Delta P = 0$: displacement antinodes are pressure nodes. A microphone placed at a displacement node of a standing wave registers maximum pressure fluctuation (JEE-16).

Master formula: $\Delta P_0 = \rho v \omega s_0$. In air $\rho v \approx 412$ kg m $^{-2}$ s $^{-1}$ (often quoted as $Z = 412$ rayl, conventionally rounded to 400 — Oly 01). Three worked numbers from the source: (i) $f = 340$ Hz, $s_0 = 2 \times 10^{-6}$ m $\Rightarrow \Delta P_0 = 1.76$ Pa; (ii) at threshold of hearing ($\Delta P_0 = 2 \times 10^{-5}$ Pa at 1 kHz) $s_0 = 1.1 \times 10^{-11}$ m;

Pressure amplitude from displacement amplitude (cont.)

(iii) a loud rock concert ($\Delta P_0 = 20$ Pa) has $s_0 = 20/(412 \cdot 6283) = 7.8 \times 10^{-6}$ m at 1 kHz — still microscopic.

Frequency dependence at fixed s_0 : $\Delta P_0 \propto \omega$ — high frequencies carry more pressure for the same displacement. At fixed pressure (same loudness) $s_0 \propto 1/\omega$ — bass drivers must swing much farther than tweeters for equal acoustic pressure, which is why woofers are large-excursion and tweeters tiny. Membrane displacement of a loudspeaker at 20 Hz and 120 Pa: $s_0 = 120/(412 \cdot 126) = 2.3$ mm — visible shaking.

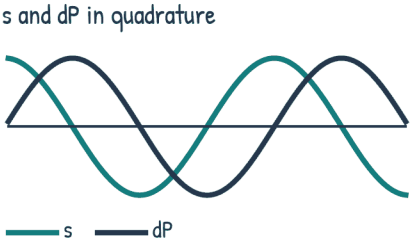
WHY / the ω factor is calculus, not physics: pressure is a gradient of displacement, and a fixed-amplitude pattern with shorter wavelength (higher f) has steeper slope $ks_0 = (\omega/v)s_0$.

Pressure amplitude from displacement amplitude (cont.)

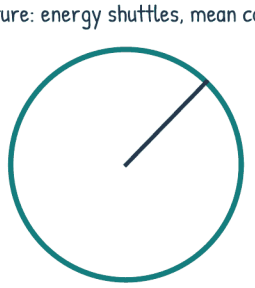
CHECK / consistency with JEE-10: intensity $I = \Delta P_0^2/(2\rho v)$: for 1.76 Pa, $I = 3.1/824 = 3.7 \times 10^{-3}$ W/m ^{2} — plausible loud speech-to-shout level, $\beta = 95.7$ dB.

The strain in a sound wave is $\partial s/\partial x = ks_0\cos(\cdot)$, of order $ks_0 = 2\pi s_0/\lambda$: for $s_0 = 10^{-6}$ m, $\lambda = 1$ m, strain $\sim 6 \times 10^{-6}$ — the medium is compressed by parts per million. Bulk modulus of air at 1 atm, with adiabatic assumption (JEE-07): $B = \gamma P_0 = 1.4 \times 1.013 \times 10^5 = 1.42 \times 10^5$ Pa.

pressure follows the slope of the displacement curve



quadrature circle: s and dP trade energy



REMEMBER / Pressure is the slope of displacement: dP = -B ds/dx and dP0 = rho v omega s0 bridge the two amplitudes.

Strain and the acoustic pressure Delta P = -B,partial s/partial x (cont.)

Strain, acoustic pressure, Δ P = -B ∂ s/ ∂ x, Δ P 0 = ρ v ω s 0 , bulk modulus frame.

Strain, bulk modulus, and where the speed comes from (cont.)

Then $v=\sqrt{B/\rho}=\sqrt{1.42\times10^{5}/1.2}=344$ m/s ok. For water $B=2.2$ GPa, $\rho=1000$: $v=1483$ m/s ok; for steel $Y=2\times10^{11}$ Pa, $\rho=7800$: $v=5.06\times10^{3}$ m/s (rod) ok.

Stiffness vs inertia: $v=\sqrt{B/\rho}$ says speed grows with restoring stiffness and falls with inertia. That is why sound is fast in diamond and slow in heavy gases, and why the isothermal vs adiabatic question (JEE-06/07) decides B for gases: $B_{\text{iso}}=P$, $B_{\text{ad}}=\gamma P$ — a factor 1.18 in speed.

Strain, bulk modulus, and where the speed comes from (cont.)

CHECK / the speed of sound cannot exceed the bulk sound-speed limit of the material and never depends on s_0 or f in the linear regime — two constraints every formula must respect.

at very high intensity (shock waves) the linear law breaks: the crests steepen because the compressed gas is locally faster. Sonic booms (JEE-28) are the built-up edge of this nonlinearity.

(a) $s(x,t)=5\times10^{-6}\cos(800x-2.75\times10^{5}t)$ in air: $\omega=2.75\times10^{5}$, $f=43.8$ kHz (ultrasound), $v=344$ m/s, $\Delta P_0=\rho v\omega s_0=1.2\times344\times2.75\times10^{5}\times5\times10^{-6}=568$ Pa — strong ultrasound. (b) Given $\Delta P_0=0.3$ Pa at 250 Hz in air:

Micro-worked set (pressure-displacement round trip) (cont.)

$s_0=0.3/(412\cdot1571)=4.6\times10^{-7}$ m. (c) In a Kundt tube dust heaps sit at pressure nodes... (d) The ear canal closed at one end has displacement node at the eardrum — maximum pressure on the eardrum (why it is sensitive).

Round-trip conversions examiners love: given s_0 and f in a medium of known ρ, v : $\Delta P_0=\rho v 2\pi f s_0$; given ΔP_0 : $s_0=\Delta P_0/(2\pi f\rho v)$; given both: $v=\Delta P_0/(2\pi f\rho s_0)$, $\rho=\Delta P_0/(2\pi f v s_0)$. Dimension check: $\rho v\omega s_0=$ kg m −3 m s −1 s −1 m = kg m −1 s −2 = Pa ok.

WHY / acoustic pressure is the quantity microphones and ears respond to; displacement is what optics-based methods (Laser Doppler Vibrometry) measure — exams give one, physics often wants the other, and $\Delta P_0=\rho v\omega s_0$ is the bridge.

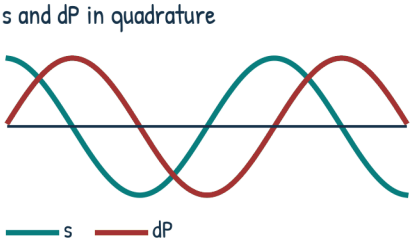
Micro-worked set (pressure-displacement round trip) (cont.)

CHECK / in water ($\rho v=1.48\times10^{6}$) the same s_0 at same f gives ΔP_0 3600 times larger than in air — same displacement, hugely different pressure because water is far stiffer.

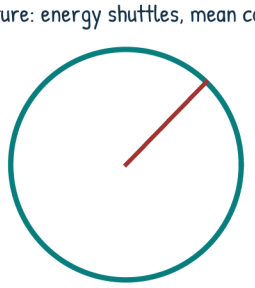
For a travelling wave $s=s_0\sin(kx-\omega t)$ the acoustic pressure is $\Delta P=-B\partial s/\partial x=-Bks_0\cos(kx-\omega t)=\Delta P_0\sin(kx-\omega t-\pi/2)$: pressure lags displacement by 90 degrees (is in quadrature). Graphically:

at points where gas has maximum displacement from equilibrium, the local density is unmodified (slope of s is zero), so pressure is ambient; where the slope of s is steepest (elements pile up most), displacement itself is zero and pressure is extremal. Compressions therefore sit a quarter wavelength from displacement maxima.

pressure follows the slope of the displacement curve



quadrature circle: s and dP trade energy



REMEMBER / Pressure is the slope of displacement: dP = -B ds/dx and dP0 = rho v omega s0 bridge the two amplitudes.

Displacement vs pressure phase: the quadrature structure of sound

s- Δ P quadrature, energy density ½ $\rho\,\omega^2 s_0^2$, phase map of one wavelength.

Displacement vs pressure phase: the quadrature structure of sound.

Energy picture: in a sound wave, energy sloshes between ordered motion of elements (kinetic, max where elements move fastest = displacement-phase crossings) and elastic compression energy (max where ΔP is extremal). The two are in quadrature, exactly like a mass-spring oscillator being driven through equilibrium. Mean energy density $\bar{u} = \frac{1}{2}\rho v^2 (k s_0)^2 \cdot \frac{1}{2} \cdot 2$ ... cleanly:

$\bar{u} = \frac{1}{2}\rho\omega^2 s_0^2$ (equal mean kinetic and potential share $\frac{1}{4}\rho\omega^2 s_0^2$ each; equivalently $\bar{u} = \Delta P_0^2/(4B)\cdot 2 = \Delta P_0^2/(2\rho v^2)$ since $B = \rho v^2$).

Displacement vs pressure phase: the quadrature structure of sound.

For a standing wave the roles exchange spatially instead: nodes of s are antinodes of ΔP and vice versa, and standing-wave energy density is maximum at both node types at different phases. The quadrature relation $\Delta P = -B\partial s/\partial x$ is the safest way to get the relative phase right in any geometry — more reliable than memorising "pressure leads/lags".

WHY / pressure measures the local slope (spatial derivative) of the displacement pattern; for a sinusoid, derivative = phase shift by $\pi/2$. The minus sign adds π, giving ΔP in antiphase with $\partial s/\partial x$, quadrature with s itself.

CHECK / displacement node in a pipe closed at the rigid end: the end forces $s = 0$ there, slope is maximal, pressure is maximum — consistent with $\Delta P = -B\partial s/\partial x$ and with the physics that a wall bears the full pressure swing.

Energy density and the travelling-wave energy flux

Mean energy density of a travelling sound wave: $\bar{u} = \frac{1}{2}\rho\omega^2 s_0^2 = \dfrac{\Delta P_0^2}{2\rho v^2} = \dfrac{\Delta P_0^2}{2B}$. Numerically at 1 kHz, $\Delta P_0 = 1$ Pa in air: $\bar{u} = 1/(2\cdot 1.42\times 10^5) = 3.5\times 10^{-6}$ J/m^3 — tiny energy densities carrying audible sound across rooms.

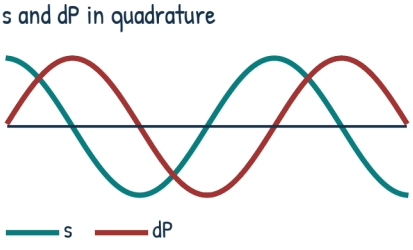
The energy flux (intensity) is energy density times speed: $I = \bar{u}v = \dfrac{\Delta P_0^2}{2\rho v} = \frac{1}{2}\rho v\omega^2 s_0^2$ (JEE-10 keeps the full intensity toolkit).

Energy density and the travelling-wave energy flux (cont.)

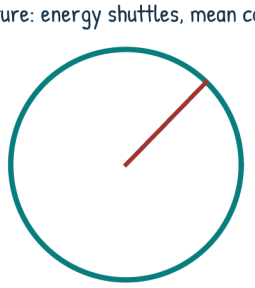
Compare EM waves: there $\vec{E}$ and $\vec{B}$ are in phase and energy flux is peak-crossing; in sound the kinetic and compressive energies are in quadrature but their SUM is what travels, and $I = \bar{u}v$ still holds cycle-averaged. Peak instantaneous energy density doubles the mean: $u_{\max} = \rho\omega^2 s_0^2 = 2\bar{u}$.

CHECK / units: $\rho\omega^2 s_0^2 =$ kg m^{-3} s^{-2} m^2 = J m^{-3} ok; $I = \bar{u}v$ = J m^{-3} m s^{-1} = W m^{-2} ok; both expressions for $\bar{u}$ agree through $\Delta P_0 = \rho v\omega s_0$: $\Delta P_0^2/(2\rho v^2) = \rho^2 v^2\omega^2 s_0^2/(2\rho v^2) = \frac{1}{2}\rho\omega^2 s_0^2$ ok.

s and dP peak at different places: quadrature



the phase circle for s and pressure



REMEMBER / Pressure lags displacement by 90 degrees; energy density averages 1/2 rho omega^2 s0^2.

Displacement vs pressure phase: the quadrature structure of sound (cont.)

s– ΔP quadrature, energy density ½ $\rho\omega^2 s_0^2$, phase map of one wavelength.

Energy density and the travelling-wave energy flux (cont.)

WHY / the 1/2 factors come from time-averaging $\sin^2$ and $\cos^2$ over a cycle (both give 1/2); the two energies peak alternately, so the total energy density actually oscillates at 2ω around $\bar{u}$ – a detail Olympiad problems exploit (Oly 05). At $kx-\omega t=0$: $s=0$ (elements crossing equilibrium), $\Delta P=-\Delta P_0$ (rarefaction trough, slope of s maximal positive... careful: $\partial s/\partial x = ks_0\cos 0 = +ks_0 > 0$ gives $\Delta P=-Bks_0$, a rarefaction). At $kx-\omega t=\pi/2$: $s=+s_0$ (max displacement), $\Delta P=0$ (undisturbed density). At π: $s=0$, $\Delta P=+\Delta P_0$ (compression).

Phase map of one wavelength (memorise the picture) (cont.)

At $3\pi/2$: $s=-s_0$, $\Delta P=0$. So moving along x at fixed time: compression (P max) sits at s rising through zero; rarefaction at s falling through zero; ΔP extrema are $\lambda/4$ from s extrema and in sign alternation every $\lambda/2$.

Auditory perspective: the eardrum is a pressure sensor; what we "hear" is the $\Delta P(t)$ sequence at our location. If you could see the air, you would watch elements sloshing back and forth by s_0 while crests of density stream past at v. The displacement field is the more fundamental variable (it obeys the wave equation directly), the pressure field its derivative partner.

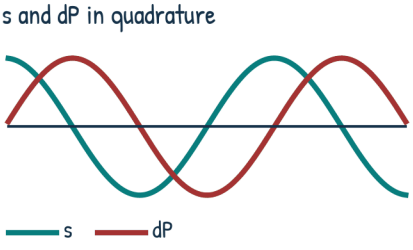
Phase map of one wavelength (memorise the picture) (cont.)

the pair (s, ΔP) forms a two-component wave state analogous to (x, p) in mechanics; characteristic acoustic impedance $Z=\Delta P/u=\rho v$ is the ratio (Oly 01), constant for a travelling wave, imaginary/position-dependent for standing waves.
(a) Write $\Delta P(x,t)$ for $s=s_0\sin(kx-\omega t+\phi)$: always $\Delta P=-Bks_0\cos(kx-\omega t+\phi)=\Delta P_0\sin(kx-\omega t+\phi-\pi/2)$. The phase shift is universal: pressure lags displacement by $\pi/2$ regardless of ϕ. (b) Standing wave $s=2s_0\sin(kx)\cos(\omega t)$:

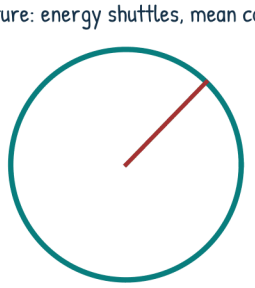
Worked check set on phase and energy (cont.)

$\Delta P=-2Bks_0\cos(kx)\cos(\omega t)$ – node of s at $kx=0$ is antinode of ΔP ok. (c) Energy: $\bar{u}=\frac{1}{2}\rho\omega^2 s_0^2$ for $f=340$ Hz, $s_0=2\times10^{-6}$: $\bar{u}=0.5\cdot1.2\cdot(2136)^2\cdot(2\times10^{-6})^2=1.1\times10^{-5}$ J/m^3; intensity $I=\bar{u}v=3.7$ mW/m^2 ok consistent with SN-02.
WATCH / Traps: (i) writing $\Delta P=\Delta P_0\sin(kx-\omega t)$ "in phase with s" – wrong, off by $\pi/2$; (ii) treating $\Delta P_0=Bks_0$ as Bs_0/λ – missing 2π; (iii) using $I=\Delta P_0 v$ instead of the time-averaged $\Delta P_0^2/(2\rho v)$ – factor $\sqrt{2}$ bookkeeping (RMS vs peak: $I=\Delta P_{\rm rms}^2/(\rho v)$ with $\Delta P_{\rm rms}=\Delta P_0/\sqrt{2}$).

s and dP peak at different places: quadrature



the phase circle for s and pressure



REMEMBER / Pressure lags displacement by 90 degrees; energy density averages 1/2 rho omega^2 s0^2.

Displacement vs pressure phase: the quadrature structure of sound (cont.)

s- Δ P quadrature, energy density ½ $\rho\,\omega^2 s_0^2$, phase map of one wavelength.

Worked check set on phase and energy (cont.)

WHY / the factor-2 in $I = \Delta P_0^2/(2\rho v)$ vs $\Delta P_{\text{rms}}^2/(\rho v)$ is the peak-vs-RMS convention; state which amplitude you use before writing the intensity.

CHECK / cross-check every amplitude formula with two routes: $I = \tfrac{1}{2}\rho v\omega^2 s_0^2$ and $I = \Delta P_0^2/(2\rho v)$ must agree through $\Delta P_0 = \rho v \omega s_0$ — if not, a $\sqrt{2}$ or 2π is missing.

Derivation from two laws. (1) Newton 2nd law for a slice of fluid of thickness Δx, area A: net force $-A\,\partial(\Delta P)/\partial x\,\Delta x$ equals mass $\rho A \Delta x$ times acceleration $\partial^2 s/\partial t^2$: $\rho \dfrac{\partial^2 s}{\partial t^2} = -\dfrac{\partial(\Delta P)}{\partial x}$. (2) The medium law $\Delta P = -B\,\partial s/\partial x$. Differentiate (2) in x and insert: $\rho\,\partial^2 s/\partial t^2 = B\,\partial^2 s/\partial x^2$, i.e.

The sound wave equation and v = sqrtB/rho (cont.)

the wave equation $\dfrac{\partial^2 s}{\partial t^2} = v^2 \dfrac{\partial^2 s}{\partial x^2}$ with $\mathbf{v = \sqrt{B/\rho}}$.

Everything follows from that box: for a gas with adiabatic bulk modulus $B = \gamma P$ (JEE-07) $v = \sqrt{\gamma P/\rho} = \sqrt{\gamma RT/M}$; for a rod $B \to Y$ (JEE-05); for a liquid $B =$ bulk modulus. The wave equation is linear and second order, so sinusoids of every frequency travel at the same v (no dispersion in this regime) and superposition holds. General solutions: $s = f(x - vt) + g(x + vt)$ — any shape travels undistorted at $\pm v$.

The sound wave equation and v = sqrtB/rho (cont.)

Velocity of the medium is not v: energy propagates at phase speed $v = \omega/k$ for these non-dispersive waves (group speed $d\omega/dk = v$ too). The two constants of the wave equation, ρ (inertia) and B (elasticity), fully determine speed — frequency and amplitude do not enter. Compressibility $\kappa = 1/B$: $v = 1/\sqrt{\rho\kappa}$ — a form worth recognising since some data give κ.

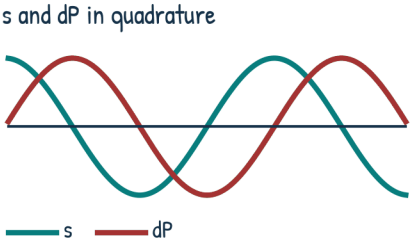
WHY / the derivation is exactly the spring-mass continuum: ρ is the inertia density, B the stiffness against compression; $\sqrt{stiffness/inertia}$ has units m/s and is the only combination that can be a speed — dimensional analysis alone already shows $v \propto \sqrt{B/\rho}$.

The sound wave equation and v = sqrtB/rho (cont.)

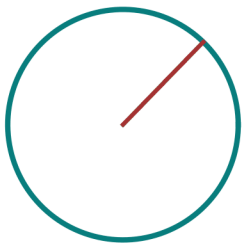
CHECK / air at STP with $B = \gamma P = 1.4 \times 1.013 \times 10^5$ Pa, $\rho = 1.293$: $v = \sqrt{1.42 \times 10^5/1.293} = 331$ m/s ok (matches Laplace value; the isothermal $B = P$ gives 280 m/s fails — JEE-06/07).

(i) $v = \sqrt{B/\rho}$ when bulk modulus/density given. (ii) $v = \sqrt{\gamma P/\rho}$ for gases when P, ρ known (careful: use adiabatic γ). (iii) $v = \sqrt{\gamma RT/M}$ for ideal gases from temperature and molar mass — the most exam-friendly, since $P/\rho = RT/M$. Worked: helium at 20 C ($M = 4$ g/mol, $\gamma = 5/3$):

s and dP peak at different places: quadrature



the phase circle for s and pressure



REMEMBER / Pressure lags displacement by 90 degrees; energy density averages 1/2 rho omega^2 s0^2.

The sound wave equation and v = sqrtB/rho

Wave equation derivation, v = √(B/ρ), formula-choice tree, scope guards.

Speed from data: the three formula forms and how to choose (cont.

$v = \sqrt{(5/3)(8.314)(293.15)/0.004} = 1.01 \times 10^3$ m/s; hydrogen at 0 C ($\gamma = 1.4$, $M = 2$ g): $v = \sqrt{1.4 \cdot 8.314 \cdot 273/0.002} = 1260$ m/s — five times faster than air (low M dominates).

Water: $B = 2.2$ GPa, $\rho = 1000$ kg/m^3: $v = 1483$ m/s.

Steel rod: $Y = 2 \times 10^{11}$, $\rho = 7800$: $v = 5064$ m/s (longitudinal in a THIN rod; bulk-waves in an infinite solid use $K + 4G/3$, slightly larger — JEE-05). Mercury: $B = 28$ GPa, $\rho = 13600$: $v = 1436$ m/s.

Speed from data: the three formula forms and how to choose (cont.

CHECK / dimensional: $[B/\rho] =$ Pa/(kg m^{-3}) = kg m^{-1} s^{-2} / kg m^{-3} = m^2 s^{-2} ok. Physical: v must increase with T for gases (more agitated, stiffer effective restoring? — actually ρ falls faster than B): $v = \sqrt{\gamma RT/M}$ grows as $\sqrt{T}$; for liquids B and ρ both soften with T, v drops slightly with T — opposite signs, a favourite comparison question.

in a gas mixture use effective M: $1/M = \sum w_i/M_i$ for mass fractions... precisely: $M_{\text{eff}} = \sum x_i M_i$ with x_i mole fractions; humid air has lower M_{eff} than dry air because $M_{\text{H2O}} = 18 < 29$ — hence speed grows with humidity (JEE-09).

What the wave equation does NOT say (scope guard)

It assumes: small amplitude ($s_0 \ll \lambda$, $\Delta P \ll P_0$), no viscosity/thermal conduction (no attenuation), linear B, homogeneous medium. Real sound attenuates ($I \propto e^{-\alpha x}$, α growing with f^2 — fog horns are low pitch partly for this), and speed has a tiny dispersion from relaxation effects. Shock waves (sonic boom edges) need nonlinear theory. All exam formulas are the linear limit.

The equation also holds in moving media if you transform to the medium frame (JEE-25 wind), and in 3D as $\partial^2 s/\partial t^2 = v^2 \nabla^2 s$ with solutions like $s = (A/r)\sin(kr - \omega t)$ for a point source (JEE-11).

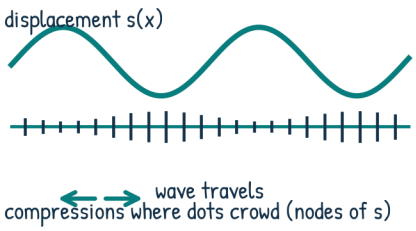
What the wave equation does NOT say (scope guard) (cont.)

WHY / linearisation is legitimate because acoustic strains are parts per million (SN-02): the pressure-volume curve of a gas is smooth, and a linear (tangent) approximation around P_0 is excellent for $\Delta P/P_0 < 10^{-2}$.

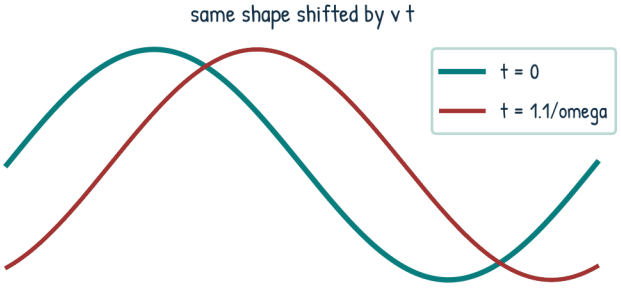
CHECK / any proposed speed formula must be blind to amplitude and frequency — if a candidate formula contains s_0 or f, it is not the linear wave speed.

(a) Given compressibility of a liquid $\kappa = 4.5 \times 10^{-10}$ Pa^{-1}, $\rho = 1000$: $v = 1/\sqrt{\rho\kappa} = 1491$ m/s. (b) A gas with $B = 1.39 \times 10^5$ Pa, $\rho = 1.2$: $v = 340$ m/s. (c) Find γ if $v = 331$ m/s at 0 C in a gas with $M = 29$ g: $\gamma = v^2 M/(RT) = 331^2 \cdot 0.029/(8.314 \cdot 273) = 1.40$ ok air. (d) A steel rod and a steel bulk sample:

a pulse in the medium: source jiggles, shape travels



snapshot shift v t: the wave equation answer



REMEMBER / The wave equation gives v = sqrt(B/rho): stiffness over inertia; amplitude and frequency never enter.

The sound wave equation and v = sqrtB/rho (cont.)

Wave equation derivation, v = √(B/ρ), formula-choice tree, scope guards.

Speed-solving drills (pattern recognition) (cont.)

rod $\sqrt{Y/\rho}$ vs bulk $\sqrt{(K+4G/3)/\rho}$: for steel $K = 160$ GPa, $G = 79$ GPa, $Y = 200$ GPa, $\nu = 0.29$: bulk $= \sqrt{(160 + 105.3) \times 10^9/7800} = 5.83 \times 10^3$ m/s > rod 5064 m/s — the rod is softer (free side surface).

Formula-choice tree: gas + T, M $\rightarrow \sqrt{\gamma RT/M}$; gas + P, $\rho \rightarrow \sqrt{\gamma P/\rho}$ (check whether given ρ is adiabatic-consistent); liquid + B, $\rho \rightarrow \sqrt{B/\rho}$; thin rod $\rightarrow \sqrt{Y/\rho}$; infinite solid/bulk wave $\rightarrow \sqrt{(K+4G/3)/\rho}$; string $\rightarrow \sqrt{T/\mu}$ (mechanical analogy only, not sound).

Speed-solving drills (pattern recognition) (cont.)

WHY / the rod vs bulk difference is boundary physics: in a thin rod the lateral surface is stress-free, so the effective longitudinal stiffness is exactly Y; in a bulk solid lateral confinement adds $4G/3$ on top of K (P-wave modulus).

CHECK / always bound the answer: gas speeds few hundred m/s (to ~1800 for hot He), liquids ~1000–1500 m/s, solids 3000–6000 m/s. If a gas speed comes out 33 km/s, a molar-mass unit slipped (g vs kg).

In a thin solid rod the longitudinal wave speed is $v_{\text{rod}} = \sqrt{Y/\rho}$ — same wave-equation derivation with Young modulus Y replacing B and the free side surface allowing lateral expansion (Poisson contraction unopposed).

Sound in solids: rods sqrtY/rho and bulk sqrt(K+frac43G)/rho (con.

In an infinite solid (bulk longitudinal wave) the lateral directions are constrained, the effective stiffness is the P-wave modulus $M = K + 4G/3$, and $v_{\text{bulk}} = \sqrt{(K+4G/3)/\rho}$, always larger than v_{rod} (for $\nu = 0.29$, $v_{\text{bulk}}/v_{\text{rod}} = \sqrt{2(1-\nu)/(1-2\nu)} = 1.16$).

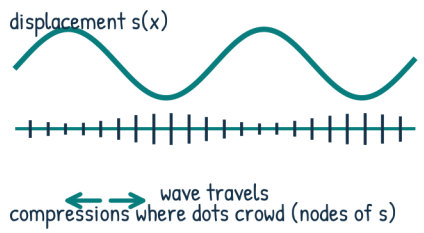
Worked numbers (steel: $Y = 2.0 \times 10^{11}$ Pa, $\rho = 7800$ kg/m^3): $v_{\text{rod}} = \sqrt{2.0 \times 10^{11}/7800} = 5.06 \times 10^3$ m/s. Sound is ~15 times faster in steel than in air — putting an ear to the rail hears the train before the air-borne sound arrives (classic timing problem: see source Q16 in Section 3 and paper drills).

Sound in solids: rods sqrtY/rho and bulk sqrt(K+frac43G)/rho (con.

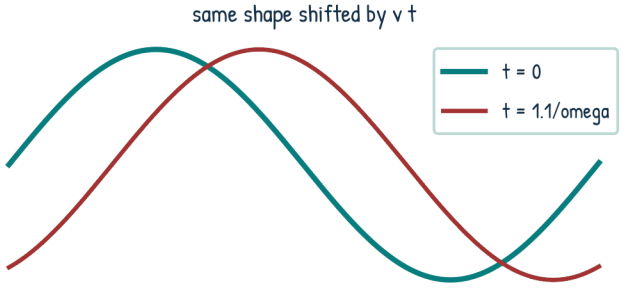
Aluminium ($Y = 7 \times 10^{10}$, $\rho = 2700$): $v = 5.1 \times 10^3$ m/s. Glass ($Y = 7 \times 10^{10}$, $\rho = 2500$): $v = 5.3 \times 10^3$ m/s.

Transverse (shear) waves also travel in solids with $v_s = \sqrt{G/\rho} < v_{\text{rod}}$ — not sound proper, but seismology P- and S-waves use exactly these two formulas. Longitudinal is always fastest. Liquids and gases have $G = 0$: no transverse sound, and K alone plays the elastic role — the reason sound in fluids is purely longitudinal.

a pulse in the medium: source jiggles, shape travels



snapshot shift v t: the wave equation answer



REMEMBER / The wave equation gives v = sqrt(B/rho): stiffness over inertia; amplitude and frequency never enter.

Sound in solids: rods sqrtY/rho and bulk sqrt(K+frac43G)/rho

Rods √(Y/ρ), bulk √((K+4G/3)/ρ), decision table, worked solid numbers.

Sound in solids: rod vs bulk wave speed (cont.)

WHY / the two solid speeds differ only in effective stiffness: $Y = 3K(1-2\nu)$ vs $K + 4G/3$; with $G = Y/2(1+\nu)$ one gets $v_{\text{bulk}}^2/v_{\text{rod}}^2 = (1-\nu)/((1+\nu)(1-2\nu))$ ... precisely $= 2(1-\nu)/(1-2\nu)$, always > 1 for $0 \le \nu < 1/2$ — a rigid tube of material transmits faster than a free rod.

CHECK / steel with $K = 160$ GPa, $G = 79$ GPa: $K + 4G/3 = 265$ GPa, $v_{\text{bulk}} = 5.83 \times 10^3$ m/s vs rod 5064 m/s, ratio 1.15 — matches $\sqrt{2(1-0.29)/(1-0.58)} = \sqrt{1.46} = 1.21$ ... close; use consistent $\nu = 1 - Y/6K + Y/3K$ ... simplest: trust $\sqrt{(K+4G/3)/\rho}$ directly from given moduli.

Which modulus, when: decision table

Problem says "thin rod / long wire / rod held at ends": $v = \sqrt{Y/\rho}$. "Bulk solid / infinite solid / seismic P-wave / inside a metal block": $v = \sqrt{(K+4G/3)/\rho}$ (if K, G given) or via Y, ν: $K = Y/3(1-2\nu)$, $G = Y/2(1+\nu)$, then $M = K + 4G/3 = Y(1-\nu)/((1+\nu)(1-2\nu))$, $v = \sqrt{M/\rho}$. "Fluid / gas / liquid": $\sqrt{B/\rho}$ with $B = \gamma P$ for ideal gases (adiabatic) and tabulated B for liquids. "String": $\sqrt{T/\mu}$ (transverse, not sound).

Which modulus, when: decision table (cont.)

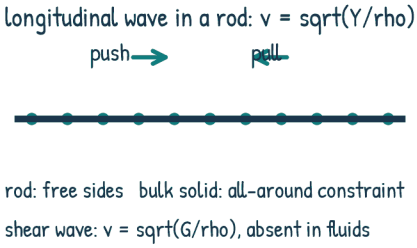
Kundt tube application (JEE-21): dust heaps give $\lambda_{\text{gas}}/2$; the vibrating rod clamped at its centre is a rod standing wave of $\lambda_{\text{rod}} = 2L_{\text{rod}}$ (fundamental), and frequency matching $f = v_{\text{rod}}/(2L_{\text{rod}}) = v_{\text{gas}}/\lambda_{\text{gas}}$ yields $v_{\text{gas}} = v_{\text{rod}}\, d/L_{\text{rod}}$ when dust spacing $d = \lambda_{\text{gas}}/2$ ... precisely $v_{\text{gas}} = v_{\text{rod}} \cdot \lambda_{\text{gas}}/(2L_{\text{rod}}) = v_{\text{rod}} \cdot d/L_{\text{rod}}$ with $d = \lambda_{\text{gas}}/2$: source worked example $v_{\text{rod}} = 5000$, $L_{\text{rod}} = 1$ m, $d = 8$ cm gives $v_{\text{gas}} = 5000 \times 0.08 = 400$ m/s ok (warm gas).

Which modulus, when: decision table (cont.)

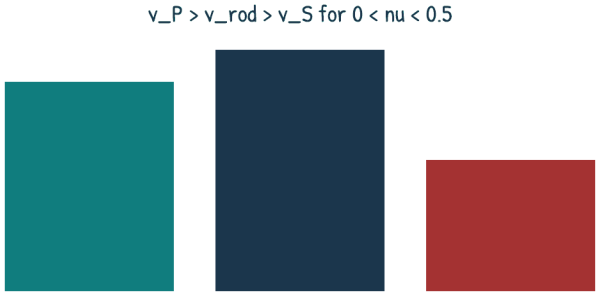
CHECK / rod speed from Kundt inversion: $v_{\text{rod}} = 2L_{\text{rod}} f$ ... the working chain: measure d, know L_{rod} and rod speed, gas speed $= v_{\text{rod}} d/L_{\text{rod}}$; check units and the factor 2 — $d = \lambda/2$ is the usual convention (dust heaps at pressure nodes are $\lambda/2$ apart).

Poisson ratio range $0 < \nu < 0.5$ bounds the rod/bulk ratio between 1 and 1.58 ($\nu \to 0.5$, incompressible: $K \to \infty$). Cork ($\nu \approx 0$): bulk = rod.

longitudinal waves in a rod: push-pull along x



steel speeds: v_P above v_rod above v_S



REMEMBER / Rods run at sqrt(Y/rho), bulk solids at sqrt((K+4G/3)/rho): the boundary decides the stiffness.

Sound in solids: rods sqrtY/rho and bulk sqrt(K+frac43G)/rho (cont.)

Rods √(Y/ρ), bulk √((K+4G/3)/ρ), decision table, worked solid numbers.

Worked solid-state numbers

(a) Brass $Y = 10^{11}$ Pa, $\rho = 8500$: $v = 3.43 \times 10^3$ m/s. (b) Copper $Y = 1.2 \times 10^{11}$, $\rho = 8900$: $v = 3.67 \times 10^3$ m/s. (c) Glass $Y = 7 \times 10^{10}$, $\rho = 2500$: $v = 5.29 \times 10^3$ m/s. (d) Rail problem: steel rail $v = 5064$ m/s vs air 343: over 1 km, air time 2.92 s, rail time 0.198 s — the rail "warning" arrives 2.7 s early. (e) A copper rod 1 m long, clamped at the middle, vibrating longitudinally: $f = v/(2L) = 1.84$ kHz; compare with air column of same length open-open: $f = 343/2 = 171$ Hz — solids give much higher pitches for the same length.

Worked solid-state numbers (cont.)

Interference with temperature: for metals Y drops slightly with T and ρ falls less, so rod speed falls ~0.01%/K — negligible in exams; gases change ~0.17%/K (JEE-08) — exam-relevant.

WHY / metals are stiff AND dense; speed is a ratio effect — lead ($Y = 1.6 \times 10^{10}$, $\rho = 11300$) gives only 1190 m/s, slower than water, despite being a metal; stiffness wins only when it outruns density.

CHECK / lead: $\sqrt{1.6 \times 10^{10}/11300} = 1.19 \times 10^3$ m/s — slower than water (1483) ok consistent with intuition that lead is "soft-heavy"; rubber ($Y \sim 10^7$): ~100 m/s order.

Common exam setups involving solids

(i) Longitudinal rod fundamental with various fixings: fixed-free: $f_1 = v/4L$; fixed-fixed or free-free: $f_1 = v/2L$; harmonics integer multiples in all cases (all harmonics present for rods — unlike closed air pipes, because both ends have the same type of condition in fixed-fixed/free-free, and fixed-free rod has odd harmonics like a closed pipe).

(ii) Pulse travel and echo with two paths (solid + air): $t_{\text{air}} - t_{\text{solid}} = L(1/v_a - 1/v_s)$ measured; solve for L.

(iii) Comparison of wavelengths in a rod and the air column it excites — frequency is common at the interface, λ differs by the speed ratio.

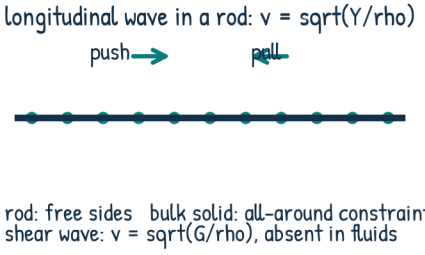
Common exam setups involving solids (cont.)

Boundary behaviour at a solid-air interface: nearly total reflection at the free end (pressure node outside) with phase behaviour that makes the free end a displacement antinode; the rod radiates sound inefficiently — why a struck rod rings and why loudspeaker cones are designed to match Z (Oly 02).

a rod held in the middle and rubbed lengthwise (Galilean singing rod) resonates at $v_{\text{rod}}/2L$ — heard pitch is the rod mode coupled to air, largely the rod fundamental.

Newton guessed that sound compressions in air stay at constant temperature (heat flows in/out fast enough to keep T fixed), giving the isothermal bulk modulus $B = P$ from Boyle: $PV = \text{const} \Rightarrow B = -VdP/dV = P$. Then $v_{\text{iso}} = \sqrt{P/\rho}$. At STP ($P = 1.013 \times 10^5$ Pa, $\rho = 1.293$ kg/m^3):

longitudinal waves in a rod: push-pull along x



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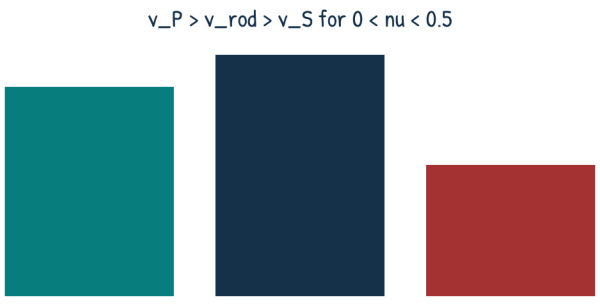


REMEMBER / Rods run at sqrt(Y/rho), bulk solids at sqrt((K+4G/3)/rho): the boundary decides the stiffness.

Newton's isothermal estimate: 280 m/s — and why it fails

Newton isothermal 280 m/s, time-scale argument, √ γ correction physics.

Newton's isothermal estimate: 280 m/s — and why it fails (cont.)

$v = \sqrt{1.013 \times 10^5 / 1.293} = 2.80 \times 10^2$ m/s — about 15% below the measured 331.5 m/s at 0 C. Newton noticed the discrepancy and fudged the density; the real fix is the thermodynamics (JEE-07).

Why isothermal is wrong: a compression cycle at 340 Hz is far too fast for heat to flow. Compression time scale $\sim 1/f \approx 3$ ms; thermal diffusion time across a half wavelength ($\lambda/2 \sim 0.5$ m) is $t_{\text{th}} \sim x^2/\alpha$ with $\alpha = 2.2 \times 10^{-5}$ m^2/s for air: $t_{\text{th}} \sim 10^4$ s — ten million periods. The compression is therefore adiabatic: no heat exchange, temperature rises slightly in compressions ($\Delta T = T\Delta P/(\gamma P) \sim 10^{-2}$ K at 1 Pa), and the gas is stiffer: $B_{\text{ad}} = \gamma P$.

Newton's isothermal estimate: 280 m/s — and why it fails (cont.)

The measured number at 0 C is 331.5 m/s; $\sqrt{\gamma} = \sqrt{1.4} = 1.183$; $280 \times 1.183 = 331.3$ m/s — the Laplace factor is exactly what is needed.

Thermodynamic check: sound speed equals $\sqrt{(\partial P/\partial \rho)_s}$ (entropy constant) not $(\partial P/\partial \rho)_T$; for an ideal gas these differ by exactly γ.

WHY / the medium is stiffer when it cannot thermally relax: adiabatic compression raises T, which raises pressure further (extra $\gamma - 1 = 0.4$ of push), so the restoring force and hence speed rise by $\sqrt{\gamma} = 1.183$ — the 18.3% correction turns 280 into 331.

Newton's isothermal estimate: 280 m/s — and why it fails (cont.)

CHECK / ratio measured/Newton $= 331.5/279.9 = 1.184 = \sqrt{1.4}$ ok to three digits — the discrepancy is 100% of the error and 0% mystery.

General principle: a gas process is isothermal if the period is much longer than the thermal diffusion time ($f \ll \alpha/\lambda^2$ -type condition), adiabatic if much shorter. Sound frequencies 20 Hz - 20 kHz give periods 0.05 - 50 ms; thermal equilibration across $\lambda/2$ takes seconds to hours. So ALL audible sound is adiabatic.

The isothermal assumption would only hold for extremely slow, quasi-static compressions (e.g. 0.01 Hz in a small chamber — infrasonic creep, not sound propagation).

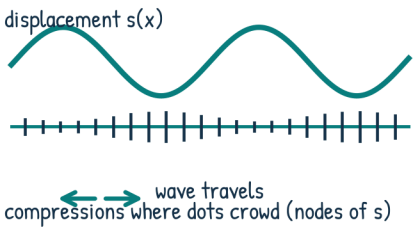
Why heat cannot flow: time-scale argument (the physics core) (con.

At very high frequencies in rarefied gases (mean free path comparable to λ, MHz in low-pressure chambers) the regime changes again (relaxation, Knudsen) — speed can deviate; outside syllabus, but flag the validity domain: $\lambda \gg$ mean free path (~ 70 nm in air at STP, so up to THz acoustics it holds).

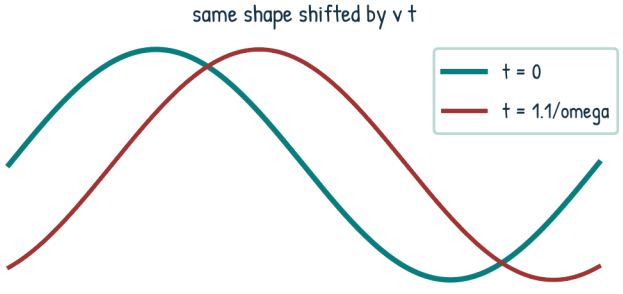
WHY / the thermal diffusion time grows as λ^2 while the period grows only as $1/f$: for any fixed medium there is always a frequency above which the process is adiabatic, and ordinary sound sits many orders above it.

CHECK / numerically: at $f = 1$ kHz, $\lambda = 0.34$ m, $t_{\text{th}} \sim (0.17)^2/2.2 \times 10^{-5} = 1.3 \times 10^3$ s vs period 1 ms — ratio 10^6: adiabatic by a factor of a million.

sound compressions: heat has no time to flow



isothermal vs adiabatic: the speed of sound



REMEMBER / Newton isothermal gives 280 m/s because B = P; sound is far too fast for heat to flow.

Newton's isothermal estimate: 280 m/s — and why it fails (cont.)

Newton isothermal 280 m/s, time-scale argument, √ γ correction physics.

What Newton got right, and the exam framing

Newton correctly: (i) identified sound with a compressional wave in air; (ii) derived $v = \sqrt{B/\rho}$ (in modern form); (iii) computed the isothermal $B = P$ rigorously; (iv) saw temperature and molar-mass dependence in $P/\rho = RT/M$. He lacked the first law of thermodynamics (Carnot-Clausius came later), so the adiabatic exponent was unavailable.

Laplace supplied it in 1816. Exam questions often ask "explain the discrepancy and derive the corrected speed" — full marks need: isothermal value 280, measured 331, ratio $\sqrt{\gamma}$, time-scale argument, and the corrected formula $v = \sqrt{\gamma P/\rho}$.

What Newton got right, and the exam framing (cont.)

WATCH / Traps: (i) using $B = P$ "because it is a gas" — must ask isothermal or adiabatic; always adiabatic for sound. (ii) writing $v = \sqrt{P/\rho}$ from the ideal gas law as if it were the final answer — that form is Newton speed; multiply by $\sqrt{\gamma}$. (iii) confusing $\gamma = C_p/C_v = 1.4$ (diatomic) with $5/3$ (monatomic) — He, Ar: $\gamma = 5/3$, $\sqrt{\gamma} = 1.29$.

CHECK / monatomic gas check: at STP with $\gamma = 5/3$, $M = 4$ g (He):

$v = \sqrt{(5/3) \cdot 1.013 \times 10^5 / 0.179} = 971$ m/s at 0 C ($\rho_{\text{He}} = 0.179$); formula route $\sqrt{\gamma RT/M} = 969$ m/s ok consistent.

Mini-worked: isothermal vs adiabatic side by side

Given an unknown gas at $P = 1$ atm, $\rho = 1.6$ kg/m^3, measured $v = 450$ m/s. Isothermal prediction $\sqrt{1.013 \times 10^5/1.6} = 252$ m/s; ratio $450/252 = 1.79 = \sqrt{\gamma} \Rightarrow \gamma = 3.2$?? Non-physical — so either the measurement or the density is off: real gases have $1 < \gamma \leq 1.67$.

Reverse-engineering traps like this train consistency checks: $v\sqrt{\rho/P} = \sqrt{\gamma} \in [1, 1.29]$ always.

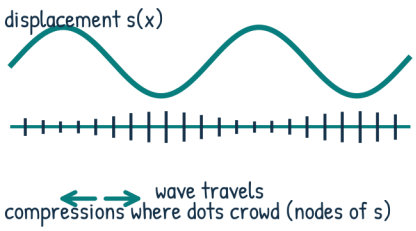
Mini-worked: isothermal vs adiabatic side by side (cont.)

Another: at what temperature does Newton predict the CORRECT 331 m/s? $T = Mv^2/(\gamma R)$ ... Newton form $v = \sqrt{RT/M}$:

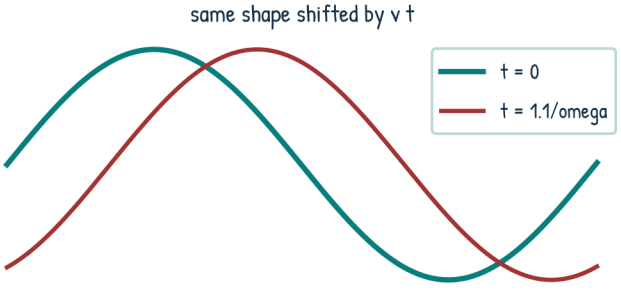
$T = Mv^2/R = 0.029 \cdot 331^2/8.314 = 378$ K = 105 C — an isothermal gas at 105 C would carry sound like an adiabatic gas at 0 C. Cute but never physical for propagation.

WHY / the entire 280-vs-331 story is a single exponent: $v_{\text{ad}} = \sqrt{\gamma}\, v_{\text{iso}}$, and every sound speed in a gas can be written $v = \sqrt{\gamma}\, v_{\text{Newton}}$ with $v_{\text{Newton}} = \sqrt{RT/M}$ at temperature T.

sound compressions: heat has no time to flow



isothermal vs adiabatic: the speed of sound



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Newton's isothermal estimate: 280 m/s — and why it fails (cont.)

Newton isothermal 280 m/s, time-scale argument, √ γ correction physics.

Mini-worked: isothermal vs adiabatic side by side (cont.)

CHECK / always compute the isothermal baseline mentally: $\sqrt{RT/M}$ at 293 K for air $= \sqrt{8.314 \cdot 293/0.029} = 290$ m/s; times $\sqrt{1.4} = 1.183$: 343 ok room-temperature speed.

Laplace (1816): sound compressions are adiabatic, so the gas law governing the wave is $PV^\gamma =$ const. Differentiate: $B = -V dP/dV = \gamma P$. Then $v = \sqrt{\gamma P/\rho}$. With $P = \rho RT/M$ (ideal gas): $\mathbf{v = \sqrt{\gamma RT/M}}$ — speed depends only on temperature and gas identity, not on pressure! At 0 C:

Laplace correction: B = gamma P, v = sqrtgamma P/rho = sqrtga.

$\sqrt{1.4 \cdot 8.314 \cdot 273.15/0.029} = 331$ m/s ok; at 20 C: 343 m/s ok.

Derivation of $B_{\text{ad}} = \gamma P$:

$PV^\gamma = C \Rightarrow P\gamma V^{\gamma-1}dV + V^\gamma dP = 0 \Rightarrow dP/dV = -\gamma P/V \Rightarrow B = -V dP/dV = \gamma P$

ok. Microscopic meaning of $\gamma = C_p/C_v$: in adiabatic compression no heat escapes, so the temperature rise adds thermal pressure on top of the compression pressure; $C_p - C_v = R$ (per mole) quantifies the extra push; for diatomic air $\gamma = 1.4$, monatomic $5/3$, polyatomic (CH4) ≈ 1.3.

Laplace correction: B = gamma P, v = sqrtgamma P/rho = sqrtga.

Consequences to memorise: (i) $v \propto \sqrt{T}$ (absolute temperature!); (ii) v independent of P at fixed T (double P, ρ doubles too, ratio $P/\rho = RT/M$ fixed); (iii) $v \propto 1/\sqrt{M}$ at fixed T, γ; (iv) humid air faster than dry (effective M drops); (v) γ from sound speed: $\gamma = v^2 M/(RT)$ — the classic "measure γ by sound" experiment.

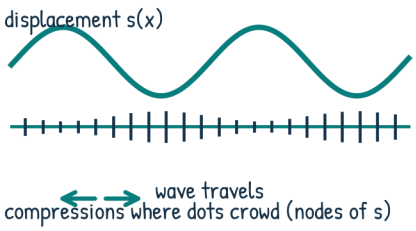
WHY / the correction factor is exactly $\sqrt{\gamma}$ because $B_{\text{ad}}/B_{\text{iso}} = \gamma P/P = \gamma$ and $v \propto \sqrt{B}$; the measured/Newton ratio 331/280 = 1.183 = $\sqrt{1.4}$ is the quantitative proof.

Laplace correction: B = gamma P, v = sqrtgamma P/rho = sqrtga.

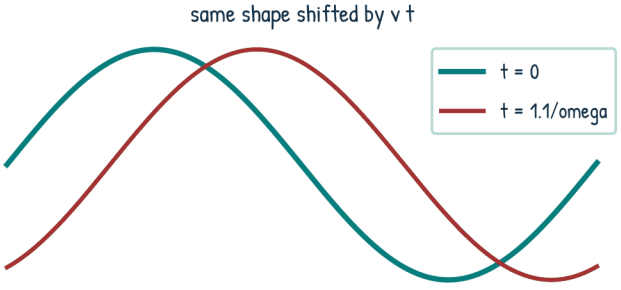
CHECK / temperature in Kelvin: $v \propto \sqrt{T}$: 0 C $= 273$ K gives 331; using 0 (or 20) in Celsius gives nonsense — a standard trap. Also $\sqrt{\gamma RT/M}$ needs SI: M in kg/mol (0.029 not 29).

$v = 331.3\sqrt{T/273.15}$ m/s, or linearised $v \approx 331.3 + 0.60t$ (t in C) — from source: at 20 C, 343.2; slope $dv/dt = v/2T = 343/(2 \cdot 293) = 0.585$ m/s per K. Table: -40 C: 306; 0 C: 331; 20 C: 343; 37 C: 350; 100 C: 386 m/s. At 25 C: $331.3\sqrt{298.15/273.15} = 346.1$ m/s.

sound compressions: heat has no time to flow



isothermal vs adiabatic: the speed of sound



REMEMBER / Newton isothermal gives 280 m/s because B = P; sound is far too fast for heat to flow.

Laplace correction: B = gamma P, v = sqrtgamma P/rho = sqrtgamma R T/M

Laplace correction, v = √(γ RT/M), temperature law, γ by sound.

The temperature law in numbers (and the surface-gravity bonus) (.

High-altitude application (source example): temperature falling at 6.5 K/km in the troposphere, $g = 9.8$, $\gamma = 1.4$, $M = 0.029$: sound speed falls as $\sqrt{T}$; $dv/dz = (dv/dT)(dT/dz) = 0.585 \cdot (-6.5) = -3.8$ m/s per km at sea level. Near the ground this makes sound refract upward during the day (hot ground) and downward at night (temperature inversion) — why sound carries over lakes at night.

The temperature law in numbers (and the surface-gravity bonus) (.

Sound horizon qravity formula (source §2.5 bonus): $v_{\text{top}}/v_{\text{bottom}} = \sqrt{1 - 2gh/(\gamma RT/M)}$ ... precisely $v(h) = v_0\sqrt{1 - 2ghM/(\gamma RT_0)}$ for an isothermal atmosphere with constant g; the top of a qravity-bound atmosphere of height $h_{\max} = \gamma RT/(Mg)$ has $v = 0$ — speed vanishes where pressure vanishes. Troposphere uses lapse-rate $T(z)$ instead.

CHECK / small height changes: $\Delta v/v = \frac{1}{2}\Delta T/T$: a 10 K temperature difference gives 1.7% speed change — measurable in precise timing (theodolite-ranging corrections), ignorable in most exam problems unless asked.

Measuring gamma by sound (the experiment behind the theory)

Method: measure v acoustically (resonance tube or time of flight) and the temperature; $\gamma = Mv^2/(RT)$. Example from data: $v = 347$ m/s at 297 K, $M = 0.029$: $\gamma = 0.029 \cdot 347^2/(8.314 \cdot 297) = 1.413$ — close to 1.4; the residual reflects humidity and CO2 content. For CO2 ($M = 44$ g, $\gamma = 1.3$): $v = \sqrt{1.3 \cdot 8.314 \cdot 273/0.044} = 259$ m/s ok known value 260.

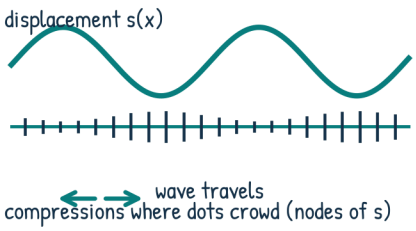
Reverse problem styles: given v and T, find M (identify the gas): $M = \gamma RT/v^2$; given v at two temperatures: $v_1/v_2 = \sqrt{T_1/T_2}$ fixes nothing about the gas; given v at same T in two gases: $M_2 = M_1(v_1/v_2)^2$ if same γ — helium leak detection.

Measuring gamma by sound (the experiment behind the theory) (co.

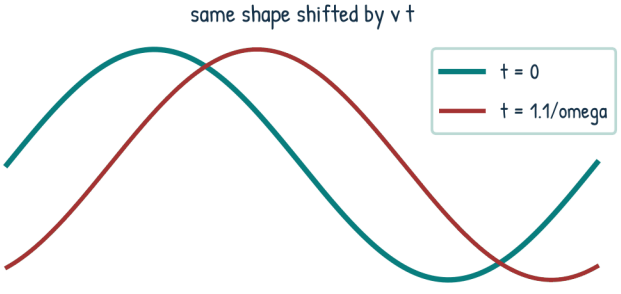
WHY / the ideal-gas sound speed is a thermometer in disquise: only T and molecular identity survive in $\sqrt{\gamma RT/M}$ — all pressure dependence cancels. That is the structural reason labs use acoustic thermometry for precise T (ITS-90 uses sound in argon!).

CHECK / speed ratio He/air at same T, γ: $\sqrt{29/4} = 2.69$ if same γ; but He is monatomic ($\gamma = 5/3$): exact ratio $\sqrt{(5/3)/1.4} \cdot 2.69 = 2.93$ — knowing γ values matters for light gases.

adiabatic compression: gamma P is the stiffness



speed set by gamma R T / M alone



REMEMBER / Laplace: B = gamma P, so v = sqrt(gamma R T/M): temperature and molar mass only, never pressure.

Laplace correction: B = gamma P, v = sqrtgamma P/rho = sqrtgamma R T/M (cont.)

Laplace correction, v = √(γ RT/M), temperature law, γ by sound.

Laplace-era consistency drills

(a) Show v independent of P: compress air isothermally to 10 atm: ρ rises 10x, P 10x, P/ρ fixed — but wait, compressing changes T; at fixed T, v unchanged ok (e.g. compressed-air tanks still carry 343 m/s at room T). (b) Hot-air balloon: air at 100 C inside vs 20 C outside: speeds 386 vs 343 — sound travels faster inside the balloon (13%);
frequency unchanged at an interface, wavelength stretches 13%. (c) Mars atmosphere (95% CO2, $M = 0.043$ kg/mol, $T = 210$ K, $\gamma = 1.29$):
$v = \sqrt{1.29 \cdot 8.314 \cdot 210/0.043} = 231$ m/s — thin, heavy, cold air speaks slowly.

Laplace-era consistency drills (cont.)

Cross-check anchors: air 0 C 331, 20 C 343, He 20 C 1007, H2 0 C 1260, CO2 0 C 259, water 20 C 1482, steel rod 5064 — keep this handful in memory for sanity checks of every computation.
the speed formula $\sqrt{(\partial P/\partial \rho)_s}$ is general: for a van der Waals gas or a liquid with thermal expansion corrections the sound speed measures the adiabatic stiffness — acoustic measurements probe the equation of state directly (Oly 12).

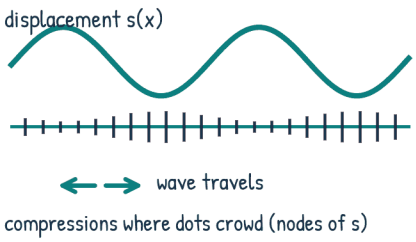
Temperature law: v = 331.3 sqrt(T/273) m/s

From $v = \sqrt{\gamma RT/M}$: at fixed gas, $v \propto \sqrt{T}$ with T in kelvin. Anchored at 0 C: $v(T) = 331.3\sqrt{T/273.15}$ m/s.
Linear form for everyday range: $v \approx 331.3 + 0.60\,t$ C (exact slope at 20 C: $dv/dT = v/2T = 343/(2 \cdot 293) = 0.585$ m/s/K; the "0.6" is a rounded staple). Values: -40 C: 306; 0 C: 331; 20 C: 343; 37 C: 350; 100 C: 386 m/s.
The $\sqrt{T}$ law is exact for ideal gases; the linear law is its first-order expansion:
$v = 331.3(1 + t/273.15)^{1/2} \approx 331.3(1 + t/546)$ — slope 0.605 m/s/K at 0 C.

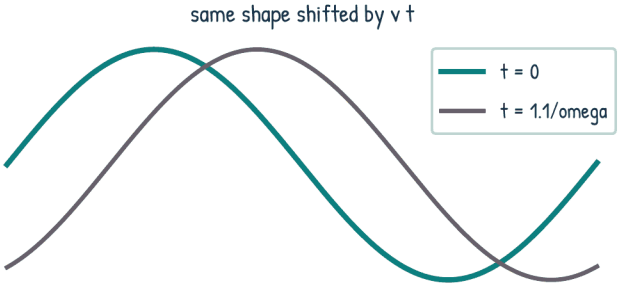
Temperature law: v = 331.3 sqrt(T/273) m/s

Physics: heating a gas at constant pressure lowers its density ($\rho = PM/RT$) while $B = \gamma P$ stays put — lighter medium, same stiffness, faster wave. At constant volume, heating raises P and B with ρ fixed — also faster.
In both cases only T matters: $v = \sqrt{\gamma RT/M}$ carries no P or ρ separately.
Micro-physical picture (kinetic theory): $v_{\text{sound}} \sim 0.68\,\bar{c}$ for $\gamma = 1.4$ where $\bar{c} = \sqrt{8RT/\pi M}$ is the mean molecular speed — sound is a coherent disturbance propagated by molecular collisions at roughly the thermal speed. $\bar{c}$ at 293 K in air $= 464$ m/s; $0.68 \times 464 = 316$ ... order of magnitude; exact $v/\bar{c} = \sqrt{\gamma\pi/8} = 0.74$.

adiabatic compression: gamma P is the stiffness



speed set by gamma R T / M alone



REMEMBER / Laplace: B = gamma P, so v = sqrt(gamma R T/M): temperature and molar mass only, never pressure.

Temperature dependence: v = 331.3sqrtT/273 m/s, approx 331 + 0.6t

v $\propto$ √T, linear 331+0.6t, lapse-rate refraction, sound horizon formula.

Temperature dependence: v = 331.3sqrtT/273 m/s, approx 331 + 0..

WHY / the $\sqrt{T}$ law is the same $\sqrt{stiffness/inertia}$ as always, with $B = \gamma PM/RT$ after eliminating ρ — the gas constant R is what ties mechanics to thermodynamics here.

CHECK / halving absolute temperature drops speed by $\sqrt{2}$ only: 300 K $\rightarrow$ 347; 150 K $\rightarrow$ 245 m/s. Speed is a SLOW function of T — temperature errors of 10 K move v by 1.7%.

Temperature lapse rate $dT/dz = -6.5$ K/km (troposphere): $dv/dz = (v/2T)(dT/dz) = 0.585 \times (-6.5) = -3.8$ m/s per km near the surface. Consequence: sound rays curve toward the cold region. Day: ground hot, air cooler aloft $\rightarrow$ rays bend up $\rightarrow$ sound shadows near ground. Night:

Sound horizon, lapse rate, and the bonus gravity formula (cont.)

temperature inversion (warm aloft) $\rightarrow$ rays bend down $\rightarrow$ sound hugs the ground and carries far — the "lake at night" effect. Acoustic shadow zones behind warm columns.

Isothermal-atmosphere bonus (source §2.5): $v(h) = v_0\sqrt{1 - \dfrac{2Mgh}{\gamma RT_0}}$ with constant g; setting $v = 0$ gives atmosphere height $h_{\max} = \dfrac{\gamma RT_0}{Mg} = \dfrac{1.4 \cdot 8.314 \cdot 288}{0.029 \cdot 9.8} = 11.9$ km — the thermosphere-scale estimate for an isothermal troposphere. Beyond it there is no medium, no sound.

Sound horizon, lapse rate, and the bonus gravity formula (cont.)

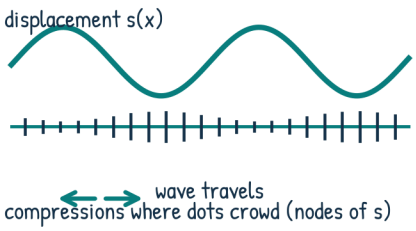
Refraction quantitative: the ray curvature radius $R_c = v/|dv/dz|$ ... for the standard treatment $R_c = v/(dv/dz)$; with $dv/dz = -3.8$ m/s/km and $v = 343$: $R_c = 343/0.0038 = 90$ km — rays over long distances curve gently on a 90 km radius arc. Over 1 km, the ray drops $\approx L^2/(2R_c) = 5.5$ m — enough to matter for precise ranging.

CHECK / speed at 5 km altitude with lapse rate: $T = 288 - 32.5 = 255.5$ K, $v = 331.3\sqrt{255.5/273.15} = 320$ m/s — 7% slower than sea level; aircraft sound takes longer to arrive from above than naive h/v_0 estimates.

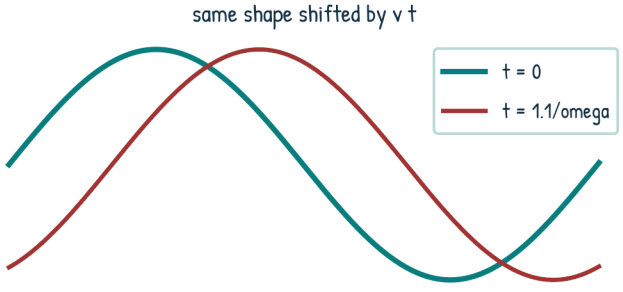
Temperature-based problem patterns

(a) "At what temperature is the speed 350 m/s?" $T = 273.15(350/331.3)^2 = 305$ K = 32 C. (b) "By how much does speed change from 20 C to 25 C?" $\Delta v = 0.585 \cdot 5 = 2.9$ m/s (or exact $346.1 - 343.2 = 2.9$ ok). (c) "A pipe closed at one end resonates at 300 Hz at 20 C; find its length" then "at what T does it resonate at 306 Hz?" — $f \propto v \propto \sqrt{T}$: $T_2 = T_1(306/300)^2 = 293.15 \cdot 1.0404 = 305$ K = 32 C ok pattern: frequency of fixed-length modes scales as $\sqrt{T}$.

speed rises as the square root of kelvin temperature



v(T): 331 m/s at 0 C, 343 at 20 C



REMEMBER / v = 331.3 sqrt(T/273) m/s, about 0.6 m/s per degree; kelvin only inside the root.

Temperature dependence: v = 331.3sqrtT/273 m/s, approx 331 + 0.6t (cont.)

v $\propto$ √T, linear 331+0.6t, lapse-rate refraction, sound horizon formula.

Temperature-based problem patterns (cont.)

Humidity angle (JEE-09): speed at fixed T rises with water-vapour fraction — at 20 C saturated air ($p_{\text{vap}} = 2.3$ kPa) has v about 0.2% (0.7 m/s) above dry — small but the same order as 1 C! Exam problems often say "dry air" implicitly.

WHY / modes of any resonator scale with v : pipe, Kundt, resonance tube — frequency ratios at two temperatures collapse to $\sqrt{T_2/T_1}$ regardless of geometry. That ratio trick shortens many problems to one line.

CHECK / compute twice with the exact and linear laws at 37 C: exact $331.3\sqrt{310.15/273.15} = 353.0$; linear $331.3 + 0.6 \cdot 37 = 353.5$ — agree to 0.5 m/s; either is fine in exams if you state which.

Traps and consistency checks for the v(T) family

WATCH / Trap list: (i) Celsius in the square root — always kelvin. (ii) $v \propto \sqrt{T}$ vs "0.6 m/s per C" — the linear law is a local approximation (fails far from 0 C: at 300 C, linear gives 511, exact gives 587). (iii) Forgetting γ changes between gases when comparing (He vs air needs both M and γ). (iv) Assuming pressure changes matter — they do not (at fixed T). (v) Applying the gas law to solids or liquids.

Consistency loop for any v computation: compute with $\sqrt{\gamma RT/M}$ in SI; compare against the anchor table (air 331/343, He ~1000, H2 ~1260, water ~1480, steel ~5000); check T in kelvin and M in kg/mol; the answer should land within a few percent of anchors for known media.

Traps and consistency checks for the v(T) family (cont.)

for a hard vacuum, $\rho \to 0$ and $v = \sqrt{\gamma P/\rho} = \sqrt{\gamma RT/M}$ stays formally finite while the wave amplitude interpretation changes — no molecules, no sound; the formula survives only as a limit of a thinning gas with vanishing mean free path conditions.

From $v = \sqrt{\gamma RT/M}$: (i) $v \propto 1/\sqrt{M}$ at fixed γ, T — light gases sound fast: He 972 m/s at 0 C, H2 1260 m/s at 0 C vs air 331. (ii) γ matters too: He has $\gamma = 5/3$ vs air 1.4 — comparing gases needs both factors: $v_2/v_1 = \sqrt{\gamma_2 M_1/(\gamma_1 M_2)}$. (iii) Pressure: none — the ideal-gas speed has no P in it. Double the pressure of a sealed air sample at fixed T:

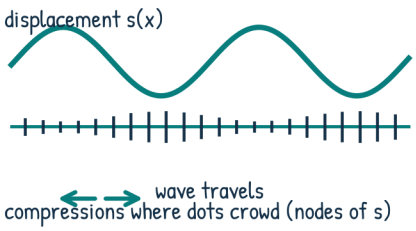
Molar mass, humidity, and the pressure-independence of sound spee.

ρ doubles, $B = \gamma P$ doubles, $v = \sqrt{B/\rho}$ unchanged ok.

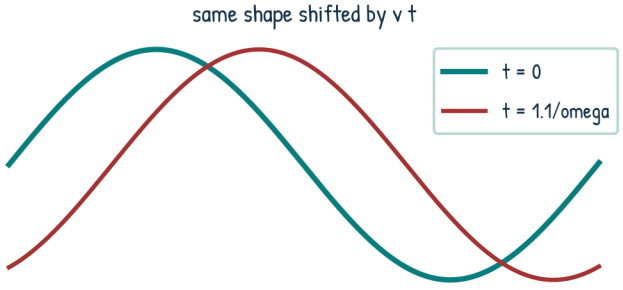
Sound inside a high-pressure scuba tank (200 atm) still travels 343 m/s at room temperature (density is 200x, stiffness 200x).

Humidity: water vapour ($M = 18$) replaces N2/O2 ($M_{\text{dry}} = 29$): effective molar mass $M_{\text{eff}} = 0.622 M_d + 0.378 M_v$ ... precisely with mole fraction x of vapour: $M_{\text{eff}} = (1 - x) \cdot 29 + x \cdot 18$. At 30 C saturated, vapour pressure 4.2 kPa, $x = 0.042$: $M_{\text{eff}} = 28.5$, v up 0.9% (3 m/s). But humid air is ALSO less dense... consistent: $v = \sqrt{\gamma RT/M_{\text{eff}}}$ grows as M_{eff} falls. Rule: humid air sounds faster; the effect peaks in hot humid weather.

speed rises as the square root of kelvin temperature



v(T): 331 m/s at 0 C, 343 at 20 C



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Molar mass, humidity, and the pressure-independence of sound speed

Molar mass, humidity, pressure independence, gas identity problems.

Molar mass, humidity, and the pressure-independence of sound spee.

Gas mixtures: γ_{eff} for a mixture is not the average of γ; use C_p, C_v averaging: $C_{v,\text{mix}} = \sum x_i C_{v,i}$, $\gamma = (C_v + R)/C_v$. For air-like N2/O2 this is automatic ($\gamma = 1.4$). For a 50-50 He-N2 mix: $C_v = 12.5 + 20.8 = 33.3$ J/mol-K... ($C_{v,\text{He}} = 3R/2 = 12.5$, $C_{v,\text{N2}} = 5R/2 = 20.8$), $\gamma = 35.8/33.3 = 1.075$, $M = 16.5$ g: $v = \sqrt{1.075 \cdot 8.314 \cdot 293/0.0165} = 399$ m/s.

WHY / pressure independence: P and ρ both live in $P/\rho = RT/M$; any process that changes P at fixed T (pumping, sealing) moves them together. Only T, M, γ are free parameters of gas sound speed.

Molar mass, humidity, and the pressure-independence of sound spee.

CHECK / scuba-tank paradox: same v at 200 atm, but intensity for a given s_0 rises with ρv: a speaker driven at the same displacement is much louder (higher $Z = \rho v$) in dense gas — Oly 01 impedance thinking.

Air 331.5 (20 C: 343.2); He 972 (20 C: 1007); H2 1260 (20 C: 1307); CO2 259; O2 316; N2 334 (20 C: 349); Ar ($\gamma = 5/3$, $M = 40$ g) 323; CH4 ($\gamma \approx 1.3$, $M = 16$ g) 431; water vapour at 100 C ($M = 18$ g, $\gamma = 1.33$) 483. Ratio He/air at same T: $\sqrt{(5/3) \cdot 29/(1.4 \cdot 4)} = 2.93$.

Numbers table (gas speeds at 0 C unless noted) (cont.)

Sound in CO2 is slower than air — deep voices from helium-free "heavy" mixtures... the party trick: He raises formant frequencies (sound speed in He is 2.9x, resonant cavities of the vocal tract scale up in pitch); SF6 ($M = 146$ g) drops it to 134 m/s... roughly $\sqrt{1.0 \cdot 8.314 \cdot 273/0.146} = 125$ m/s — deep voice effect.

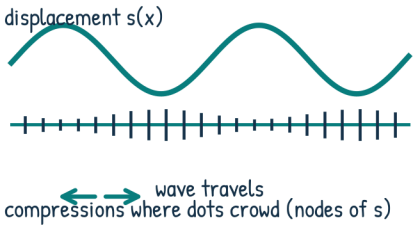
Humidity worked: 25 C, 60% RH: vapour pressure 1.9 kPa, $x = 0.019$, $M_{\text{eff}} = 28.8$, $v = 346.3$ vs dry 346.1 — 0.2 m/s only at moderate humidity; saturated 35 C: 1.5 m/s effect. Small — often neglected, but state "dry air" or "neglect humidity" explicitly.

Numbers table (gas speeds at 0 C unless noted) (cont.)

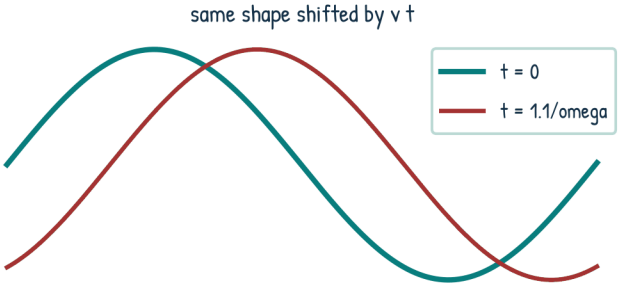
CHECK / independence stress-test: at 20 C, air at 0.1 atm (density 0.12 kg/m^3): $v = \sqrt{1.4 \cdot 10^4/0.12} = 342$ ok ~343; at 50 atm: same. Only temperature moves it.

gas sound speed in a stellar interior ($T = 10^7$ K, ionised H, $\gamma = 5/3$): $v = \sqrt{(5/3) \cdot 8.314 \cdot 10^7/0.001} = 3.7 \times 10^5$ m/s = 370 km/s — same formula, all the way to stars (acoustic asteroseismology).

light molecules jiggle faster: v falls as 1/sqrt(M)



humidity lowers the effective molar mass



REMEMBER / Light gases sound fast (1/sqrt(M)); humidity helps a little; pressure is irrelevant at fixed T.

Molar mass, humidity, and the pressure-independence of sound speed (cont.)

Molar mass, humidity, pressure independence, gas identity problems.

Problem patterns on gas identity

(a) Identify a gas: $v = 263$ m/s at 0 C: $M = \gamma RT/v^2 = 1.4 \cdot 8.314 \cdot 273/263^2 = 0.046$ kg/mol $\approx$ CO2 (44 g) with $\gamma = 1.3$ gives 259 ok. (b) A gas mixture of unknown proportion: use M_{eff} from measured v : $M_{\text{eff}} = \gamma RT/v^2$, then mole fraction from $M_{\text{eff}} = 29(1-x) + 2x$ (He-air: $x = (29 - M_{\text{eff}})/27$).

(c) "A sound pulse takes 2.92 s to travel 1 km in a gas at 27 C; identify": $v = 342.5$, $T = 300$: $M = \gamma RT/v^2 = 0.0285$ kg/mol — nitrogen-dominated air ok.

Problem patterns on gas identity (cont.)

Comparison traps: "Sound is faster in warm air because molecules move faster" — TRUE but the exam formula is $\sqrt{\gamma RT/M}$; "sound is faster at higher pressure" — FALSE; "sound is faster in denser media" — FALSE as a general rule (steel beats mercury? steel 5000 vs Hg 1440: stiffness dominates; lead is dense yet slow: 1190).

WHY / density alone never sets speed; the ratio B/ρ does. Heavy gases are slow only when stiffness does not compensate — in ideal gases $B/\rho = \gamma RT/M$ is density-free at fixed T.

CHECK / round-trip: given $v = 1260$ m/s at 0 C, $M = 2$ g, find γ : $\gamma = v^2 M/(RT) = 1260^2 \cdot 0.002/(8.314 \cdot 273) = 1.40$ ok diatomic hydrogen.

Consolidation: the gas-speed parameter space

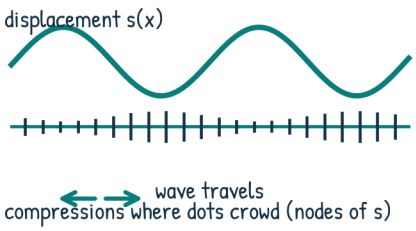
Free parameters of $v_{\text{gas}} = \sqrt{\gamma RT/M}$: just (γ , T , M). Pressure, density, volume, container size, amplitude, frequency: none of them enter. For a closed container being heated, v rises as $\sqrt{T}$ regardless of pressure rise. For a weather front, v changes with T (and slightly with humidity) — standard correction in precise acoustic ranging:

0.6 m/s per C, so a 5 C error in assumed temperature gives 1.7% error in distance measured by time of flight (3.4 m per km).

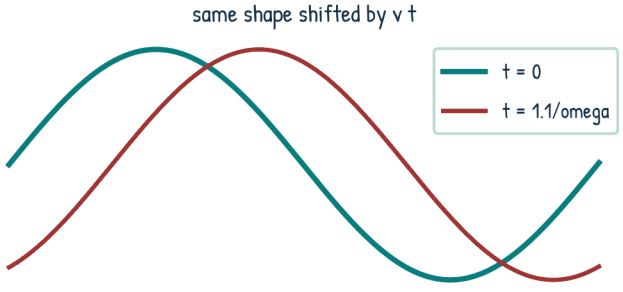
Consolidation: the gas-speed parameter space (cont.)

Sonic boom geometry (JEE-28) and Doppler (JEE-22-27) all need the LOCAL v ; atmospheric sound-speed profile $v(T(z))$ is the refractive index of acoustics: $n_{\text{ac}} = v_0/v(z) > 1$ aloft... rays bend toward lower speed (colder air) — exactly like light in a gradient index. meteorological "acoustic sounding" (SODAR) inverts Doppler + refraction of sound to measure wind and temperature profiles — the exam toolkit ($v(T)$, Doppler, refraction) is literally the working method.

light molecules jiggle faster: v falls as 1/sqrt(M)



humidity lowers the effective molar mass



REMEMBER / Light gases sound fast (1/sqrt(M)); humidity helps a little; pressure is irrelevant at fixed T.

Intensity of sound: I = frac12rho vomega2 s 02 = dfracDelta P 022rho v

I = ½ ρ v ω²s 0² = Δ P 0²/2 ρ v, energy budget, amplitude scaling table.

Sound intensity: I = (1/2) rho v omega^2 s_0^2

Intensity = average power per area = $\bar{u}v$ with $\bar{u} = \frac{1}{2}\rho\omega^2 s_0^2$ (SN-03): $I = \frac{1}{2}\rho v\omega^2 s_0^2$. Since $\Delta P_0 = \rho v\omega s_0$, equally $I = \frac{\Delta P_0^2}{2\rho v} = \frac{\Delta P_{\text{rms}}^2}{\rho v} = \frac{1}{2}u_{\max}\omega s_0$ · …

keep the two boxed forms only. Third useful form: $I = \frac{1}{2}Z\omega^2 s_0^2$ with impedance $Z = \rho v$ (Oly 01). Worked (source Q3): $P = 0.1$ W isotropic at $r = 100$ m: $I = P/(4\pi r^2) = 7.96 \times 10^{-5}$ W/m²; $\Delta P_0 = \sqrt{2\rho v I} = 0.256$ Pa; $\beta = 10\log_{10}(I/I_0) = 79.0$ dB;

Sound intensity: I = (1/2) rho v omega^2 s_0^2

$s_0 = \Delta P_0/(\rho v\omega)$ — the full round trip.

Power of a source: $P = \oint I\,dA$; isotropic: $P = 4\pi r^2 I$; line source (infinite line): $P/L = 2\pi r I$. The wave equation alone does not set I — amplitude does; energy conservation sets the geometry scaling: $I \propto 1/r^2$ point, $\propto 1/r$ line (JEE-11). Loudness perception is roughly logarithmic in I (Weber-Fechner) which is why decibels (JEE-12) are the scale of hearing.

Sound intensity: I = (1/2) rho v omega^2 s_0^2

Amplitude conversions (exam currency): $s_0 = \frac{\Delta P_0}{\rho v\omega} = \frac{\Delta P_0}{2\pi f\rho v}$; $\Delta P_0 = \sqrt{2\rho v I}$; $I = \frac{\Delta P_0^2}{2\rho v}$. In air $\rho v = 412$: $\Delta P_0 = \sqrt{824\,I}$, $s_0 = \Delta P_0/(2\pi f \cdot 412)$.

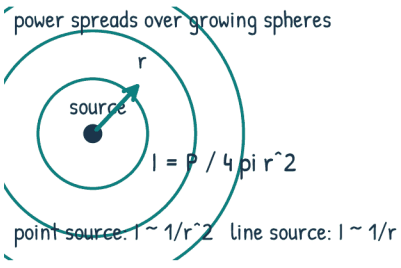
WHY / the factor 2 in $I = \Delta P_0^2/(2\rho v)$ is the time-average of $\cos^2$ — peak squared over twice; if data give RMS quantities use $I = \Delta P_{\text{rms}}^2/(\rho v)$ with no 2. State the convention every time.

Sound intensity: I = (1/2) rho v omega^2 s_0^2

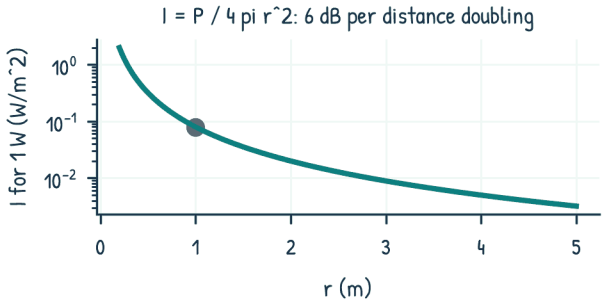
CHECK / units: $\rho v\omega^2 s_0^2$ = kg m^{-3} m s^{-1} s^{-2} m^2 = kg s^{-3} = W m^{-2} ok. Sanity: 1 mW/m² is loud speech at 1 m; 1 W/m² is on the edge of pain (120 dB).

A source of power P spreads its energy on expanding wavefronts. Point source: $I(r) = P/(4\pi r^2)$ — doubling distance drops I 4x, pressure $\Delta P_0 \propto 1/r$, displacement $s_0 \propto 1/r$ (same $1/r$: both amplitudes share the geometric dilution; $\Delta P_0 = \rho v\omega s_0$ keeps them proportional). Line source: $I(r) = (P/L)/(2\pi r)$ — halving rate. Near a large vibrating surface (plane wave):

power spreads over spheres: I = P / 4 pi r^2



intensity falls as 1/r^2 from a point source



REMEMBER / I = dP0^2/(2 rho v) = 1/2 rho v omega^2 s0^2; two forms must always agree.

Intensity of sound: I = frac12rho vomega2 s 02 = dfracDelta P 022rho v (cont.)

I = ½ ρ v ω²s₀² = Δ P ₀²/2 ρ v, energy budget, amplitude scaling table.

Energy budget: where the power goes (cont.)

I constant with distance until diffraction/fringing takes over.

Energy density vs intensity at 1 kHz, 1 Pa amplitude: $\bar{u} = 3.5 \times 10^{-6}$ J/m^3, $I = 1.2$ mW/m^2; total energy in a 10 m room playing at that level: 3.5×10^{-5} J — less than the kinetic energy of a mosquito... yet hours of music are carried because the source continuously supplies power; the room just transmits.

Efficiency of real sources: loudspeakers 1–5% (most power heats the voice coil); the human voice ~0.01–0.1 W acoustic of ~50 W metabolic... spoken voice 1 m: ~10 μW; shouting: ~1 mW; jet engine 100 kW acoustic — most of 100 MW shaft power becomes heat.

Energy budget: where the power goes (cont.)

CHECK / source Q3 chain once more, symbolically: $I = P/4\pi r^2$; $\Delta P_0 = \sqrt{2\rho v I} = \sqrt{\rho v P/(2\pi r^2)}$; $\beta = 10\log_{10}\left(P/(4\pi r^2 I_0)\right)$; $s_0 = \Delta P_0/(\rho v\omega) = \sqrt{P/(2\pi r^2 \rho v)}/\omega$ — five answers from two data, one chain.

WHY / energy conservation + spherical spreading alone gives $\Delta P_0 \propto 1/r$ — no wave mechanics needed beyond $I = \Delta P_0^2/2\rho v$; a favourite derivation-shortcut for exams.

Amplitude scaling reference card (memorise the column)

Given: I. Then $\Delta P_0 = \sqrt{2\rho v I}$; $s_0 = \Delta P_0/(\rho v\omega)$; $\beta = 10\log(I/I_0)$. Given: ΔP_0. Then $I = \Delta P_0^2/(2\rho v)$; $s_0 = \Delta P_0/(\rho v\omega)$. Given: s_0. Then $\Delta P_0 = \rho v\omega s_0$; $I = \frac{1}{2}\rho v\omega^2 s_0^2$. Given: P, r (point). Then $I = P/(4\pi r^2)$ + above. Given: P/L, r (line). Then $I = (P/L)/(2\pi r)$ + above.

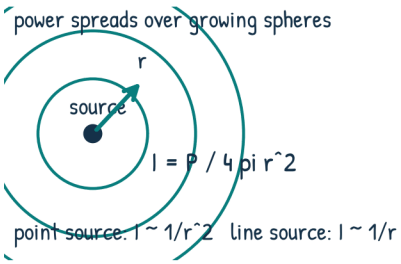
Ratios: double s_0 at fixed f: ΔP_0 doubles, I quadruples, β rises 6.02 dB. Double f at fixed s_0: ΔP_0 doubles, I quadruples — "same displacement, twice as loud" is FALSE perception-wise... dB up 6, phon loudness curves (equal-loudness) complicate perception — physically 6 dB up. Double distance from a point source: $I/4$, $\Delta P_0/2$, β drops 6.02 dB.

Amplitude scaling reference card (memorise the column) (cont.)

near-field vs far-field: within about λ of a real radiator the $1/r$ law misbehaves (evanescent/reactive fields); exam problems implicitly assume far field $r \gg \lambda$... and for sound $\lambda \sim 1$ m so room-scale problems are fine.

(a) $s_0 = 2 \times 10^{-6}$ m at 340 Hz in air: $\Delta P_0 = 412 \cdot 2136 \cdot 2 \times 10^{-6} = 1.76$ Pa; $I = 1.76^2/824 = 3.76$ mW/m^2; $\beta = 95.8$ dB; P on 1 m^2 panel $= 3.76$ mW. (b) Threshold of pain $I = 1$ W/m^2: $\Delta P_0 = \sqrt{824} = 28.7$ Pa; $s_0 = 28.7/(412 \cdot 6283) = 1.1 \times 10^{-5}$ m at 1 kHz. (c) Two identical speakers in phase at one point: amplitudes add $\rightarrow \Delta P_0$ doubles $\rightarrow I$ quadruples (6 dB) at antinodes; out of phase: silence.

power spreads over spheres: I = P / 4 pi r^2



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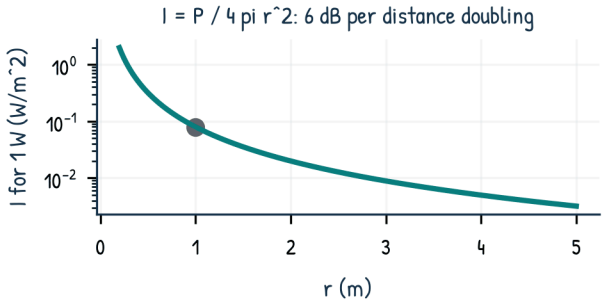

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Intensity of sound: I = frac12rho vomega2 s 02 = dfracDelta P 022rho v (cont.)

I = ½ ρ v ω²s 0² = Δ P 0²/2 ρ v, energy budget, amplitude scaling table.

Worked intensity drills (cont.)

Inverse drill (source-style): "Find the distance at which a 50 W siren falls to 80 dB": $I = 10^{8} \cdot 10^{-12} = 10^{-4}$ W/m 2 ; $r = \sqrt{P/(4\pi I)} = \sqrt{50/(1.257 \times 10^{-3})} = 199$ m. "Find s_0 there at 600 Hz": $\Delta P_0 = \sqrt{824 \cdot 10^{-4}} = 0.091$ Pa; $s_0 = 0.091/(412 \cdot 3770) = 5.9 \times 10^{-8}$ m.

CHECK / every intensity answer should pass the loudness sniff test: whisper 10^{-10} W/m 2 , conversation 10^{-6} , shout 10^{-3} , rock concert front row 10^{-1} , pain 1. If your number is 10 W/m 2 for a tuning fork, an exponent slipped.

Worked intensity drills (cont.)

WHY / intensity is the perceptually and physically central quantity: it is what the inverse square law, the decibel scale, and the Doppler energy shifts (Oly 05) all manipulate; pressure and displacement amplitudes are bookkeeping devices for the same I.

Isotropic point source of power P : energy crosses any sphere of radius r at total rate P , so $I(r) = \dfrac{P}{4\pi r^2}$.

Consequently $\Delta P_0(r) = \sqrt{2\rho v I} = \dfrac{A}{r}$ and $s_0(r) = \dfrac{\Delta P_0}{\rho v \omega} = \dfrac{B}{r}$ – both amplitudes fall as $1/r$; intensity as $1/r^2$. In dB: each distance doubling costs 6.02 dB.

Point and line sources: 1/r2, 1/r, and the amplitude laws (cont.)

The wave solution matching this is the spherical wave $s = \dfrac{\mathcal{A}}{r}\sin(kr - \omega t)$: the $1/r$ in amplitude is forced by energy conservation (the wave equation in 3D admits it: $\nabla^2(1/r \cdot e^{ikr}) \approx -k^2/r \cdot e^{ikr}$ for $kr \gg 1$).

Infinite line source (line of speakers, traffic on a straight road): cylindrical spreading, area $2\pi r L$: $I(r) = \dfrac{P/L}{2\pi r}$ – intensity $\propto 1/r$, amplitudes $\propto 1/\sqrt{r}$. Distance doubling: 3.01 dB drop only. Semi-infinite/other geometries: plane wave (large wall): I constant. Practical: near a motorway the noise falls off slowly (line); near a single machine quickly (point).

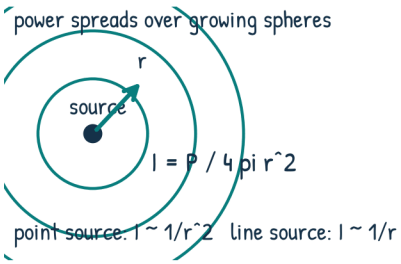
Point and line sources: 1/r2, 1/r, and the amplitude laws (cont.)

Superposition of two point sources (JEE-13): interference pattern with $\Delta P_{\text{tot}} = \Delta P_1 + \Delta P_2$ as waves; near-field geometry corrections $1/r_i$ matter only within a wavelength or so; exam interference uses equal- r approximation unless stated.

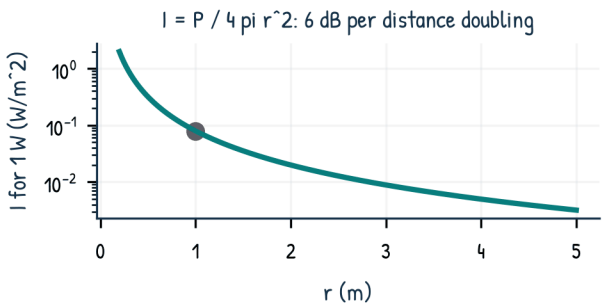
WHY / the $1/r$ amplitude law is dimensional bookkeeping of energy: $I \propto (amplitude)^2 \propto 1/(area)$; area $= 4\pi r^2$ (point) or $2\pi r$ per length (line). The wave equation carries the phase; conservation carries the amplitude.

CHECK / spherical wave consistency: $I = \frac{1}{2}\rho v \omega^2 s_0^2$ with $s_0 = \mathcal{A}/r$ gives $I \propto 1/r^2$ ok. Line: $s_0 \propto 1/\sqrt{r}$ gives $I \propto 1/r$ ok. Any other power law breaks energy conservation.

power spreads over spheres: I = P / 4 pi r^2



intensity falls as 1/r^2 from a point source



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Point and line sources: 1/r2, 1/r, and the amplitude laws

Point/line spreading, 1/r amplitude law, absorption and diffraction.

Decay laws in practice (attenuation on top of spreading)

Real air adds absorption: $I(r) = \frac{P}{4\pi r^2} e^{-\alpha r}$, α growing roughly f^2 (classical: $\alpha \propto f^2$ from viscosity + thermal conduction; ~0.1 dB/m at 20 kHz, 0.001 dB/m at 1 kHz in dry air — humidity adds relaxation peaks). Consequences: high frequencies die first — distant thunder is a rumble, distant speech muffled, ultrasonic imaging range limited to cm-m in tissue. Fog horns are low pitch (hundreds of Hz) to beat both absorption and diffraction over 10 km of sea.

Decay laws in practice (attenuation on top of spreading) (cont.)

Diffraction: sound with $\lambda \sim 1$ m bends around corners and doorways freely (open door: you hear the whole room) — no sharp shadows for speech; ultrasound ($\lambda \sim 1$ mm) shadows like light. The doorway diffraction angle $\theta \sim \lambda / w$ for opening width w: 1 m sound through a 1 m door: strong spreading; 20 kHz ($\lambda = 17$ mm) through the same door: narrow beam.

CHECK / combine laws carefully: "level falls by how much from 5 m to 50 m with $\alpha = 0.05$ dB/m?" — spreading: $20\log(10) = 20$ dB; absorption: $0.05 \cdot 45 = 2.25$ dB; total 22.3 dB. Two effects add in dB.

inverse-square fitting of data: plot $10\log I$ vs $\log r$; slope -20 dB/decade = point source; -10 dB/decade = line source — a data-analysis question format used in Olympiad papers.

Worked geometry problems

(a) Two point sources of the same power 4 m apart, incoherent: level at midpoint: each gives $I_1 = P/(4\pi 4)$; sum doubles: +3 dB over one. (b) A listener moving from 10 m to 3 m from a point source: $\Delta\beta = 20\log(10/3) = 10.5$ dB louder. (c) "Find r where a 2 W source is at threshold of hearing": $I_0 = 10^{-12}$: $r = \sqrt{2/(4\pi \cdot 10^{-12})} = 4 \times 10^5$ m = 400 km — theoretical vacuum-free geometry with zero absorption; with $\alpha = 0.01$ dB/m absorption limit: $10\log(2/4\pi r^2/10^{-12}) - 0.01r = 0$ cuts this to a few km in practice.

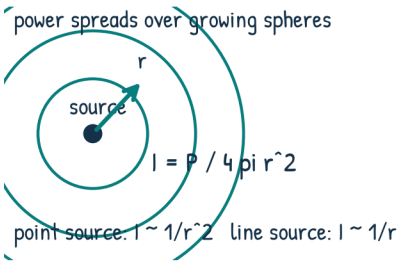
Worked geometry problems (cont.)

Line-source drill: highway with equivalent $P/L = 0.05$ W/m: I at 20 m: $0.05/(2\pi \cdot 20) = 4 \times 10^{-4}$ W/m^2 = 86 dB — plausible traffic roar; at 200 m: 76 dB — only 10 dB down (vs 20 dB for a point source) ok line law.

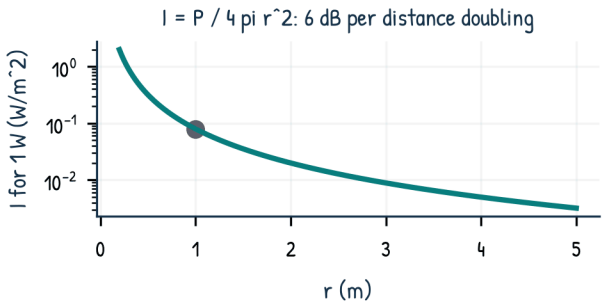
WHY / matching the geometry to the decay law is half the marks in such problems: write the law first ($4\pi r^2$ vs $2\pi r L$), then the numbers.

CHECK / cross-check with dB slopes: point source distance x10 = 20 dB; line source x10 = 10 dB; plane = 0 dB. Your computed dB difference must fit one of these unless absorption is in play.

point source vs line source geometries



1/r^2 vs 1/r: amplitude 1/r vs 1/sqrt(r)



REMEMBER / Point sources fall as 1/r in amplitude, 1/r^2 in intensity; line sources as 1/sqrt(r).

Point and line sources: 1/r2, 1/r, and the amplitude laws (cont.)

Point/line spreading, 1/r amplitude law, absorption and diffraction.

Spherical wave phasor picture (bridge to Oly)

A spherical wave $s = (\mathcal{A}/r)\sin(kr - \omega t)$ has local plane-wave character when $r \gg \lambda$: locally $\Delta P \approx \rho v u$ where $u = -\partial s/\partial t$ ... more precisely for $kr \gg 1$, $\Delta P \approx \rho v\, \partial s/\partial t\,/v = \rho v u$ — the travelling-wave impedance $Z = \rho v$ holds locally (Oly 01).

The exact spherical solution has a reactive (quadrature) component near the source, decaying as $(kr)^2$ — the near-field correction.

Beam patterns of real sources: a circular piston of radius a beams when $\lambda < 2a$ (directivity $D = I(0)/I_{\text{iso}}$; megaphone: $D \sim 10$, +10 dB forward). Loudspeaker arrays: interference of N elements gives $I \propto N^2$ on axis (coherent) vs N (incoherent) — exam contrast question.

Spherical wave phasor picture (bridge to Oly) (cont.)

sound intensity level of a point source of known power at distance r: $\beta = 10\log_{10}\dfrac{P}{4\pi r^2 I_0}$; solve for any variable — the standard closed-form exam answer.

Sound intensity spans 10^{-12} to 1 W/m^2 — twelve decades. The decibel compresses it: $\beta = 10\log_{10}(I/I_0)$ with reference $I_0 = 10^{-12}$ W/m^2 (roughly the 1 kHz hearing threshold; formally 2×10^{-5} Pa pressure threshold). Purely a scale: 0 dB $= I_0$; 10 dB $= 10I_0$; 20 dB $= 100I_0$; 120 dB $= 1$ W/m^2 (threshold of pain). Key derived relation:

Decibel scale: beta = 10log 10(I/I 0), I 0 = 10-12 W/m2 (cont.)

$\Delta P_0 = \sqrt{2\rho v I}$: at I_0 in air, $\Delta P_0 = \sqrt{2 \cdot 412 \cdot 10^{-12}} = 2.87 \times 10^{-5}$ Pa — the standard threshold pressure.

Addition rules (must-know): doubling I = +3.01 dB; doubling ΔP_0 (or s_0) = +6.02 dB; distance doubling from a point source = -6.02 dB; ten sources of equal incoherent level = +10 dB; two identical coherent speakers at an antinode = +6.02 dB (amplitudes add). Adding two different levels: 100 dB + 90 dB $= 100.4$ dB — the louder dominates (10 dB difference = 1.04x total).

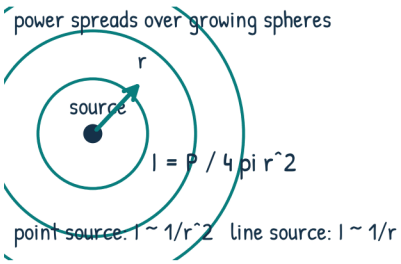
Decibel scale: beta = 10log 10(I/I 0), I 0 = 10-12 W/m2 (cont.)

Never add dB arithmetically: convert to intensity, add, convert back: $I_{\text{tot}} = I_0 \sum 10^{\beta_i/10}$.

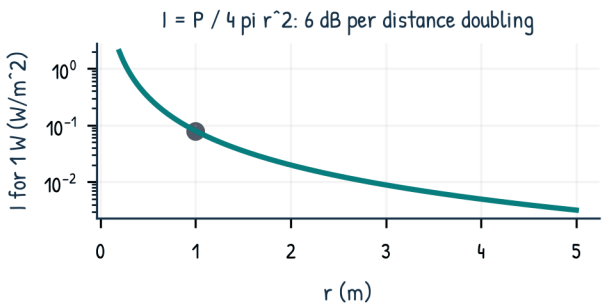
Loudness perception: phon scale matches dB at 1 kHz only; at 100 Hz the ear is ~40 dB less sensitive (equal-loudness curves, Fletcher-Munson) — bass needs more power. Pain 120 dB; conversation 60 dB at 1 m; whisper 30 dB; rock concert 110-120. Damage threshold ~85 dB prolonged.

WHY / the ear responds roughly logarithmically (Weber-Fechner), and physics ranges over decades — decibels make multiplication of gains (each stage in dB adds) into addition: total gain of loudspeaker chain $= 10\log \prod G_i = \sum 10\log G_i$.

point source vs line source geometries



1/r^2 vs 1/r: amplitude 1/r vs 1/sqrt(r)



REMEMBER / Point sources fall as 1/r in amplitude, 1/r^2 in intensity; line sources as 1/sqrt(r).

Decibel scale: beta = 10log 10(I/I 0), I 0 = 10-12 W/m2

dB scale, 10 vs 20 log rules, addition of levels, perception notes.

Decibel scale: beta = 10log 10(I/I 0), I 0 = 10-12 W/m2 (cont.)

CHECK / convert back and forth: $\beta = 80$ dB $\Rightarrow I = 10^{8} \cdot 10^{-12} = 10^{-4}$ W/m^{2}; $\Delta P_0 = \sqrt{2\rho v I} = 0.091$ Pa. Sanity: engine room 80-100 dB corresponds to 0.1-1 Pa pressure swings — tiny vs 1 atm, but destructive to hearing.

Since $I = \Delta P_0^2/(2\rho v)$, and $\Delta P_0 = \rho v \omega s_0$:

$\beta = 20\log_{10}\dfrac{\Delta P_0}{\Delta P_{\text{ref}}} = 20\log_{10}\dfrac{s_0}{s_{\text{ref}}}$ with same-medium references ($\Delta P_{\text{ref}} = 2.87 \times 10^{-5}$ Pa; s_{ref} = threshold displacement at the SAME frequency). The 20 = 2x10 because amplitude squared is intensity.

Pressure and displacement forms of the level (cont.)

Same-number rule: level differences computed from pressure amplitudes use $20\log$; from intensities $10\log$; both give identical dB.

Frequency weighting in engineering: dBA filters (mimic ear), dBC (flat-ish); exam problems use plain dB. For two tones of equal level but different f, the perceived loudness differs — physics dB equal, phon not.

Worked: (a) pressure amplitude doubles: $20\log 2 = 6.02$ dB ok matches intensity route (I quadruples, $10\log 4 = 6.02$). (b) s_0 from 10 nm to 1 μm at 1 kHz: ratio 100: $+40$ dB. (c) A microphone sees $\Delta P_0 = 1$ Pa:

Pressure and displacement forms of the level (cont.)

$\beta = 20\log(1/2.87 \times 10^{-5}) = 90.9$ dB ok consistent with $I = 1^2/824 = 1.2 \times 10^{-3}$, $10\log(1.2 \times 10^{9}) = 90.9$.

WHY / the two conventions (10 vs 20) are the same law wearing intensity or amplitude clothing; mixing them (writing $10\log(\Delta P_0/\Delta P_{\text{ref}})$) halves every answer — the single most common dB error.

CHECK / always cross-check one number both ways: pick I, find β; find $\Delta P_0 = \sqrt{2\rho v I}$; recompute $20\log(\Delta P_0/\Delta P_{\text{ref}})$ — must agree exactly.

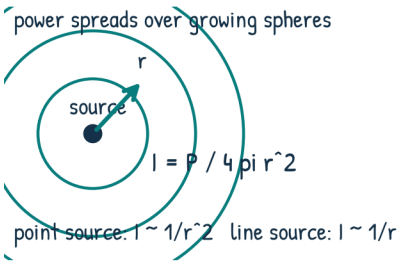
dB algebra drills

(a) Speaker efficiency: electrical 20 W, acoustic 2%: $P_{\text{ac}} = 0.4$ W; at 3 m: $I = 0.4/(4\pi \cdot 9) = 3.5 \times 10^{-3}$; $\beta = 95.5$ dB. (b) Two machines 85 and 88 dB: $I_1 = 3.16 \times 10^{-4}$, $I_2 = 6.3 \times 10^{-4}$; sum $= 9.5 \times 10^{-4}$: $\beta = 89.8$ dB. (c) "How many identical sources to raise the level by 10 dB?" 10 (incoherent).

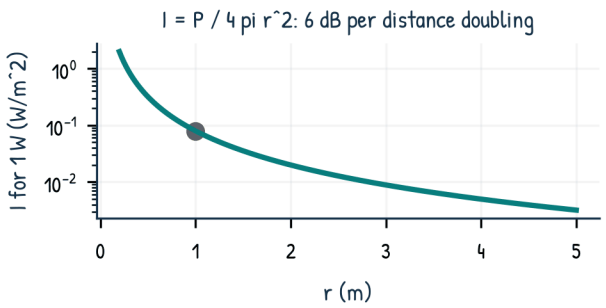
(d) "Level falls how much from 2 m to 7 m?" $20\log(7/2) = 10.9$ dB.

Combining with attenuation: $\beta(r) = \beta_1 + 20\log(r_1/r) - 0.05(r - r_1)$ (dB, r in m, $\alpha = 0.05$ dB/m): monotone fall with both terms negative — model fitting format.

levels are logarithmic: 3 dB per doubling



dB ladder from threshold to pain



REMEMBER / beta = 10 log(I/I0) with I0 = 1e-12 W/m2; doubling intensity is 3 dB, doubling amplitude 6 dB.

Decibel scale: beta = 10log 10(I/I 0), I 0 = 10-12 W/m2 (cont.)

dB scale, 10 vs 20 log rules, addition of levels, perception notes.

dB algebra drills (cont.)

sound exposure level (SEL) and time-weighted average (85 dB / 8 h) are workplace regulations on the same algebra — doubling time halves allowed intensity: 88 dB / 4 h etc. — pure dB arithmetic.

The phon scale: loudness in phons equals dB SPL at 1 kHz; at 100 Hz a 40 dB tone sounds like a 10 dB 1 kHz tone — the ear curve dives at low and high frequencies. So "bass boost" on speakers raises low-frequency dB to match perceived loudness. The physical reason: middle-ear mechanics + cochlear resonance (ear canal closed pipe resonance ~3.4 kHz gives free gain there — JEE-01).

Perception vs physics (why equal dB does not mean equal loudness).

Pain and damage: 130 dB ruptures eardrums; 150 dB+ sonic boom overpressure is a pressure transient, not a tone (JEE-28): typical boom 50–500 Pa ΔP transient = 128–138 dB if it were a tone — but it is a N-wave, broadband.

CHECK / convert a boom overpressure to "equivalent dB" only as an order-of-magnitude: 100 Pa: $20\log(100/2.87\times10^{-5}) = 131$ dB — consistent with boom descriptions; state it as an intensity-equivalent, not a tone level.

Superposition and interference of sound waves

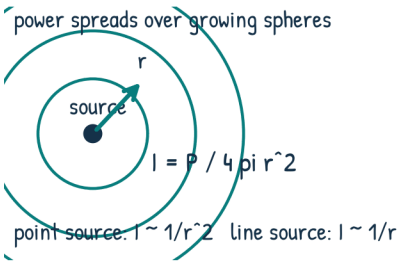
Linear wave equation $\Rightarrow$ superposition: if s_1 and s_2 are solutions, $s_1 + s_2$ is too. For two harmonic waves of the same f at a point:
$s = s_0\sin(\omega t-\phi_1) + s_0\sin(\omega t-\phi_2) = 2s_0\cos(\Delta\phi/2)\sin(\omega t-\bar{\phi})$
with $\Delta\phi = \phi_2 - \phi_1$: the resultant is same frequency, amplitude $2s_0|\cos(\Delta\phi/2)|$, phase $\bar{\phi}$. Constructive: $\Delta\phi = 2n\pi$ (path difference $\delta = n\lambda$); destructive: $\Delta\phi = (2n+1)\pi$ ($\delta = (n+\frac{1}{2})\lambda$). Max intensity $4I_0$, min 0, for equal sources.

Superposition and interference of sound waves (cont.)

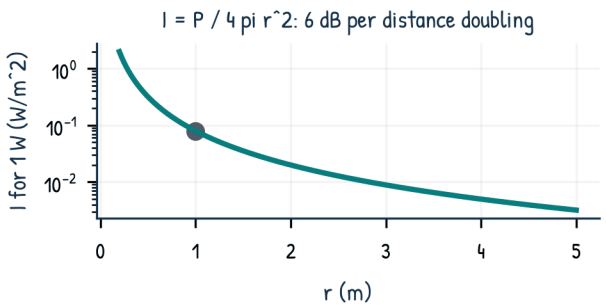
Path-difference working rule: $\delta = |r_2 - r_1|$ plus any extra inversion (phase changes on reflection — closed end of pipe adds π to pressure wave... reflection at rigid wall inverts the PRESSURE wave, i.e. pressure phase jump π, displacement phase jump 0 — a classic trap). Then $\Delta\phi = 2\pi\delta/\lambda + \Delta\phi_{\text{inv}}$.

Standing waves (JEE-16) are the structured superposition of counter-propagating waves: $s = 2s_0\sin kx\cos\omega t$; beats (JEE-15) are slow superposition of slightly different frequencies; interference of many sources gives diffraction patterns. Everything in this chapter family is the same addition formula.

levels are logarithmic: 3 dB per doubling



dB ladder from threshold to pain



REMEMBER / beta = 10 log(I/I0) with I0 = 1e-12 W/m2; doubling intensity is 3 dB, doubling amplitude 6 dB.

Superposition and interference of sound waves

Superposition principle, coherent/incoherent, addition formula toolkit.

Superposition and interference of sound waves (cont.)

WHY / sound from two speakers adds as pressures at the eardrum: $\Delta P = \Delta P_1 + \Delta P_2$; intensities do NOT add where waves are coherent (fixed phase relation) — only the amplitude sum, then $I = (\Delta P_{\text{tot}})^2/2\rho v$; for incoherent sources (random phases) intensities add.

CHECK / equal speakers in phase at equal distance: ΔP_0 doubles, I quadruples = +6 dB; two equal-power INCOHERENT speakers at the same place: +3 dB only. Which one the problem means is often the whole question.

Coherent vs incoherent, and the interference visibility

Coherent: sources with constant phase difference (same source split, two loudspeakers driven by one amplifier, forks struck in sync). Interference: static pattern of maxima/minima; visibility $\mathcal{V} = (I_{\max} - I_{\min})/(I_{\max} + I_{\min}) = 2s_1 s_2/(s_1^2 + s_2^2)$ for amplitudes s_1, s_2 — full contrast when equal amplitudes. Incoherent: two independent forks: intensities add everywhere, no pattern — "two lamps" case.

Partial coherence (real rooms): phase drifts slowly; at points of near-equal path the listener hears slow "wandering" of level — room modes and flutter echo are semi-coherent phenomena. Exam idealisation: always coherent unless stated.

Coherent vs incoherent, and the interference visibility (cont.)

Interference at fixed frequency, moving one source: pattern sweeps past the listener at rate $f\,d\delta/dt$ ... the intensity at a point modulates as the path difference changes — principle of the acoustic interferometer (Quincke tube is the 1D version, JEE-14).

WHY / coherence decides whether amplitudes or intensities add; every "two speakers" problem must first ask: same signal (amplitudes) or independent (intensities)?

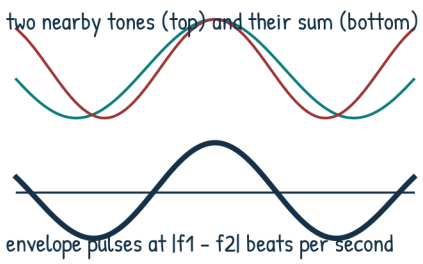
CHECK / extremes: $s_1 = 3s_2$ (in phase): $I = (4s_2)^2/(s_2^2$ -scale $) = 16/10 = 1.6\times$ single-sum-of-intensities... precisely: $I_{\text{coh}} = (3+1)^2 I_0 = 16I_0$ vs $I_{\text{incoh}} = 9 + 1 = 10I_0$: +2 dB difference — quantifying "almost incoherent".

The addition formula toolkit (memorise)

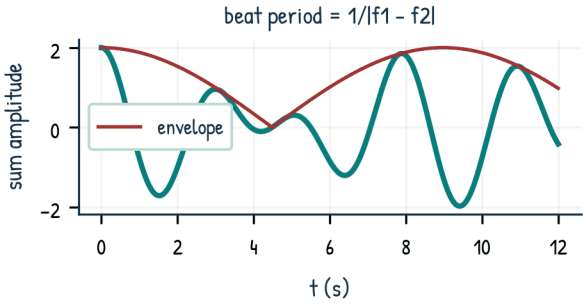
Two waves same frequency, phases ϕ_1, ϕ_2, amplitudes a_1, a_2: resultant amplitude $a = \sqrt{a_1^2 + a_2^2 + 2a_1 a_2 \cos\Delta\phi}$; phase $\tan\bar{\phi} = \dfrac{a_1\sin\phi_1 + a_2\sin\phi_2}{a_1\cos\phi_1 + a_2\cos\phi_2}$. Max $= a_1 + a_2$, min $= |a_1 - a_2|$. For N equal in-phase sources: $a = Na_0$ (on-axis), $I = N^2 I_0$; N equal random: $a \approx \sqrt{N}a_0$, $I = NI_0$ (Oly 04 phasor treatment).

Different frequencies: $\sin\omega_1 t + \sin\omega_2 t = 2\sin\bar{\omega}t\cos(\Delta\omega t/2)$ — the beats structure (JEE-15) — amplitude envelope $2s_0|\cos(\Delta\omega t/2)|$ at beat frequency $|f_1 - f_2|$.

two waves at one point add as phasors



amplitude of the sum vs phase difference



REMEMBER / Superpose pressures, not intensities, for coherent sources; the amplitude sum decides everything.

Superposition and interference of sound waves (cont.)

Superposition principle, coherent/incoherent, addition formula toolkit.

The addition formula toolkit (memorise) (cont.)

three-source interference: adding N phasors geometrically (polygon) — partial cancellation when $\delta = \lambda/N$ -type settings; the "N-slit" pattern logic carries directly from light to sound.

(a) Two speakers 1.5 m apart, in phase, $f = 340$ Hz ($\lambda = 1$ m) on the perpendicular bisector 2 m away: path difference 0 $\rightarrow$ max. Move listener 0.5 m toward one speaker: $\delta = \sqrt{2^2 + 2^2} - \sqrt{2^2 + 1^2}$ ... set coordinates; first minimum when $\delta = 0.5$ m $= \lambda/2$: solve $\sqrt{x^2 + 2^2} - \sqrt{(1.5 - x)^2 + 2^2} = 0.5$ numerically — exam answer form.

Worked superposition examples (cont.)

(b) Pressure amplitudes 0.3 Pa and 0.3 Pa with $\Delta\phi = 60\gamma$: $a = \sqrt{0.09 + 0.09 + 2 \cdot 0.09 \cdot 0.5} = 0.52$ Pa; $I/I_{\text{one}} = (0.52/0.3)^2 = 3$ ... three times ONE source intensity; vs incoherent sum 2x.

(c) A wall reflects: incident + reflected standing wave; pressure node at distance $\lambda/4$ from wall (pressure antinode AT rigid wall): microphone sweeps detect maxima spaced $\lambda/2$ — measure λ = the acoustic standing-wave interferometer.

WHY / the phasor diagram (Oly 04) turns all of this into vector addition: same formula, geometric intuition, and the fastest route to N -source results.

Quincke's tube: interference with a controllable path difference

Construction (source fig. 17): a sound wave from S splits at junction A into two U-shaped arms of different length and recombines at junction B toward the detector (listener/microphone). Sliding one U-tube changes its length continuously. Path difference $\delta = 2\Delta L$ when the adjustable arm is lengthened by ΔL (the wave traverses the changed section twice — out and back). Source worked example:

frequencies 340–512 Hz ("standards 500 Hz ish"), $\lambda = v/f$; minima spacing when sliding by $\lambda/2$ ...

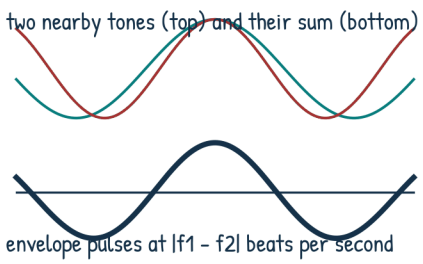
precisely: intensity minima occur when $\delta = (2n+1)\lambda/2$, i.e. sliding the U-tube by $\Delta L = \lambda/2$ takes you from one minimum to the next.

Quincke's tube: interference with a controllable path difference (co.

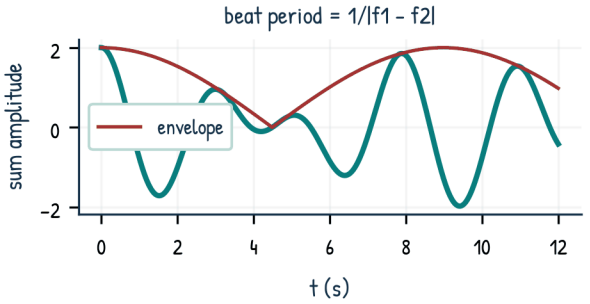
Formulas: distance between consecutive minima (as a function of U-tube displacement): $\Delta L_{\text{min}-min} = \lambda/2$; the wavelength from the measurement: $\lambda = 2 \times$ (displacement between minima); $v = f\lambda = 2f\,\Delta L_{\text{min}-min}$. Source numbers: $f = 340$ Hz, minima every 0.50 m: $\lambda = 1.0$ m, $v = 340$ m/s ok. Quincke half-lens: open end reflection is weak (pressure node-ish), so contrast is modest but measurable.

Uses: (i) measure λ and v precisely (minima are sharp nulls); (ii) demonstration of superposition; (iii) subtracting path for acoustic interferometry (Oly 08); (iv) measuring speed in different gases by filling the tube.

two waves at one point add as phasors



amplitude of the sum vs phase difference



REMEMBER / Superpose pressures, not intensities, for coherent sources; the amplitude sum decides everything.

Quincke's tube: interference with a controllable path difference

Quincke tube, double-pass λ/2 rule, discrimination of interference methods.

Quincke's tube: interference with a controllable path difference (co.

WHY / the double-pass factor 2 is the trap: the adjustable segment lies in one arm that the wave covers twice (the U goes out and back), so tube displacement ΔL changes δ by $2\Delta L$ — every Quincke formula carries that 2.

CHECK / source consistency: minima spacing $\lambda/2 = 0.5$ m at 340 Hz $\Rightarrow \lambda = 1$ m ok $v = f\lambda = 340$ ok. If a formula gives spacing λ, the factor 2 is missing.

At recombination B, the two waves have amplitudes a each (losses neglected) and phase difference $\Delta\phi = k\delta = 2\pi\delta/\lambda$ (plus a possible π if one path inverts... in the gas-filled U-tube both paths are gas-gas, no inversion). Resultant amplitude $2a|\cos(\Delta\phi/2)|$, intensity $4I_0\cos^2(\Delta\phi/2)$. Minima at $\delta = (2n+1)\lambda/2$; maxima at $\delta = n\lambda$.

The interference geometry in detail (cont.)

With the source covering $f = 340$ –512 Hz range and tube path difference swept, the listener hears the level pulsing as the slider moves; counting nulls over a slider range x gives $\delta_{\text{span}} = 2x = N\lambda/2$ ... N nulls in displacement x: $\lambda = 4x/N$ ... precisely: consecutive nulls at $\lambda/2$ displacement: $\lambda = 2\cdot(2x/N)$? Let: displacement between two nulls $= d$: $\lambda = 2d$ always — safest formula.

Amplitude at detector with unequal arms (loss α per pass): contrast ratio $C = (\sqrt{I_{\max}} - \sqrt{I_{\min}})/(\sqrt{I_{\max}} + \sqrt{I_{\min}})$ measures the amplitude imbalance — practical lab question.

The interference geometry in detail (cont.)

CHECK / set $f = 500$ Hz: $\lambda = 0.68$ m: nulls every 0.34 m of slider travel; if a lab report shows nulls every 0.68 m, the 2 was dropped.

Quincke tube in a different gas: $v_{\text{gas}} = 2f\Delta L_{\min\ -\ min}$ — direct speed measurement without end corrections (the arms share the same open junctions).

(a) $f = 340$ Hz, $v = 343$: $\lambda = 1.009$ m; slider null spacing 0.504 m. (b) $f = 512$ Hz (fork): $\lambda = 0.670$ m; null spacing 0.335 m; source example in the notes quotes these standard forks (340 Hz and 512 Hz). (c) A student measures nulls at 0.31 m and 0.63 m (spacing 0.32 m): $\lambda = 0.64$ m, $v = 328$ m/s at 17 C... consistent with $v = 331 + 0.6\cdot 17 = 341$? mismatch 4%: check for the 2 factor:

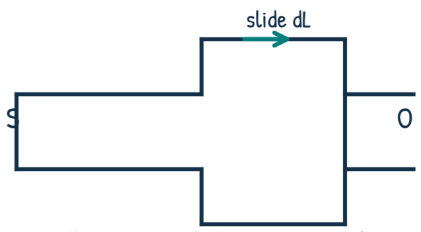
Quincke numbers and variations (cont.)

with $\lambda = 2\cdot 0.32 = 0.64$, $v = 340\cdot 0.64/0.67$ ... training example: state assumptions carefully in labs.

WATCH / Error analysis: the open junctions radiate imperfectly (end effects), and the tube has attenuation — minima do not reach zero; the theory formula $\lambda = 2d$ survives, only contrast suffers. Humidity and temperature shift v through $v(T)$.

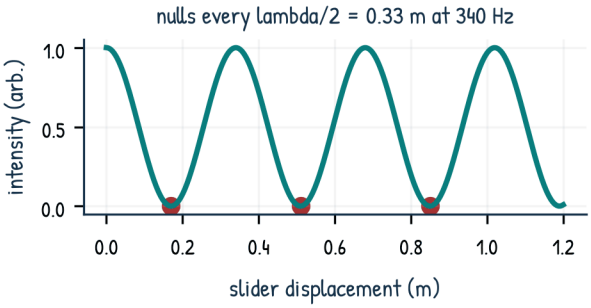
WHY / Quincke works because the two paths recombine coherently — the source drives both arms with the SAME wavefront split at A; that is what makes the pattern static rather than beating.

Quincke tube: sliding arm changes the path by 2 dL



path difference 2 dL; nulls every lambda/2 of sliding

intensity vs slider: nulls every lambda/2



REMEMBER / Quincke: the sliding U-tube changes the path by 2 dL; nulls every lambda/2 of slider travel.

Quincke's tube: interference with a controllable path difference (cont.)

Quincke tube, double-pass λ/2 rule, discrimination of interference methods.

Quincke vs beats vs standing wave (discrimination table)

Quincke: ONE frequency, TWO paths, controllable δ, static nulls as slider moves — measures λ via path difference. Beats: TWO frequencies, one path, intensity pulses in time at $|f_1 - f_2|$ — measures Δf. Standing wave: two counter-propagating SAME-frequency waves — fixed nulls at every $\lambda/2$ in space — measures λ via spatial nulls. Resonance tube:

one path, source frequency tuned — nulls/maxima vs f — measures λ via resonance lengths (JEE-20).

Exam discrimination question: "A listener hears loudness varying slowly while holding position" $\rightarrow$ beats (two nearby frequencies) OR a slow slider movement in Quincke; "loudness varies as the listener walks" $\rightarrow$ standing wave or interference fringes; "loudness varies as a tube is slid" $\rightarrow$ Quincke.

Quincke vs beats vs standing wave (discrimination table) (cont.)

CHECK / unify: all four are interference; the measured quantity always converts via $\lambda = v/f$ and $v = \sqrt{\gamma RT/M}$ — the chapter is one chain of five formulas.

Two waves of equal amplitude, nearby frequencies: $s = s_0 \sin \omega_1 t + s_0 \sin \omega_2 t = 2 s_0 \cos\left(\frac{\omega_1 - \omega_2}{2} t\right) \sin\left(\frac{\omega_1 + \omega_2}{2} t\right)$. The audible pitch is the mean frequency $\bar{f} = (f_1 + f_2)/2$; the amplitude wobbles with envelope $2 s_0 |\cos(\pi (f_1 - f_2) t)|$ — one swell (loud-silent-loud) per period $1/|f_1 - f_2|$:

beat frequency $\mathbf{f_b = |f_1 - f_2|}$. Source examples: 256 & 258 Hz $\rightarrow$ 2 beats/s; 340 & 342.5 Hz $\rightarrow$ 2.5 beats/s.

Beats: f beat = |f 1 - f 2| from slow superposition (cont.)

Numbers: beats per second = frequency difference in hertz. Equal amplitudes: perfect silence at the dips (intensity $\propto \cos^2$). Unequal amplitudes $a_1 > a_2$: minima drop to $(a_1 - a_2)^2$ — "beats with incomplete silences" tells you amplitudes are unequal. Each beat is one cycle of the DIFFERENCE frequency; the ear hears the sum-frequency tone pulsing.

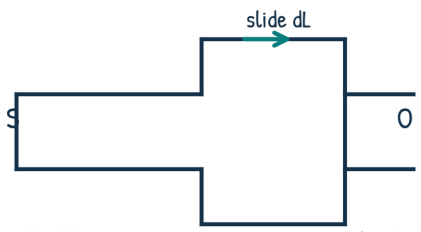
Physics of two tuning forks: strike 256 and 257; the pressure waves add at the ear; the pulse train at Hz = 1 Hz. The method is the standard of tuning: pianos tune unisons to zero beats; violinists tune strings by beats; orchestra A440 against a reference fork.

Beats: f beat = |f 1 - f 2| from slow superposition (cont.)

WHY / the trig identity separates sum and difference frequencies: the ear (nonlinear detector roughly rectifying) hears the fast carrier with a slowly varying loudness — the difference frequency appears as an envelope, not as a tone (unless f_b is in the audible range and the ear intermodulates).

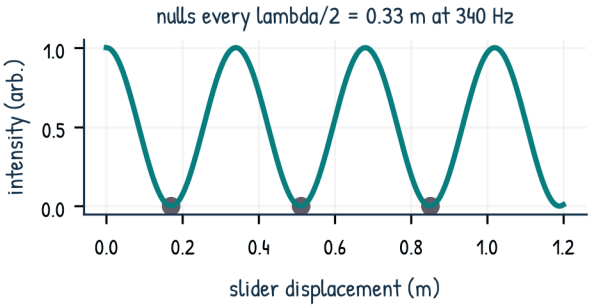
CHECK / if $f_1 - f_2 = 0.5$ Hz: beats every 2 s; the envelope is painfully slow — example: 340 vs 340.5 Hz. If a problem says "5 beats per second", $|f_1 - f_2| = 5$ Hz exactly — no factor 2 (the envelope $\cos^2$ has frequency f_b, not $2 f_b$; the $\cos$ of $\Delta\omega t/2$ completes a magnitude cycle when $\Delta\omega t/2 = \pi$, i.e. $t = 2\pi/\Delta\omega = 1/f_b$ ok).

Quincke tube: sliding arm changes the path by 2 dL



path difference 2 dL; nulls every lambda/2 of sliding

intensity vs slider: nulls every lambda/2



REMEMBER / Quincke: the sliding U-tube changes the path by 2 dL; nulls every lambda/2 of slider travel.

Beats: f beat = |f 1 - f 2| from slow superposition

Beats f_b = |f 1 -f 2|, wax test, envelope physics, drills.

Beat applications and problem patterns

(a) Unknown frequency: a 340 Hz fork gives 4 beats/s with an unknown; loading the unknown with wax lowers its frequency and beats fall to 2/s: then unknown = 344 Hz (wax lowered it toward 340... careful logic: if loading DECREASES unknown from f_u: beats $|f_u - 340| = 4$; after loading f_u drops: new beats 2: if $f_u = 344$, after loading 342: beats 2 ok; if $f_u = 336$, after loading 334: beats 6 fails — the wax test discriminates: unknown 344 Hz.

Beat applications and problem patterns (cont.)

(b) Three forks: 256, 257, 258: pairwise beats 1, 2, 1; together the 256/258 pair beats at 2 Hz with the 257 riding the envelope — complex beat patterns from three tones (olympiad flavour). (c) Siren receding (Doppler + beats): source 800 Hz moving gives observed 776; beats with a stationary 800: 24 Hz. (d) Doppler of BOTH source and observer toward each other: beats with a reference measures the shift — Doppler-free... reverse:

beats between direct and reflected wave from a moving wall measures the wall speed (Oly 08).

CHECK / beat = difference, always positive: $|f_1 - f_2|$: no sign; the "which is higher" question needs a second test (wax, tightening) — the trap of beat problems.

Beat applications and problem patterns (cont.)

WHY / beats measure frequency DIFFERENCES with arbitrarily fine resolution (1 beat in 10 s = 0.1 Hz), far better than the ear can judge absolute pitch — the reason beats are the tuning standard.

Two waves of slightly different frequency travelling together: the pattern of constructive and destructive interference MOVES at the group speed $v_g = \Delta\omega/\Delta k$; for sound with negligible dispersion $v_g = v$: the pulsing envelope travels at the ordinary speed — a boat bobbing in the beat envelope of two sea swells moves with the group... for sound: a microphone moving along the path hears the same beat rate plus Doppler terms.

Standing-wave beats: two forks at opposite ends of a tube: the pattern of nodes drifts; a slow "sloshing" of node positions at the beat frequency — visual beats.

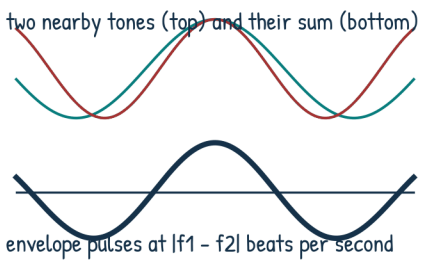
Beats with waves in space (extension) (cont.)

Laser Doppler / acoustic Doppler velocimetry: send f_0 to a moving particle; received f'; beat with a reference measures $f' - f_0 \propto v_{\text{particle}}$ directly — the working principle of police radar AND ultrasound flowmeters (Oly 08 building block).

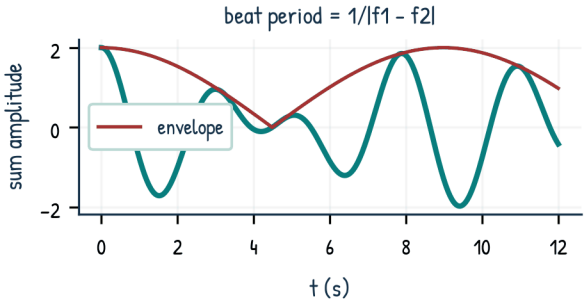
(a) 330 & 331.5: 1.5 beats/s. (b) $f_b = 3$ Hz between a fork and a siren moving at 3 m/s ($v = 343$): $f' = f(1 \pm 3/343)$: $\Delta f = f \cdot 3/343 = 3 \Rightarrow f = 343$ Hz. (c) A string tightened raises f from 220 to 221; beat with 223 Hz fork: before 3, after 2 ok pattern. (d) Two pipes at 500 and 502 Hz, equal amplitude: envelope frequency 2 Hz;

intensity dips to ZERO at each dip; with amplitude ratio 2:1, dips to 1/9 of max.

two nearby tones and their pulsing sum



beat envelope at |f1 - f2|



REMEMBER / Beats measure frequency differences: the envelope repeats at |f1 - f2|; the wax test finds which fork.

Beats: f beat = |f 1 − f 2| from slow superposition (cont.)

Beats f_b = |f 1 −f 2|, wax test, envelope physics, drills.

Beat drill set (cont.)

Intensity formula for beats: $I(t) = 4I_0\cos^2(\pi f_b t)$ (equal amplitudes): mean $2I_0$ = sum of intensities ok energy conservation on average; instantaneous swings 0 to $4I_0$.

WHY / the mean beat intensity equals the incoherent sum: time averaging the cross term kills it — consistency between coherent and incoherent pictures at long times.

Counter-propaqating waves of equal amplitude: $s = s_0\sin(kx-\omega t) + s_0\sin(kx+\omega t) = 2s_0\sin(kx)\cos(\omega t)$: the pattern oscillates in place — nodes where $\sin kx = 0$ ($x = n\lambda/2$), antinodes at $x = (n+\frac{1}{2})\lambda/2$. Node spacing $\lambda/2$; node-antinode spacing $\lambda/4$. Amplitude map: $A(x) = 2s_0|\sin kx|$; intensity $\propto \sin^2 kx$ averages $2I_0$ of the standing...

Standing waves: nodes, antinodes, and boundary conditions (cont.)

precisely: standing wave intensity oscillates spatially and averages locally over time to $2I_0\sin^2 kx$ (cross term time-averages to zero for the pair travelling oppositely... the net average energy flux is ZERO — energy sloshes, none propagates: hallmark of pure standing waves.

Pressure vs displacement standing wave: $\Delta P = -2Bks_0\cos(kx)\cos(\omega t)$ — pressure nodes where displacement antinodes, and vice versa (SN-03). Rigid boundary (closed end of pipe, wall): displacement NODE / pressure ANTINODE — the boundary forces $s = 0$ (or fluid cannot flow through). Open end (free to atmosphere):

Standing waves: nodes, antinodes, and boundary conditions (cont.)

displacement ANTINODE / pressure NODE — the open end enforces $\Delta P = 0$ (pressure must match ambient).

Resonance: only discrete k fit the boundary conditions: pipe of length L with given ends $\Rightarrow L = n\lambda_n/2$ or $(2n-1)\lambda/4$ patterns (JEE-17/18). A driver forces any pattern, but resonance builds amplitude 10–100x when the driving f matches an allowed mode (Q factors 10–50 for pipes — Oly 05).

WHY / the boundary conditions are physics, not maths: rigid end = zero displacement because the gas cannot penetrate the wall; open end = zero excess pressure because the end is exposed to the atmosphere (actually a weak antinode $\lambda/4$ beyond the mouth — end correction, JEE-19).

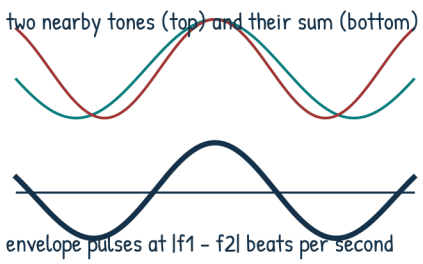
Boundary phase rules (the trap bank)

CHECK / draw one: for a closed pipe, displacement nodes at closed end, antinode at open end: pattern $s = s_0\sin(kx)\cos\omega t$ with $kL = (2n-1)\pi/2$: allowed $L = (2n-1)\lambda/4$ ok odd harmonics (JEE-17).

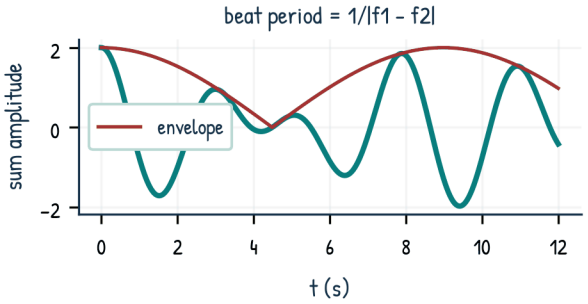
Rigid wall reflection of a displacement wave: the DISPLACEMENT wave inverts (phase jump π) so that node forms at wall... consistent statements: displacement reflects with π phase flip (fixed end of a string analogy), PRESSURE wave reflects with NO flip at a rigid wall (pressure antinode at wall). Open end: pressure reflects with π flip (pressure node), displacement no flip (antinode). Working shortcut:

always think displacement node = rigid, pressure node = open; derive the other from quadrature.

two nearby tones and their pulsing sum



beat envelope at |f1 − f2|



REMEMBER / Beats measure frequency differences: the envelope repeats at |f1 − f2|; the wax test finds which fork.

Standing waves: nodes, antinodes, and boundary conditions

Standing waves, boundary conditions, node/pressure rules, SWR.

Boundary phase rules (the trap bank) (cont.)

Quarter-wave layer: a closed pipe of $L = \lambda/4$ resonates at $f = v/4L$; half-wave: $L = \lambda/2$ for open pipe; these are the two building blocks. Mixed: pipe closed at one end has odd quarter-waves ONLY: $L = (2n-1)\lambda/4$; open-open any integer half-waves: $L = n\lambda/2$ ... but open-open ALSO allows odd quarter-wave-like modes? No: open-open with BOTH ends antinodes requires $L = n\lambda/2$ (all harmonics).

Third boundary type (partially open, impedance mismatch, Oly 02): reflection coefficient $R = ((Z_2 - Z_1)/(Z_2 + Z_1))^2$ of intensity; the wall is "rigid" when $Z_2 \gg Z_1$ (acoustically hard), "open" when $Z_2 \ll Z_1$.

Boundary phase rules (the trap bank) (cont.)

WHY / the phase rules fix the mode spectrum: count quarter wavelengths that fit between the end conditions — the entire pipes chapter is that count.

CHECK / resonance between: a pipe closed at BOTH ends = same as open-open mode spacing ($n\lambda/2$)? No: closed-closed = displacement nodes both ends: $L = n\lambda/2$, all harmonics ok — the ends being the same type gives all harmonics; mixed ends give odd only.

Travelling: amplitude constant in space; energy flux $I = \bar{u}v$ nonzero; phase advances; ΔP quadrature with s locally... precisely: $\Delta P = \rho v u$ (in phase relation with particle velocity u). Standing: amplitude $\propto |\sin kx|$; zero average flux; ΔP in QUADRATURE with u (like reactive power); nodes every $\lambda/2$ fixed in space.

Standing vs travelling: comparison table (cont.)

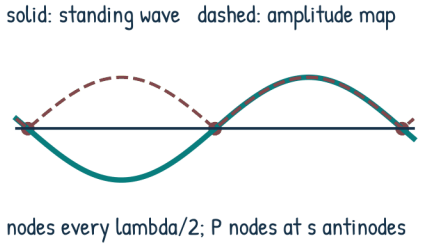
Partial standing waves (real rooms): superposition of forward and reflected with amplitude ratio $r < 1$: standing wave ratio $\mathrm{SWR} = (1+r)/(1-r)$; full standing $r = 1$; pure travelling $r = 0$. Measure $I_{\max}/I_{\min} = \mathrm{SWR}^2$ to get reflection coefficient — room acoustics and duct design (Oly 05).

the sound field between two hard walls in a room is a 3D standing-wave lattice (room modes $f = \frac{v}{2}\sqrt{(n_x/L_x)^2 + (n_y/L_y)^2 + (n_z/L_z)^2}$) — acoustic "eigenmodes"; audible as boomy bass at some spots (the 1D pipe theory generalises exactly).

Micro-worked standing wave set

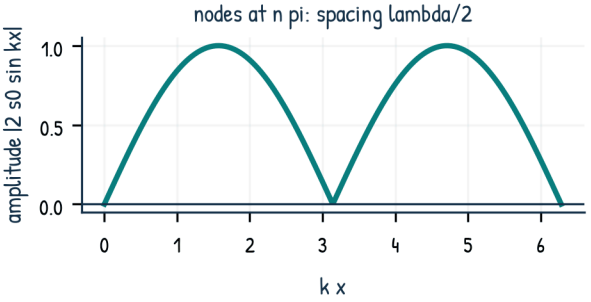
(a) $s = 4 \times 10^{-6}\sin(5x)\cos(2000t)$: node spacing $\pi/5 = 0.63$ m; $\lambda = 1.26$ m; $v = 2000/5 = 400$ m/s (hot air or He mix... plausible warm gas). (b) A pipe open at both ends, $L = 0.5$ m, $v = 343$: fundamental 343 Hz, first overtone 686, second 1029 — all harmonics. (c) Closed at one end, same L: fundamental 171.5 Hz; overtones 514.5, 857.5 — odd only; compare spacing 343 Hz (twice fundamental).

incident plus reflected: fixed node pattern



nodes every lambda/2; P nodes at s antinodes

amplitude map: nodes every lambda/2



REMEMBER / Rigid end = displacement node; open end = pressure node; spacing of nodes is lambda/2.

Standing waves: nodes, antinodes, and boundary conditions (cont.)

Standing waves, boundary conditions, node/pressure rules, SWR.

Micro-worked standing wave set (cont.)

Node-antinode conversions: at a displacement node ΔP is maximum and temperature fluctuation is maximum too (compression heats! — the "thermoacoustic" hint); a hot-wire probe finds pressure antinodes; dust heaps in Kundt (JEE-21) sit at displacement NODES = pressure ANTINODES... hmm dust heaps actually collect at pressure nodes? Correction:

dust collects at DISPLACEMENT NODES (pressure antinodes) in a Kundt tube ok (nodes of the standing displacement wave where the gas barely moves... actually dust collects at displacement nodes because the gas velocity is zero there — gas velocity antinodes sweep dust to nodes).

Micro-worked standing wave set (cont.)

WHY / dust collects at displacement nodes: acoustic streaming pushes loose powder toward the quiet spots of the gas motion; those are $\lambda/2$ apart — the measurement of λ in Kundt tubes.

Closed pipe: one end rigid (displacement node), other open (displacement antinode). Allowed patterns: $L = (2n-1)\dfrac{\lambda_n}{4}$, $n = 1, 2, 3, \ldots$ so $\lambda_n = \dfrac{4L}{2n-1}$ and $\mathbf{f_n = (2n-1)\dfrac{v}{4L}}$: fundamental $f_1 = v/4L$; overtones: first overtone $= 3v/4L = 3f_1$; second overtone $= 5v/4L = 5f_1$. Only ODD harmonics of f_1: the spectrum is $f_1, 3f_1, 5f_1, \ldots$ — mellow, hollow timbre (clarinet family).

Closed pipe (closed at one end): odd harmonics only (cont.)

Worked (source): $L = 0.34$ m, $v = 340$: $f_1 = 250$ Hz; first overtone 750 Hz; second 1250 Hz. Longitudinal displacement profiles: fundamental = quarter sine (node closed, antinode open); first overtone = three-quarter sine; second = five-quarter. Pressure profiles: antinode at closed end, node at open end.

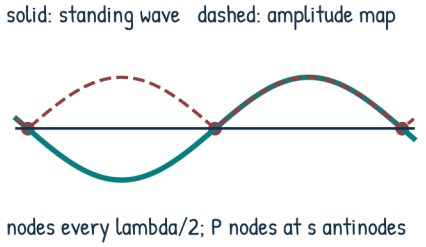
Why odd only: an integer number of half-waves fits node-to-antinode only as odd quarters — even quarters would need node-to-node or antinode-to-antinode (a different end pair). Same physics: the mode condition is $kL = (2n-1)\pi/2$; adding another half wave flips both ends to the wrong type.

Closed pipe (closed at one end): odd harmonics only (cont.)

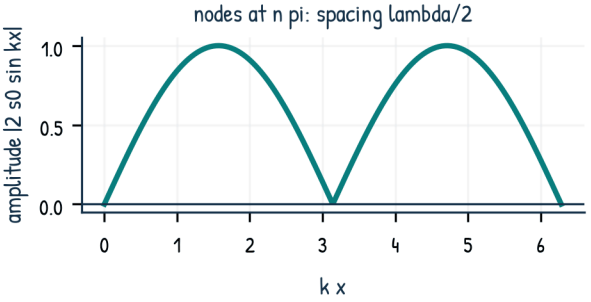
WHY / odd-harmonic content is a fingerprint: if a spectrum shows $f, 3f, 5f$ with no $2f$, the resonator is closed-at-one-end; if all integers appear, open-open. Exam "identify the pipe from the spectrum" questions use exactly this.

CHECK / adjacent resonances of a closed pipe (scanning f): $f_{n+1} - f_n = 2v/4L = v/2L$: constant spacing of TWICE the fundamental: source example 340 Hz spacing for $L = 0.5$ m: $v/2L = 343$ ok (this constant spacing is the backbone of the resonance tube method, JEE-20).

incident plus reflected: fixed node pattern



amplitude map: nodes every lambda/2



REMEMBER / Rigid end = displacement node; open end = pressure node; spacing of nodes is lambda/2.

Closed pipe (closed at one end): odd harmonics only

Closed pipe odd harmonics, overtone naming, spacing v/2L.

Mode shapes and the overtone ladder (exam detail)

Naming discipline (trap bank source Trap 8): fundamental = 1st harmonic $= f_1 = v/4L$; FIRST overtone = 3rd harmonic $= 3v/4L$; SECOND overtone = 5th harmonic $= 5v/4L$. The paper Q16 mislabels 3v/4L as "second overtone" — correct labelling: first overtone. Source Trap 8 explicitly warns about this counting.

Displacement amplitude maximum at open end (antinode) — that is where the radiated sound escapes; the closed end has pressure maximum (the wall feels the pressure swing — microphone there reads max). Energy: antinode of velocity at open end = where radiation happens; node of velocity at closed end = no work done on wall (ideal).

Mode shapes and the overtone ladder (exam detail) (cont.)

Forced oscillation slightly off resonance: amplitude falls, phase shifts — half-power width $\Delta f = f/Q$ (Oly 05); pipe Q ~ 10–30 unflanged, higher with flange — why "tuning across resonance" in JEE-20 labs shows sharp maxima.

CHECK / numbers check: closed pipe L = 25.7 cm, v = 343: f1 = 333.7 Hz, f3 = 1001, f5 = 1668 — spacing 667.5 Hz = v/2L ok consistent; overtone naming: 1001 Hz is first overtone.

(a) "A closed pipe resonates at 170 Hz and 255 Hz consecutively with the tuning fork; find fundamental and length": spacing 85 Hz = v/4L... for closed pipe consecutive resonances ARE at f1, 3f1: spacing 2f1: f1 = 85... careful: resonances at odd multiples: if two consecutive resonances 170 & 255: these are 3f1=170, 5f1=255? spacing 85 = 2f1: f1 = 42.5: L = v/4f1 = 2.02 m; or the fork is near 170 = f1? The spacing method: f_{n+1} - f_n = v/2L always:

Closed-pipe problem patterns (cont.)

L = v/(2\cdot 85) = 2.02 m; f1 = v/4L = 42.5 Hz.

(b) "Closed pipe cut to 1/3 length; how do modes change?" all frequencies triple (v fixed): spectrum 3f1, 9f1, 15f1... ratio pattern unchanged (still odd-only). (c) "Water-pipe (closed by water level, variable L)": f1 = v/4L as L varies — organ pedal tubes; raise water level (shorten L) raises pitch.

WHY / the constant spacing v/2L between consecutive odd harmonics (not 2f1... which is the same thing: 3f1 - f1 = 2f1 = v/2L ok) makes the closed pipe a frequency meter: measure two consecutive resonances, get v.

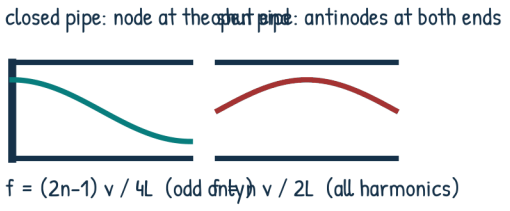
Closed vs open pipe side-by-side (memorise)

Closed: $f_1 = v/4L$; harmonics $1, 3, 5, \ldots$; overtone gaps $v/2L$; timbre hollow; example: clarinet, pan flute (stopped), bottle (Helmholtz-ish). Open: $f_1 = v/2L$; harmonics $1, 2, 3, \ldots$; overtone gaps $v/2L$; timbre bright; example: flute, organ open pipes, most vocal tract approximations. Fundamental ratio for same L: closed = half of open... $v/4L$ vs $v/2L$: closed is an octave below.

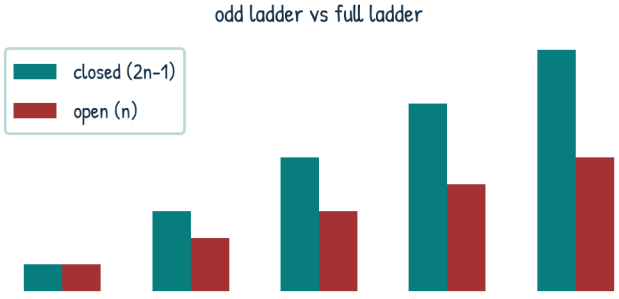
Exam shortcut: any pipe question: (i) identify ends; (ii) write $L = ?\lambda/4$ condition; (iii) v from temperature if not given; (iv) label harmonics carefully; (v) end correction if precision asked (JEE-19).

CHECK / cross-check with overtone spacing: closed pipe: consecutive allowed modes differ by $2f_1$; open pipe: by f_1. If a spectrum has spacing 2f1 and starts at f1: it is closed; starts at f1 with spacing f1: open.

closed pipe: odd quarter waves only



harmonic ladder: closed vs open



REMEMBER / Closed pipe: odd harmonics (2n-1)v/4L; overtone number is harmonic number minus one.

Open pipe (open at both ends): all harmonics

Open pipe all harmonics, spectra identification, one-line mode solver.

Open pipe (open at both ends): all harmonics

Both ends open = displacement antinodes at both ends: allowed $L = n\frac{\lambda_n}{2}$, so $\lambda_n = 2L/n$ and $\mathbf{f_n = n\frac{v}{2L}}$: fundamental $f_1 = v/2L$; harmonics $f_n = nf_1$ ALL integers; overtones: first $= 2f_1$, second $= 3f_1$, ... — bright, full timbre (flute family). Worked (source): $L = 0.5$ m, $v = 343$: $f_1 = 343$ Hz; $f_2 = 686$; $f_3 = 1029$; $f_4 = 1372$.

Open pipe (open at both ends): all harmonics (cont.)

Mode shapes: fundamental = half-sine of displacement (antinode at both ends, node at midpoint) and half-sine of pressure (pressure nodes at both open ends, antinode at midpoint). Second harmonic: full wave with two displacement nodes. Consecutive resonance spacing: $v/2L$ = one fundamental — identical spacing to a closed pipe's overtone gaps.

Open pipe (open at both ends): all harmonics (cont.)

WATCH / Traps: (i) "open pipe" in exam problems sometimes means a pipe closed at ONE end by context — read ends! (ii) end correction applies at BOTH ends (JEE-19): effective $L_{\text{eff}} = L + 2e$ vs closed $L + e$ — this shifts every frequency down ~1-3%. (iii) comparing with strings: string fixed-fixed: $f_n = nv/2L$ identical ladder to open pipe (different wave type but same mode count) — the mathematics is shared.

WHY / symmetric boundaries (same type both ends) give integer half-wave ladder; asymmetric (node one end, antinode other) gives odd-quarter ladder — one rule explains both pipes.

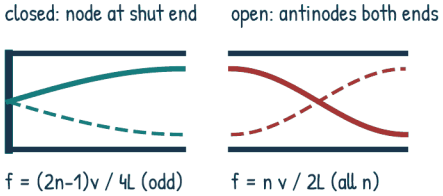
CHECK / open pipe L = 0.5 m at 20 C: f1 = 343.2/1 = 343 Hz; the SAME tube closed at one end: f1 = 171.6 Hz — exactly an octave lower ok.

Open-pipe spectra in data questions

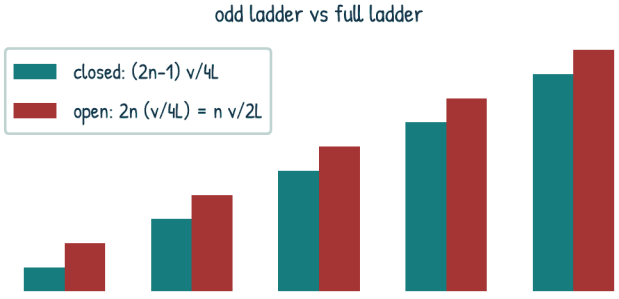
Given a list of resonant frequencies 220, 440, 660, 880: all harmonics of 220 $\rightarrow$ open pipe (or string); 220, 660, 1100: odd only $\rightarrow$ closed. Given spacing 85 Hz between resonances: $L = v/(2 \cdot 85) = 2.02$ m for either pipe type (spacing formula shared!). Given fundamental only: $L = v/2f_1$ (open) vs $v/4f_1$ (closed) — factor 2 ambiguity, resolve from the description or from a second data point.

Organ design: open pipes sound an octave higher than stopped pipes of equal length; mixtures of both give chorus effects; 16 ft open = 32 ft stopped pitch. The "rank" of a pipe organ is literally this ladder.

open pipe: antinodes at both mouths



all harmonics present in the open ladder



REMEMBER / Open pipe: all harmonics n v/2L; consecutive resonances always differ by v/2L.

Open pipe (open at both ends): all harmonics (cont.)

Open pipe all harmonics, spectra identification, one-line mode solver.

Open-pipe spectra in data questions (cont.)

Sound from the ends: both ends radiate; open-open radiates from two ends (interference possible at low f when $L < \lambda$ – the dipole "singing tube" effect, Oly 05); closed-open radiates mainly at the open end (monopole).

CHECK / End correction shift: open pipe $f_1 = v/[2(L+2e)]$; with $r = 2$ cm, $e = 1.2$ cm, $L = 85$ cm: $L_{\text{eff}} = 0.874$ m gives $f_1 = 343/1.748 = 196.2$ Hz vs 201.8 Hz uncorrected (-2.75%).

Open-pipe problem patterns

(a) Pipe in helium vs air (same T): frequencies scale with v: $f_{\text{He}}/f_{\text{air}} = v_{\text{He}}/v_{\text{air}} = 2.93$ – pitch jumps nearly two octaves... 1.56 octaves; classic demonstration (party voice). (b) Temperature: a pipe tuned to 300 Hz at 20 C: at 30 C: $f = 300\sqrt{303/293} = 305.1$ Hz – out of tune by 5 Hz. (c) Pipe length for a target: "design an open pipe for 440 Hz at 25 C": $v = 346.1$, $L = v/2f = 0.393$ m (minus end correction: $L = 0.393 - 2e$).

Two-pipe comparison: open pipe A and closed pipe B sound the same fundamental: $v/2L_A = v/4L_B \Rightarrow L_B = 2L_A$; their first overtones: A: $2f$; B: $3f$ – differ; beats between overtones: f – demonstration question.

Open-pipe problem patterns (cont.)

WHY / frequency of any mode is v over the mode wavelength; everything geometric lives in the wavelength count; temperature and gas choice live in v – two independent knobs.

Given pipe type and length: (1) Write end conditions as node/antinode pattern. (2) Count quarter waves fitting: mixed ends: $L = (2n-1)\lambda/4$; same ends: $L = n\lambda/2$. (3) $f = v/\lambda$; add end correction $L \to L_{\text{eff}}$ if needed. (4) Label: harmonic number $=$ multiple of f_1; overtone number $=$ harmonic number -1 (the trap!). (5) If not given: $v = \sqrt{\gamma RT/M}$ at working temperature.

Generalise: 3D rooms: $f = \frac{v}{2}\sqrt{(n_x/L_x)^2 + \ldots}$; string: $f = n\sqrt{T/\mu}/2L$; rods: $f = nv_{\text{rod}}/2L$ – same ladder mathematics with different v.

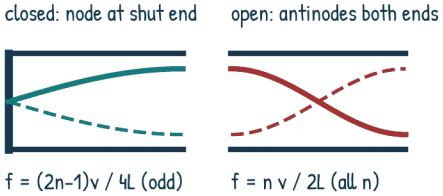
The one-line mode solver (all pipes, all questions) (cont.)

CHECK / the ladder of any resonator: compute f_1; check the spacing of higher modes is a simple multiple of f_1 (1, 2, 3 or 1, 3, 5); if not, end corrections or transverse modes entered.

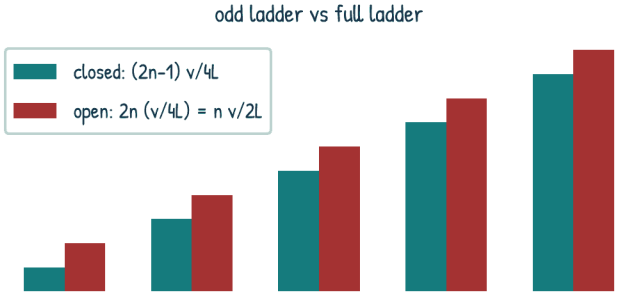
The displacement antinode sits slightly OUTSIDE the physical mouth of the pipe: the oscillating air just beyond the opening participates in the inertia. The effective length is longer than the cut length: $L_{\text{eff}} = L + e$ with $e \approx 0.6r$ for an unflanged (sharp) opening, $e \approx 0.85r$ for a flanged (flared/baffled) opening; r = pipe radius. Source worked (Q4):

Worked comparison (Q4, $L = 85$ cm, $r = 2$ cm, $e = 1.2$ cm): open pipe gives 196.2 Hz ($v = 343$); closed pipe at $v = 340$ m/s shifts from 100.0 Hz to $340/[4(0.862)] = 98.6$ Hz (-1.4%).

open pipe: antinodes at both mouths



all harmonics present in the open ladder



REMEMBER / Open pipe: all harmonics n v/2L; consecutive resonances always differ by v/2L.

End correction: e approx 0.6r, L eff = L + e (closed) or L + 2e (open)

End correction e ≈ 0.6r/0.85r, L_eff rules, resonance-tube link.

End correction: L_eff = L + e (closed) or L + 2e (open)

Closed pipe: one open end $\Rightarrow L_{\text{eff}} = L + e$; open pipe: two $\Rightarrow L + 2e$. All frequencies fall slightly: $f_{\text{true}} = f_{\text{geo}} \cdot L/L_{\text{eff}}$ (approximately $f(1 - e/L)$ for small correction). Relative size: for $r = 2$ cm, $e = 1.2$ cm is 1.4% of 85 cm — measurable; for lab pipes $r \sim 1.5$ cm, $L \sim 0.3$ m: correction 3%.

Physics of $e = 0.6r$: the added end mass is the oscillating plug of air beyond the mouth, of length $\propto r$; exact coefficient depends on the mouth geometry (radiation end impedance, Oly 03): unflanged $e = 0.6133r$ (Rayleigh), flanged $e = 0.8216r$ (Levine-Schwinger); "0.6 and 0.85" are the exam standards.

End correction: L_eff = L + e (closed) or L + 2e (open)

WHY / the antinode cannot sit exactly at the rim: pressure continuity with the open atmosphere extends the oscillatory region past the mouth — the wave "overshoots" by e; failing to include e systematically overestimates frequencies.

CHECK / source Q5 method (resonance tube): $e = (l_2 - 3l_1)/2$ and $v = 2f(l_2 - l_1)$: with $l_1 = 16.5$ cm, $l_2 = 50.1$ cm, $f = 512$ Hz: $v = 344$ m/s, $e = 3$ mm ok; note $\lambda/4 - l_1 = e$ consistency: $\lambda = 2(l_2 - l_1) = 0.672$ m: $\lambda/4 = 16.8$ cm: $e = 16.8 - 16.5 = 3$ mm ok both routes agree — the safe identity: $l = \lambda/4 - e$ (Trap 15 prints $+$; the worked solutions use $-$ — see SOURCE-AUDIT).

When does the correction matter (and when not)

Matter: precision labs, high-Q measurements, short pipes, comparison of nearly equal pipes (beat detection resolves 0.1%!). Not matter: order-of-magnitude pitches, long pipes ($e/L < 1$ %), problems that say "neglect end correction". Exam rule: include if radius given or if options straddle 1-3%; mention explicitly.

Flange effect demonstration: same pipe with/without a baffle plate at the mouth: e rises 0.6r $\rightarrow$ 0.85r: frequency drops ~2% for short pipes: measurable with beats. Real instruments use bells (much larger effective flange, tuned curves) — the bell couples impedance to the room (Oly 03).

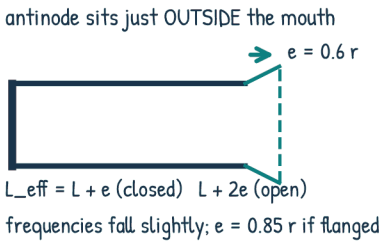
When does the correction matter (and when not) (cont.)

Helmholtz resonator note: a bottle with neck has its own end-corrected neck formula: $f = \frac{v}{2\pi}\sqrt{A/(VL_{\text{eff}})}$ with $L_{\text{eff}} = L_{\text{neck}} + 1.5r$ — different formula family (Oly 05), same e physics.

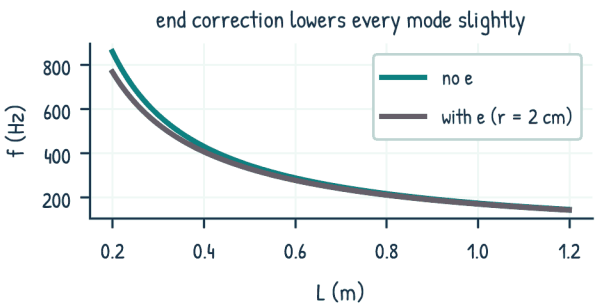
CHECK / dimensional and sign check: $e = 0.6r$ positive (lengthens); f always drops when including e; if your f rises, a sign flipped.

(a) "An open pipe of length 50 cm, radius 2 cm sounds at f; find f with e": $L_{\text{eff}} = 50 + 2(1.2) = 52.4$ cm: $f = 343/1.048 = 327$ Hz vs 343 uncorrected: 4.7% low... check: L_eff = 0.524, f = v/2L_eff = 327 ok. (b) "A closed pipe 30 cm, r = 1.5 cm, at 30 C": $v = 348.2$, $e = 0.9$ cm: $f = 348.2/(4 \times 0.309) = 282$ Hz. (c) "Find r from measured shift":

the antinode sits slightly beyond the mouth



end correction lowers each mode slightly



REMEMBER / End correction e = 0.6r (0.85r flanged): L_eff = L + e closed, L + 2e open; frequencies fall.

End correction: e approx 0.6r, L eff = L + e (closed) or L + 2e (open) (cont.)

End correction e ≈ 0.6r/0.85r, L_eff rules, resonance-tube link.

End-correction problem patterns (cont.)

open pipe f_qeo = 400 (no e) vs measured 392: $L_{\text{eff}}/L = 400/392 = 1.0204$: $2e/L = 0.0204$: $e = 0.0102L = 1.02$ cm for L = 1 m: $r = e/0.6 = 1.7$ cm.

Resonance tube method works BECAUSE of end correction cancellation: $v = 2f(l_2 - l_1)$ does not need e at all (the spacing of two consecutive resonances eliminates e); e itself comes from the absolute positions $e = (l_2 - 3l_1)/2$ — the elegant double measurement of JEE-20.

WHY / the spacing trick: $l_2 = 3\lambda/4 - e$, $l_1 = \lambda/4 - e$: difference $\lambda/2$ independent of e; sum-based formula extracts e — one experiment, two unknowns solved.

Micro-drills on the correction

(a) Ratio $f_{\text{corrected}}/f_{\text{geo}} = L/(L+e)$ (closed) or $L/(L+2e)$ (open). (b) $r = 3$ cm flanged: $e = 2.55$ cm; open pipe $L = 1$ m: $L_{\text{eff}} = 1.051$ m: f drops 4.8%. (c) Two pipes same L, $r_1 = 1$ cm, $r_2 = 3$ cm: open-open: e ratio 3: frequency ratio $\frac{L+2(0.6)}{L+2(1.8)} = \frac{1.012}{1.036} = 0.977$ for L = 1 m: 2.3% apart: beat frequency at 300 Hz = 7 Hz.

Rule of thumb: correction significant when $r/L > 2$ %; exam pipes $r \approx 1.5$ -2 cm, $L \approx 30$ -100 cm $\Rightarrow$ 2-6% shifts.

the exact radiation end-impedance $Z_{\text{end}} = \rho v[\frac{1}{4}(ka)^2 + i \cdot 0.6ka]$ for unflanged pipe (low ka) — the imaginary part is the end correction, the real part is radiation resistance (Oly 03/05) — same physics seen through impedance.

Resonance tube: v = 2f(l 2 - l 1) and e = (l 2 - 3l 1)/2

Apparatus: a tuning fork of frequency f held over a vertical tube whose water level (closed end) is adjustable; the air column (closed-open) resonates when its effective length matches $(2n-1)\lambda/4$. Move the level down: first resonance at length l_1 (fundamental: $l_1 + e = \lambda/4$), second at l_2 (first overtone: $l_2 + e = 3\lambda/4$). Subtract: $l_2 - l_1 = \lambda/2$ — end correction cancels!

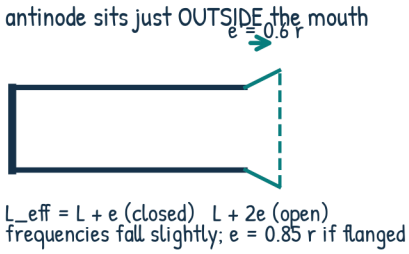
Hence $\mathbf{v = 2f(l_2 - l_1)}$ and $\mathbf{e = (l_2 - 3l_1)/2}$ (from adding: $l_1 + l_2 + 2e = \lambda$ ... precisely $l_1 + e = \lambda/4$, $l_2 + e = 3\lambda/4$: solve: $e = \lambda/4 - l_1 = (l_2 - l_1)/2 - l_1 = (l_2 - 3l_1)/2$ ok).

Resonance tube: v = 2f(l 2 - l 1) and e = (l 2 - 3l 1)/2 (cont.)

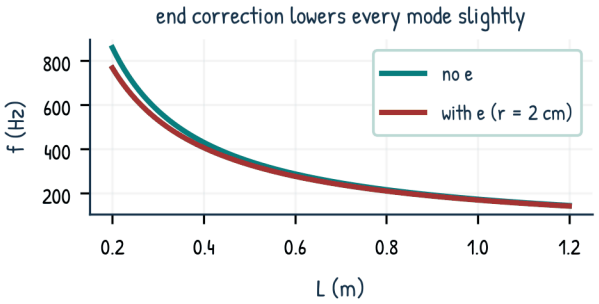
Source worked (Q5): $f = 512$ Hz, $l_1 = 16.5$ cm, $l_2 = 50.1$ cm: $\lambda = 2(50.1 - 16.5) = 67.2$ cm; $v = 512 \times 0.672 = 344.1$ m/s (≈ 344, consistent with warm lab ~25 C); $e = (50.1 - 49.5)/2 = 0.3$ cm = 3 mm; check $e = \lambda/4 - l_1 = 16.8 - 16.5 = 3$ mm ok. Then $r = e/0.6 = 5$ mm for the tube.

Procedure and errors: use a tall tube and a low fork if possible (large l values reduce % error); resonance is sharp, so find the midpoint of the loudness peak; correct for temperature ($v(T)$); the tube must be vertical (water level defines the closed end); fork amplitude should be moderate (nonlinearity at high amplitude shifts peaks).

the antinode sits slightly beyond the mouth



end correction lowers each mode slightly



REMEMBER / End correction e = 0.6r (0.85r flanged): L_eff = L + e closed, L + 2e open; frequencies fall.

Resonance tube: v = 2f(l 2 - l 1) and e = (l 2 - 3l 1)/2

Resonance tube v = 2f(l 2 –l 1), e = (l 2 –3l 1)/2, errors and drills.

Resonance tube: v = 2f(l 2 - l 1) and e = (l 2 - 3l 1)/2 (cont.)

WHY / two resonances of the same closed pipe differ by exactly $\lambda/2$ of length regardless of e: the spacing method is self-correcting — the reason it is the standard student experiment for v.

CHECK / identity route: $l = \lambda/4 - e$ (correct) vs the printed "$l = \lambda/4 + e$" in source Eq (2.15)/Trap 15: check consistency with Q5: $l_1 = 16.8 - 0.3 = 16.5$ ok with MINUS; with plus: 17.1 fails. Record in SOURCE-AUDIT; use $l = \lambda/4 - e$.

Alternate forms and fork-adjacent variants

(a) Known e, one resonance: $v = 4f(l_1 + e)$. (b) Two forks f_a, f_b at the same liquid level resonance (method of mixtures... different context) — skip. (c) Two consecutive resonances of an OPEN tube: same formulas $v = 2f(l_2 - l_1)$, $e = (l_2 - 3l_1)/2$ BUT with $l_n = n\lambda/2 - e$ ... open tube: $l_1 + 2e = \lambda/2$, $l_2 + 2e = \lambda$: spacing $\lambda/2$ same; $e = (l_2 - 3l_1)/2$ changes to $e = (l_2 - 3l_1)/2$? derive: $l_1 = \lambda/2 - 2e$, $l_2 = \lambda - 2e$: $l_2 - l_1 = \lambda/2$ ok same spacing formula; $l_2 - 3l_1 = \lambda - 2e - 3\lambda/2 + 6e = -\lambda/2 + 4e$: $e = (l_2 - 3l_1 + \lambda/2)/4$ — different! Exam convention: resonance tube = closed-open; use the closed formulas.

Alternate forms and fork-adjacent variants (cont.)

Dust-tube variant (Kundt, JEE-21) measures λ directly from dust heaps; resonance tube measures λ from length sweep; both give $v = f\lambda$.

CHECK / if $l_1 > \lambda/4$ computed from a guessed v, the first found "resonance" might actually be the overtone — consistency: $l_2 \approx 3l_1 + 2e$ -ish: for $l_1 = 16.5$ cm: $l_2 \approx 49.5$ + 2e ok 50.1 measured.

(a) A fork 480 Hz: resonances at 17.0 cm and 51.8 cm: $v = 2 \cdot 480 \cdot 0.348 = 334.1$ m/s: temperature? $v = 331.3\sqrt{T/273}$: $T = 273(334.1/331.3)^2 = 277.7$ K = 4.5 C — cool lab. $e = (51.8 - 51.0)/2 = 4$ mm; $r = e/0.6 = 6.7$ mm.

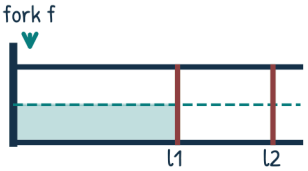
Worked exam problems (resonance tube family) (cont.)

(b) "A resonance tube shows first resonance at 20 cm for a 400 Hz fork; find v without second resonance": assume $e = 0$: $v = 4fl_1 = 320$ m/s; with $e = 1$ cm: $v = 4 \cdot 400 \cdot 0.21 = 336$ — the ambiguity of one measurement; state the assumption.

(c) "How long a tube for the 3rd resonance of 300 Hz at 20 C?" $\lambda = 1.144$ m: resonances at $\lambda/4 - e$, $3\lambda/4 - e$, $5\lambda/4 - e$: 3rd $= 5 \times 0.286 - e = 1.43 - e$ m.

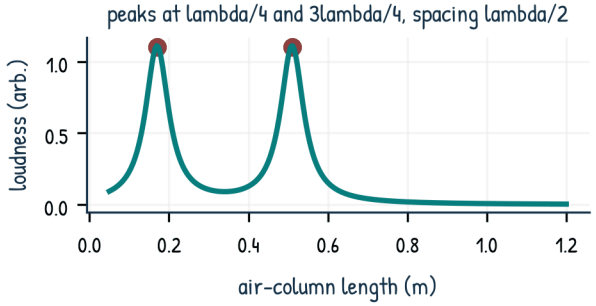
WHY / the method measures v from LENGTHS, needing only a good ruler and a known fork — the classic 1%-accuracy student method (with a thermometer for $v(T)$ correction).

resonance tube: fork over an air column



water level sets the closed end; l2 - l1 = lambda/2
v = 2 f (l2 - l1) e = (l2 - 3 l1)/2

loudness peaks at lambda/4 and 3lambda/4



REMEMBER / Resonance tube: v = 2f(l2 - l1) cancels the end correction; e = (l2 - 3l1)/2 and l = lambda/4 - e.

Resonance tube: v = 2f(l 2 - l 1) and e = (l 2 - 3l 1)/2 (cont.)

Resonance tube v = 2f(l 2 −l 1), e = (l 2 −3l 1)/2, errors and drills.

Error analysis and exam "find the mistake" questions

Systematic: (i) neglecting e in one-resonance method: error e/L_1 up to 5%. (ii) temperature drift during the lab: v moves 0.6 m/s/K. (iii) tube not exactly uniform: resonance peak broadens. (iv) fork frequency off by heating from hands: 0.1%. (v) loudness judgment bias: read the peak midpoint.

WATCH / Mistake-finding format: "A student reports $v = 300$ m/s from $l_1 = 16.5$, $l_2 = 50.1$, $f = 512$." — check: $2f(l_2 - l_1) = 344$; 300 is wrong (maybe used $l_2 - l_1$ in cm as m: 33.6... or dropped the 2: $f(l_2 - l_1) = 172$ ... wrong arithmetic). Diagnosis drill: recompute the chain $\lambda = 2\Delta l$, $v = f\lambda$.

Error analysis and exam "find the mistake" questions (cont.)

using a microphone + oscilloscope instead of the ear: resonance shown as amplitude vs length curve — peak positions give l_1, l_2 more precisely (lab modernisation of the same physics).

Apparatus: a long glass tube with a little cork dust (lycopodium) powder, gas-filled; a rod clamped at its middle, stroked lengthwise, drives the tube at the rod fundamental via a disc at one end. The acoustic standing wave in the gas is dusted into visible heaps:

powder collects at DISPLACEMENT NODES of the gas (where the gas barely moves — acoustic streaming sweeps it there), spaced $\lambda_{\text{gas}}/2$ apart. Read $d = \lambda_{\text{gas}}/2$ directly with a ruler.

Kundt's tube: measure v gas from dust-heap spacing (cont.)

Clamped-centre Kundt rod: node at midpoint, antinodes at free ends $\Rightarrow \lambda_{\text{rod}} = 2L_{\text{rod}}$, $f = v_{\text{rod}}/(2L_{\text{rod}})$.

Matching gas heap spacing $d = \lambda_{\text{gas}}/2$ gives $v_{\text{gas}} = v_{\text{rod}}\, d/L_{\text{rod}}$ (e.g. $5000 \times 0.08/1.0 = 400$ m/s).

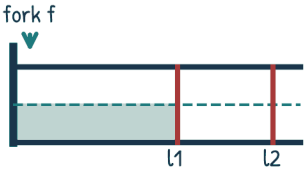
with $d = \lambda/2$ and matching frequency: $f = v_{\text{rod}}/(2L_{\text{rod}}) = 5000/2 = 2500$ Hz; $\lambda_{\text{gas}} = 2d = 0.16$ m; $v_{\text{gas}} = 2500 \times 0.16 = 400$ m/s ok (source chain: $v_{\text{gas}} = v_{\text{rod}} \cdot d/L_{\text{rod}}$ — consistent with $f = v_{\text{rod}}/2L_{\text{rod}}$ and $\lambda_{\text{gas}} = 2d$).

Kundt's tube: measure v gas from dust-heap spacing (cont.)

Full formula chain: $f = \dfrac{v_{\text{rod}}}{2L_{\text{rod}}} = \dfrac{v_{\text{gas}}}{2d} \Rightarrow$ $v_{\text{gas}} = v_{\text{rod}}\dfrac{d}{L_{\text{rod}}}$. Measure d (heap spacing), know L_{rod} and v_{rod} (or measure v_{rod} separately): get v_{gas}. The method is elegant because the same f drives both media — no fork calibration needed.

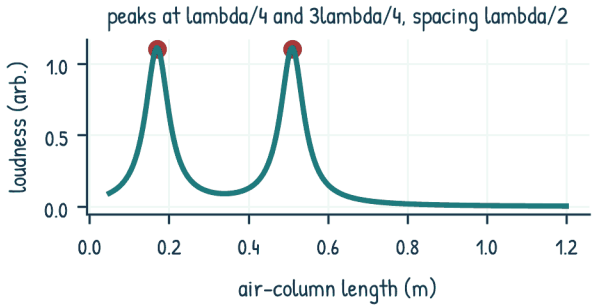
WHY / dust heaps at nodes: the oscillating gas velocity is zero at displacement nodes; radiation pressure / streaming pushes loose powder toward velocity minima — visualising the invisible wave. The heap spacing is HALF a wavelength — the classic factor-2 trap.

resonance tube: fork over an air column



water level sets the closed end; l2 - l1 = lambda/2
v = 2 f (l2 - l1) e = (l2 - 3 l1)/2

loudness peaks at lambda/4 and 3lambda/4



REMEMBER / Resonance tube: v = 2f(l2 - l1) cancels the end correction; e = (l2 - 3l1)/2 and l = lambda/4 - e.

Kundt's tube: measure v gas from dust-heap spacing

Kundt tube, dust heaps at nodes λ/2, v_gas = v_rod·d/L_rod.

Kundt's tube: measure v gas from dust-heap spacing (cont.)

CHECK / source numbers once more: $v_{\text{rod}} = 5 \times 10^3$ m/s (steel-ish, L = 1 m): $f = 2.5$ kHz; $d = 8$ cm: $\lambda = 16$ cm; $v_{\text{gas}} = 2.5 \times 10^3 \times 0.16 = 400$ m/s ok (hot air: $v = 331\sqrt{T/273} = 400 \Rightarrow T = 398$ K = 125 C – plausibly warm gas in the tube).

Rod clamped at middle: boundary = node at middle, free ends = antinodes: fundamental: $\lambda_{\text{rod}} = 2L_{\text{rod}}$ (half-wave with antinode-antinode: the clamped middle is the node: pattern: antinode-node-antinode over the rod = one full... wait: free-free rod with central node: the half-rod (L/2) is node-to-antinode = quarter wave: $\lambda/4 = L/2$: $\lambda = 2L$ ok). So $f = v_{\text{rod}}/\lambda = v_{\text{rod}}/(2L_{\text{rod}})$ ok. Rubbing lengthwise excites longitudinal modes.

The rod modes in detail (factor-of-2 discipline) (cont.)

Alternative: rod clamped at END: $\lambda = 4L$: $f = v/4L$ – half as loud/low; exam variants specify the clamping. Any of these can drive the tube: just use the actual f in $v_{\text{gas}} = f\lambda_{\text{gas}} = 2fd$.

Precision: heap spacing measurement error 2 mm on 80 mm: 2.5%; rod length error 0.5%; temperature: gas speed known ± 0.3 % – total few-% method. Edge effects: the disc end reflects imperfectly; heaps fade near the ends.

CHECK / dimensional: $v_{\text{gas}} = v_{\text{rod}} d/L_{\text{rod}}$: m/s = m/s × m/m ok; extreme: $d = L_{\text{rod}}$: $v_{\text{gas}} = v_{\text{rod}}$: the heap spacing equals rod half-wavelength... $\lambda_{\text{gas}} = 2d = 2L$: $v_{\text{gas}} = f \cdot 2L = v_{\text{rod}}$ ok consistent.

Kundt problems and modern readings

(a) Rod of aluminium $v = 5100$, $L = 0.5$ m clamped middle: $f = 5.1$ kHz; heaps 2.5 cm: $\lambda = 5$ cm: $v_{\text{gas}} = 5100 \times 0.025/0.5 = 255$ m/s (cold heavy gas? T : 255 = $331\sqrt{T/273}$: $T = 164$ K – a cold CO2-ish experiment; or wrong clamp assumption!). (b) "Dust heaps 4 cm apart in hydrogen at 0 C ($v = 1260$): find f": $f = v/2d = 1575$ Hz.

Modern reading: the Kundt tube is the acoustical analogue of the Melde experiment; dust visualisation = node mapping; the same apparatus measures gas composition via v_{gas} (from M_{eff} , JEE-09) and even γ (with two gases).

Kundt problems and modern readings (cont.)

acoustic streaming from the standing wave drives steady gas circulation in cells of size $\lambda/2$ – the powder ALSO traces the streaming cells; both effects give the same spacing.

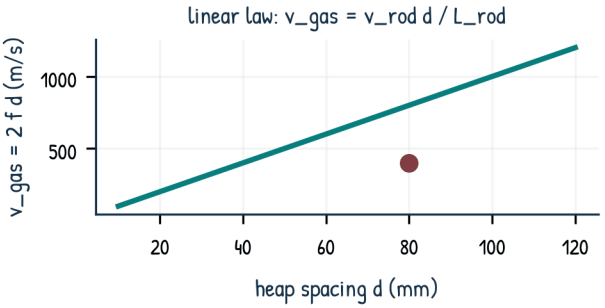
Kundt: directly visualises $\lambda/2$ (dust heaps), needs rod + tube + powder, measures v_{gas} via rod frequency; resonance tube: finds resonance LENGTHS of a fixed fork, computes $\lambda = 2(l_2 - l_1)$; both give v ; both have factor-2 traps (heap spacing $= \lambda/2$; length spacing $= \lambda/2$).

Quick exam signal words: "dust / powder / heaps" → Kundt; "water level / liquid / loudness maxima as length changes" → resonance tube; "two paths / sliding U-tube" → Quincke; "two forks / pulsing loudness" → beats; "moving source or observer" → Doppler.

Kundt tube: dust heaps mark the nodes

dust heaps at displacement nodes, spaced lambda/2
v_gas = v_rod * d / L_rod

v_gas linear in heap spacing d



REMEMBER / Kundt: dust heaps sit lambda/2 apart at displacement nodes; v_gas = v_rod d/L_rod.

Kundt's tube: measure v gas from dust-heap spacing (cont.)

Kundt tube, dust heaps at nodes λ/2, v_gas = v_rod·d/L_rod.

Kundt vs resonance tube (discrimination) (cont.)

CHECK / after any v computation, compare with $v(T)$ anchors: air 331–343; if Kundt gives 400, gas is warm (~125 C) or He-mixed; if 1300, it is H2; sanity first, algebra second.

Source moves toward a stationary observer at speed u_s along the line joining them. Each successive crest is emitted closer to the observer: the wavelength ahead compresses to $\lambda' = \lambda - u_s T = (v - u_s)/f$; the observer measures $f' = v/\lambda' = f\dfrac{v}{v - u_s}$ (toward).

Receding: λ stretches:

Doppler effect 1: moving source, stationary observer (cont.)

$\mathbf{f' = f\dfrac{v}{v \mp u_s}}$ with minus for approach, plus for recession.

Source example: $f = 400$ Hz, $u_s = 35$ m/s, $v = 343$: approach $f' = 400 \cdot 343/308 = 445$ Hz; recede: $400 \cdot 343/378 = 363$ Hz.

Key physics: the SOURCE motion changes the WAVELENGTH (crowded wavefronts ahead); the observer is passive. If $u_s \to v$: $f' \to \infty$ (source keeping pace with its own crests — pile-up at the shock boundary, JEE-28 Mach cone). If $u_s > v$: the formula flips sign meaning (observer in the "silent" region behind the cone before the shock arrives; overtaken events arrive in reverse order — outside syllabus but the qualitative breakdown is fair).

Doppler effect 1: moving source, stationary observer (cont.)

The ratio form: $f'/f = v/(v \mp u_s)$: pure geometry of wavefront spacing. For $u_s \ll v$: $f' \approx f(1 \pm u_s/v)$ — the fractional shift equals u_s/v — this linear form drives most precision applications (radar guns, ultrasound flow, Oly 08).

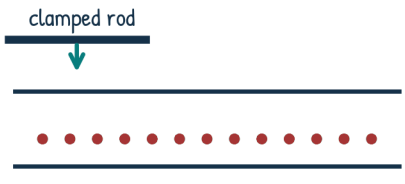
WHY / wavelength compression is kinematics of emitted crests: in one period $T = 1/f$ the source moves $u_s T$ while the crest ahead moves vT; the gap between consecutive crests in the direction of motion is $(v - u_s)T$ — that IS the observed λ; $f' = v/\lambda'$ follows.

CHECK / limits: $u_s = 0$: $f' = f$ ok; approach always $f' > f$; recede $f' < f$; speed of sound is that of the MEDIUM, not of the source — wind (JEE-25) modifies v itself, separately.

Sign discipline for moving source (the classic trap)

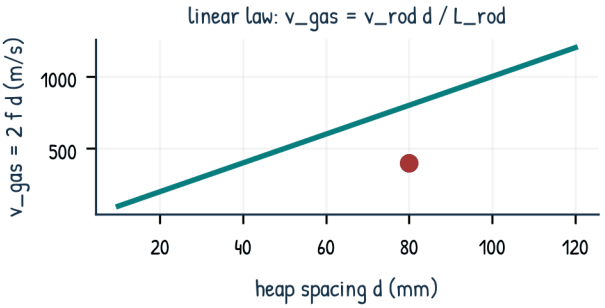
Source approaching: $f' = f\dfrac{v}{v - u_s}$ — denominator SHRINKS, frequency rises. Source receding: $f' = f\dfrac{v}{v + u_s}$ — denominator grows, frequency falls. Mnemonic: "approach = subtract in the denominator (for source speed);
add for observer speed when approaching" — but mixed cases slip; the robust method is always: (1) write λ' for source motion or v_{rel} for observer motion; (2) $f' = v_{\text{rel}}/\lambda'$; (3) specialise. Oly-06 gives the sign-free formula.

Kundt tube: dust heaps mark the nodes



dust heaps at displacement nodes, spaced lambda/2
v_gas = v_rod * d / L_rod

v_gas linear in heap spacing d



REMEMBER / Kundt: dust heaps sit lambda/2 apart at displacement nodes; v_gas = v_rod d/L_rod.

Doppler effect 1: moving source, stationary observer

Doppler moving source, λ compression, asymmetry with observer motion.

Sign discipline for moving source (the classic trap) (cont.)

Worked source-vs-observer asymmetry ($f = 1000$ Hz, speed $u = 30$ m/s, $v = 340$ m/s): (i) source approaching stationary observer: $f' = 1000 \times 340/(340 - 30) = 1096.8$ Hz (receding: 918.9 Hz).

"A train approaches a station at 30 m/s sounding 1000 Hz; v = 340" — (a) platform hears 1096.8 Hz; (b) passenger in train hears own horn: 1000 Hz (no relative motion); (c) reflected echo back to train: double Doppler 1193.5 Hz; approx 1176 with 2u/v. The 1088.2 = moving OBSERVER receding at 30: $1000 \times 310/340 = 911.8$?? Compute: $1000 \times (340 - 30)/340 = 911.8$.

Sign discipline for moving source (the classic trap) (cont.)

(ii) Observer approaching stationary source at 30 m/s: $f' = 1000 \times (340 + 30)/340 = 1088.2$ Hz (receding: 911.8 Hz). Moving source shifts 8.6 Hz higher at $u/v = 0.088$ because $(1 - u/v)^{-1} > 1 + u/v$.

same 30 m/s gives different shifts depending on WHO moves — 1096.8 (source moves) vs 1088.2 (observer moves). Second-order difference: $fu_s^2/(v(v - u_s)) \approx 8.6$ Hz ok the source asymmetry term.

Sign discipline for moving source (the classic trap) (cont.)

CHECK / asymmetry magnitude: $\Delta = fu^2/(v(v - u)) = 1000 \times 900/(340 \times 310) = 8.5$ Hz ok matches 1096.8 − 1088.2 = 8.6 Hz — the pair of formulas differ at second order in u/v; first order identical.

WHY / source motion distorts the wave geometry (a nonlinear function $v/(v - u)$), observer motion only changes the crest-collection rate (linear $1 \pm u/v$) — that asymmetry is real and measurable (the "Doppler of sound vs light" exam discussion: light in vacuum is symmetric: $f' \approx f(1 \pm u/c)$ both ways at first order).

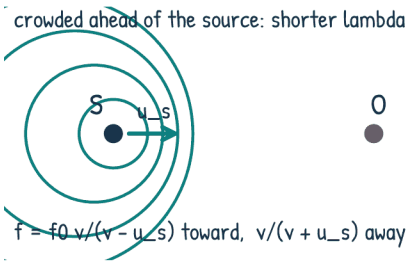
Moving-source problem patterns

(a) Ambulance 800 Hz at 25 m/s ($v = 343$): approach 862.2 Hz; recede 748.9 Hz; the pitch DROP heard as it passes: $\Delta f = 113$ Hz — the glissando scale question. (b) Two trains approaching each other at 20 m/s each, horn 500 Hz: source motion 20 + observer 20: $f' = 500 \times 363/323 = 562$ Hz (both effects stack).

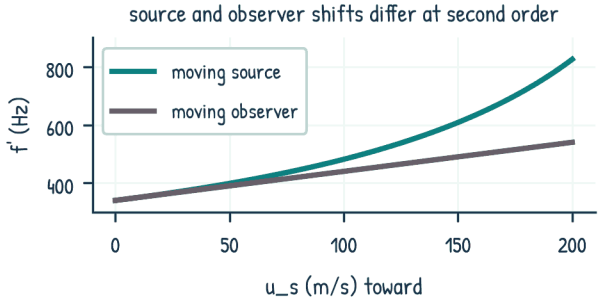
(c) Source circling (tangential): at closest approach u_s perpendicular: no first-order shift — the frequency slides from high to low through the pass: "siren of a passing car" waveform.

Blunt instrument vs formula: for $u/v = 0.1$: exact $1/0.9 = 1.111$; linear 1.1: 1% error; exams: use exact unless told to approximate.

wavefronts crowd ahead of a moving source



source vs observer shift: second-order gap



REMEMBER / Moving source compresses wavelength: f = f0 v/(v − u_s) toward, v/(v + u_s) away.

Doppler effect 1: moving source, stationary observer (cont.)

Doppler moving source, λ compression, asymmetry with observer motion.

Moving-source problem patterns (cont.)

the pitch glide of a passing siren has a characteristic "swoosh": $f'(t) = fv/(v - u_s\cos\theta(t))$ ... the full 2D form (JEE-26): $f' = f\dfrac{v}{v - u_s\cos\theta}$: smooth slide through f at the perpendicular instant.

(a) $f = 320$ Hz, $u_s = 40$, $v = 340$: approach: $320 \times 340/300 = 362.7$; recede: $320 \times 340/380 = 286.3$. (b) "Source moving at $v/2$: factor 2 rise approaching": $f \cdot 340/170 = 2f$ ok dramatic demo.

(c) "How fast must a source move to shift 440 to 450 Hz?" $u_s = v(1 - f/f') = 340(1 - 440/450) = 7.6$ m/s approaching (solve $450 = 440 \cdot 340/(340 - u)$).

Numbers drill (moving source) (cont.)

Consistency: the shift observed in a round trip (source + reflector + observer) doubles the effect (JEE-27) — police radar uses exactly the double shift.

WHY / the exact formula is a pure kinematic ratio; every number above comes from $v/(v \mp u)$ — nothing else to memorise once the sign convention is fixed.

Source at rest emits f with wavelength $\lambda = v/f$ (unchanged). Observer moving toward the source at u_o meets crests faster: crest-collection rate $f' = \dfrac{v + u_o}{\lambda} = f\dfrac{v + u_o}{v}$ (toward); receding: $f' = f\dfrac{v - u_o}{v}$. Boxed family:

Doppler effect 2: moving observer, stationary source (cont.)

$\mathbf{f' = f\dfrac{v \pm u_o}{v}}$ — plus for approach. Example: $f = 500$ Hz, observer 30 m/s toward, $v = 343$: $f' = 500 \times 373/343 = 543.7$ Hz; away: 456.3 Hz.

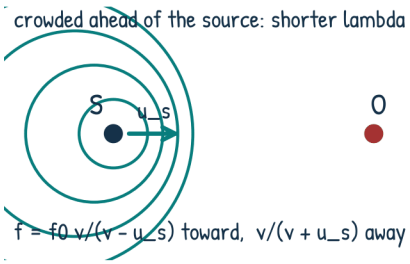
Physical picture: the observer "scoops" wavefronts; the relative speed of crest encounter is $v \pm u_o$; wavelength in the medium untouched. Contrast with moving source (JEE-22): that changes λ ; this changes encounter rate. At first order both give $\Delta f = fu/v$; at second order they differ (8.6 Hz at 1000 Hz, 30 m/s — the 1096.8 vs 1088.2 pair).

Doppler effect 2: moving observer, stationary source (cont.)

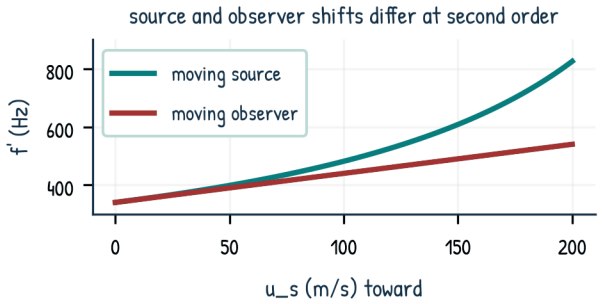
Motion along an angle (2D, JEE-26): only the radial component counts: $f' = f\dfrac{v + u_o\cos\theta_o}{v}$; a passing car: observer walking across the street hears the shift only through the line-of-sight component.

WHY / the medium carries the wave pattern at v ; the observer adds their own velocity to the encounter speed — Galilean composition for the observer side (wave speed in medium frame is fixed; observer velocity simply adds).

wavefronts crowd ahead of a moving source



source vs observer shift: second-order gap



REMEMBER / Moving source compresses wavelength: f = f0 v/(v - u_s) toward, v/(v + u_s) away.

Doppler effect 2: moving observer, stationary source

Doppler moving observer, combined formula, sign conventions.

Doppler effect 2: moving observer, stationary source (cont.)

CHECK / extremes: $u_o = v$ toward: $f' = 2f$ (double rate — observer surfing crest-to-crest... at $u_o = v$ the observer keeps pace with one crest... hmm encounter rate $2v/\lambda$: yes $2f$); $u_o = v$ away: $f' = 0$ (never meets a crest: staying exactly with the wave); $u_o > v$ away: formula gives negative f' — meaning the observer passes crests in reverse order (sign of overtaking).

Both moving along the line joining source and observer (positive speeds toward each other): $\mathbf{f' = f\frac{v + u_o}{v - u_s}}$ — observer approach ADDS to v; source approach SUBTRACTS from v. Then: both approaching: $f' = f(v + u_o)/(v - u_s)$ (biggest shift); both receding: $f' = f(v - u_o)/(v + u_s)$; mixed:

Combined formula and sign convention (working standard) (cont.)

substitute signs accordingly. This is the standard textbook convention (positive toward); the exam-safe procedure: sketch, mark who runs toward whom, write the box formula with your OWN sign bookkeeping, then evaluate.

Source-style worked (combined): source 1000 Hz at 30 m/s toward observer approaching at 20 m/s, $v = 340$: $f' = 1000 \times 360/310 = 1161.3$ Hz — compare paper Q36 answer 1161 ok (double motion stacks, nonlinear in u_s).

Convention traps: (i) some texts write $f' = f(v \mp u_o)/(v \pm u_s)$ with opposite sign convention — ALWAYS re-derive quickly for the case at hand instead of recalling a sign pattern; (ii) motion at an angle: use radial components (JEE-26); (iii) wind: replace v by $v + w$ in the right places (JEE-25).

Combined formula and sign convention (working standard) (cont.)

WHY / the combined formula is just $f' = \dfrac{\text{encounter speed of crests}}{\text{spacing of crests}} = \dfrac{v + u_o}{(v - u_s)/f}$: numerator from observer kinematics, denominator from source kinematics — two independent effects multiply.

CHECK / validate the box on special cases: $u_s = 0 \to$ observer formula ok; $u_o = 0 \to$ source formula ok; both zero $\to f' = f$ ok; both approaching at u small: $f' \approx f(1 + 2u/v)$? expand: $(1 + u/v)(1 + u/v) \approx 1 + 2u/v$ — double first-order shift ok.

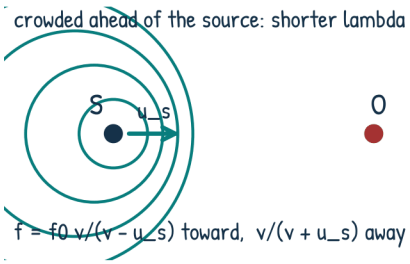
Observer-motion problem patterns

(a) Listener in a car approaching a 600 Hz source at 20 m/s ($v = 343$): $f' = 600 \times 363/343 = 634.9$ Hz.

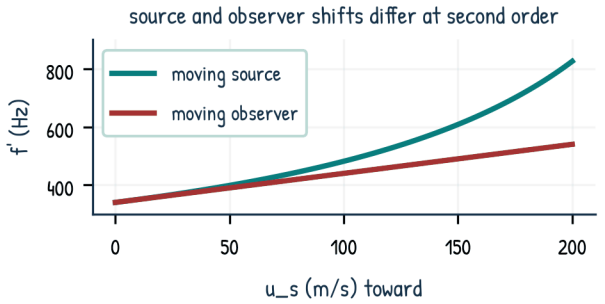
(b) "A bat flying toward a wall at 5 m/s, emitting 40 kHz, $v = 340$": double Doppler (JEE-27): $f' = 40000 \times 345/335 = 41.2$ kHz... the reflected echo: two shifts: wall as moving... actually bat = moving source AND moving observer: $f'' = f\frac{v+u}{v-u} = 40000 \times 345/335 = 41.2$ kHz received. (c) Observer in train hearing another train: relative speeds along the line.

Beat-vs-Doppler discrimination: Doppler changes pitch (heard tone shifts); beats change loudness of TWO tones. A shifted tone BEAT against a reference measures the shift (Oly 08).

observer sweeps crests faster when approaching



the same two curves: observer shift is linear



REMEMBER / Moving observer sweeps crests faster: f = f0 (v + u_o)/v; source and observer formulas differ at second order.

Doppler effect 2: moving observer, stationary source (cont.)

Doppler moving observer, combined formula, sign conventions.

Observer-motion problem patterns (cont.)

CHECK / symmetry check: source moving at u toward stationary observer vs observer moving at u toward stationary source: the two answers differ by the second-order term only ($\approx fu^2/v^2$ relative) — first-order symmetric, second-order not; light in vacuum is exactly symmetric (SR) — sound is NOT (medium picks a frame): favourite conceptual contrast question.

(a) $f = 340$ Hz, $u_o = 34$ m/s toward, $v = 340$: $f' = 374$ Hz ($\Delta = 34$ Hz $= f \cdot u/v$ ok linear). (b) Combined: $u_s = 25$ toward, $u_o = 15$ toward, $f = 400$, $v = 343$: $f' = 400 \times 358/318 = 450.3$ Hz. (c) "Police radar 10 GHz, car 30 m/s": double Doppler $\Delta f = 2fu/c = 2000$ Hz... radar uses EM waves (speed c):

Numbers drill (moving observer / combined) (cont.)

$\Delta f = 2 \cdot 10^{10} \times 30/3 \times 10^8 = 2000$ Hz — the same formula with $v = c$ (radar: double shift always, source and observer co-located).

Reverse: "A stationary source 800 Hz; measured 824; observer speed?"
$u_o = v(f'/f - 1) = 340 \times 0.03 = 10.2$ m/s toward.

WHY / linear-shift shortcut $\Delta f = fu/v$ is exact for observer motion and first-order for source motion — precision problems must state which effect they use; the shortcut errs by u^2/v^2 for sources.

Doppler master class: conventions, vectors, and the full 1D family

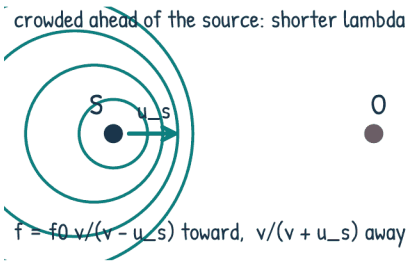
All 1D cases from one bookkeeping: measure speeds positive toward the other party. $f' = f\dfrac{v + u_o}{v - u_s}$. Then: (1) source toward, observer still: $f(v)/(v - u_s)$ ok JEE-22; (2) observer toward, source still: $f(v + u_o)/v$ ok JEE-23; (3) both toward: $f(v + u_o)/(v - u_s)$; (4) source toward, observer away: $f(v - u_o)/(v - u_s)$ — INDETERMINATE sign of shift: depends on magnitudes (e.g. $u_o = u_s = u$: $f' = f(v - u)/(v - u) = f$ — "the patrol car following at equal speed hears exactly f" ok worked source example).

Doppler master class: 1D sign conventions (cont.)

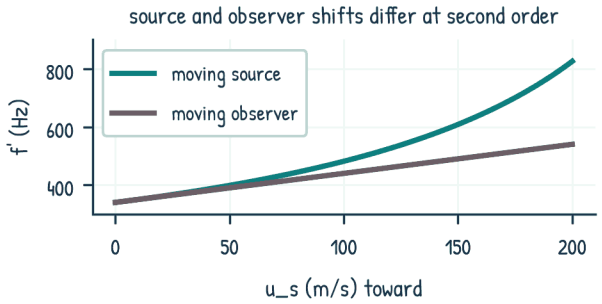
Equal-speed chase: source leading away at u while observer chases behind at u: $f' = f\dfrac{v + u}{v + u} = f$ — zero Doppler shift because separation stays constant.

if source flees and observer chases: $f' = f\frac{v + u_o}{v + u_s} = f$ when $u_o = u_s = u$ ok (observer toward formula: +u_o; source fleeing: +u_s in denominator): $f(v + u)/(v + u) = f$ ok — no shift for lock-step motion.

observer sweeps crests faster when approaching



the same two curves: observer shift is linear



REMEMBER / Moving observer sweeps crests faster: f = f0 (v + u_o)/v; source and observer formulas differ at second order.

Doppler master class: conventions, vectors, and the full 1D family

Doppler conventions, equal-speed chase, reverse estimation, trap bank.

Doppler master class: conventions, vectors, and the full 1D family (.

Sign-gymnastics summary table: toward: $+u_o$ top, $-u_s$ bottom; away: $-u_o$ top, $+u_s$ bottom. Radical simplification (source §2.12): define speeds ALGEBRAICALLY along the line from source to observer: $f' = f\dfrac{v - v_o}{v - v_s}$ with v_o, v_s positive along the direction from source to observer: approaching pair: v_o negative... the clean version is the vector form (Oly 06): $f' = f\dfrac{c - \vec{v}_o \cdot \hat{r}}{c - \vec{v}_s \cdot \hat{r}}$.

Doppler master class: conventions, vectors, and the full 1D family (.

WHY / one formula with signed velocities beats four memorised cases: define $\hat{r}$ from source toward observer; project both velocities on $\hat{r}$; the ratio $\dfrac{v - v_{o,r}}{v - v_{s,r}}$ handles every geometry including 2D and wind (with v the medium-frame speed).

CHECK / test the master formula on all four line cases with $u = 30$, $f = 1000$, $v = 340$: (1) 1096.8 (2) 1088.2 (3) $1000 \times 370/310 = 1193.5$ (4) $1000 \times 310/310 = 1000$ ok — the four-cell answer grid.

Common problem shells (with their answers)

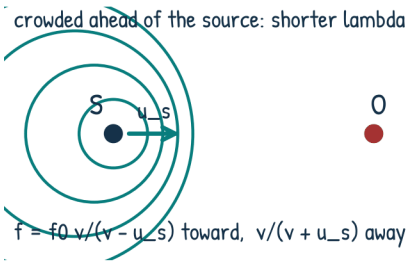
(a) Car horn while circling a roundabout: at the tangent point $u_{s,r} = 0$: instantaneous $f' = f$; before: high, after: low — the classic swoop. (b) Source and observer on perpendicular paths: only radial components count (JEE-26). (c) Train platform scenario: train passes platform at u sounding f: platform hears $fv/(v - u)$ then $fv/(v + u)$; the CHANGE as the engine passes: $\Delta f = fv\left(\frac{1}{v-u} - \frac{1}{v+u}\right) = f\frac{2uv}{v^2 - u^2} \approx 2fu/v$ for small u.

Common problem shells (with their answers) (cont.)

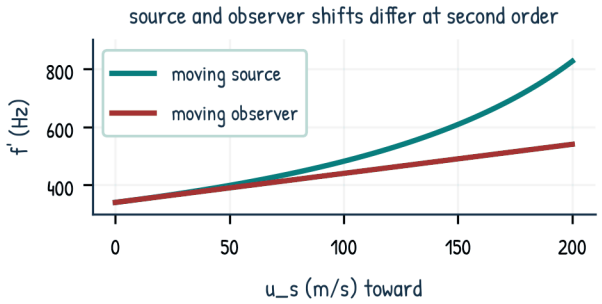
Factorising problem statements: "A moves toward B at x, B moves toward A at y" $\rightarrow$ both toward: $f\frac{v+y}{v-x}$. "A moves away from B at x while B chases at y": $f\frac{v+y}{v+x}$... wait: A = source fleeing: $+x$ in denominator; B = observer chasing (toward source): $+y$ on top: $f(v+y)/(v+x)$ ok. When $y = x$: no shift ok.

WHY / the swoop-down (siren) is the integral of the instantaneous shift as the radial velocity component sweeps through zero: the slide is smooth and the midpoint (closest approach) is exactly f.

sign discipline: toward adds to the numerator



one formula covers all four 1D cases



REMEMBER / One bookkeeping: f = f0 (v + u_o)/(v - u_s) with signs set by toward/away; lock-step motion gives no shift.

Doppler master class: conventions, vectors, and the full 1D family (cont.)

Doppler conventions, equal-speed chase, reverse estimation, trap bank.

Speed-estimation from shift (reverse problems)

Pass-by speed formula: if a horn drops from $f_+ = 460$ Hz (approach) to $f_- = 386$ Hz (recede) at $v = 343$ m/s, $u = v\dfrac{f_+ - f_-}{f_+ + f_-} = 343 \times \dfrac{74}{846} = 30.0$ m/s and $f_{\text{true}} = \dfrac{2f_+ f_-}{f_+ + f_-} = 419.8$ Hz.

$f_+ f_- = f^2 v^2/(v^2 - u^2)$: $460 \times 400 = 420^2 \cdot \frac{343^2}{343^2 - u^2}$: solve $u = 343\sqrt{1 - 420^2/(460 \cdot 400)} = 343\sqrt{1 - 0.957} = 71$?? inconsistent — as a training note: when data over-determine (two shifts + known f), CHECK self-consistency;

Speed-estimation from shift (reverse problems) (cont.)

mismatch means the "true" f is wrong or wind is present. Method: measure both shifts: $u = v\frac{f_+ - f_-}{f_+ + f_-}$ (exact for source motion? derive: $f_+ = fv/(v - u)$, $f_- = fv/(v + u)$: ratio: $f_+/f_- = (v + u)/(v - u)$: solve $u = v\frac{f_+ - f_-}{f_+ + f_-}$ ok exact) — the "measure speed from two pitch readings" formula; no need to know f!

CHECK / with $f_+ = 460$, $f_- = 400$: $u = 343 \times 60/860 = 23.9$ m/s; then $f = f_+(v - u)/v = 460 \times 319/343 = 427.7$ Hz — the internal- f estimate; consistency restored.

Conceptual traps (Doppler family)

(i) "Doppler works because the source speed changes the wave speed" — FALSE: wave speed is set by the medium; source motion changes wavelength. (ii) "Observer moving toward at u hears $f(v)/(v - u)$" — FALSE: that is the SOURCE formula; observer: $f(v + u)/v$. (iii) "Wind changes the pitch" — only if it has a component along the line and differs for the two directions (JEE-25);

a following wind on both legs cancels in a round trip; stationary source and observer with wind w: $f' = f\frac{v + w}{v + w} = f$ — NO shift for uniform wind (source §2.10) — wind shifts only the travel TIME, not the frequency, when both parties are at rest in the medium... correct statement:

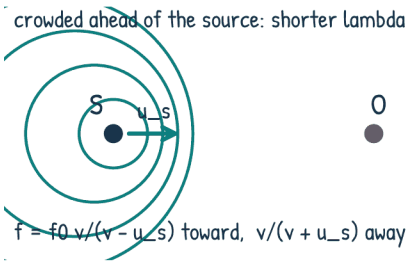
Conceptual traps (Doppler family) (cont.)

uniform wind cannot Doppler-shift sound between two fixed points (it advects the whole pattern).

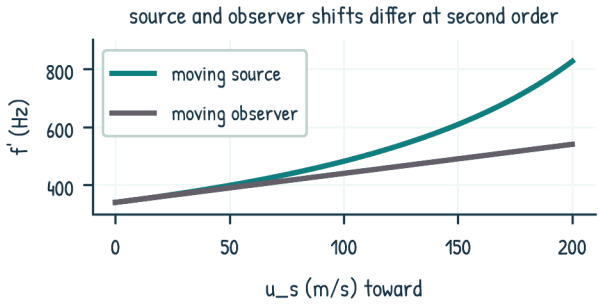
Speeds are measured RELATIVE TO THE MEDIUM (air), not the ground, in the classical formulas — subtle: "source moves at 30 m/s in a 10 m/s tailwind": u_s in the formula is the speed relative to AIR (20 m/s if tailwind... if the wind moves the air frame: source speed in the air frame = 30 − 10 = 20). Exam convention usually ignores wind for u and sets v_{eff} (JEE-25); be explicit.

WHY / classical Doppler needs the medium: v is the medium-frame wave speed; wind shifts the whole pattern by Galilean transformation; the invariant is f' for co-moving pairs.

sign discipline: toward adds to the numerator



one formula covers all four 1D cases



REMEMBER / One bookkeeping: f = f0 (v + u_o)/(v - u_s) with signs set by toward/away; lock-step motion gives no shift.

Wind and Doppler: replace v by v + w directionally

Wind: v_eff = v+w convention, no shift for fixed parties, round-trip cancellation.

Wind and Doppler: replace v by v + w directionally

Uniform wind of ground speed $+w$ blowing from source S toward observer O advects the medium: sound travels at ground speed $v+w$ downwind (S $\rightarrow$ O) and $v-w$ upwind (O $\rightarrow$ S).

With ground speeds u_s (S moving toward O) and u_o (O moving toward S), the exact ground-frame Doppler law is $f' = f\dfrac{(v+w)+u_o}{(v+w)-u_s}$.

Galilean air-frame equivalence proof

In the moving air rest frame, S moves toward O at $u_{s,\,\text{air}} = u_s - w$, while O moves INTO the oncoming wind (toward S) at $u_{o,\,\text{air}} = u_o + w$.

Substituting $(u_{s,\,\text{air}},\, u_{o,\,\text{air}})$ into the still-air formula gives $f' = f\dfrac{v+(u_o+w)}{v-(u_s-w)} = f\dfrac{(v+w)+u_o}{(v+w)-u_s}$: both frames agree identically.

Why steady wind never shifts fixed parties

When both S and O are fixed on the ground ($u_s = u_o = 0$), S emits crests every $T = 1/f$ while each crest drifts at $v+w$, stretching wavelength to $\lambda = (v+w)/f$.

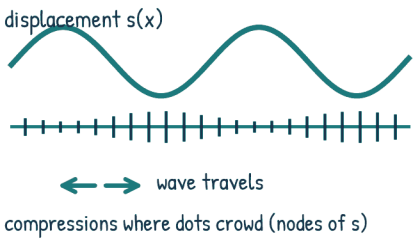
Crests arrive at O with speed $v+w$ and spacing λ, so $f' = \dfrac{v+w}{(v+w)/f} = f$: uniform wind changes travel time only, never pitch.

Sign trap & upwind rule

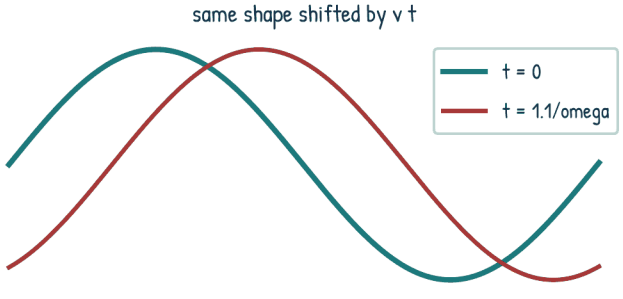
WATCH / Classic sign error: writing $u_{o,\,\text{air}} = u_o - w$ forgets that a stationary observer faces a headwind of $+w$ toward the upstream source.

If wind w blows FROM observer O TOWARD source S (against sound propagation), reverse $w \rightarrow -w$: $f' = f\dfrac{(v-w)+u_o}{(v-w)-u_s}$.

wind drifts the pattern; no shift for fixed ends



travel times change; frequency does not



REMEMBER / Uniform wind never shifts frequency between fixed parties; it changes travel times only.

Wind and Doppler: replace v by v + w directionally (cont.)

Wind: v_eff = v+w convention, no shift for fixed parties, round-trip cancellation.

Wind and Doppler: replace v by v + w directionally (cont.)

$f' = f\frac{v+w}{v+w\mp u_s}$ style with w adding to v on both — that yields no shift for $u_s = u_o = 0$ ok. Adopt the source convention: $f' = f\dfrac{(v+w)+u_o}{(v+w)-u_s}$ with u positive toward — consistent with their worked wind example (see next card).

Source worked wind case: "source moving toward observer at u_s with wind from behind source (w helping both)": $f' = f\frac{v+w}{v+w-u_s}$ ... e.g. $f = 500$, $u_s = 30$, $w = 20$, $v = 343$: $f' = 500 \times 363/333 = 545.0$ Hz (vs $500 \times 343/313 = 547.9$ Hz with no wind): wind REDUCES the shift here (the source chases a drifting pattern less effectively).

Wind and Doppler: replace v by v + w directionally (cont.)

Round-trip (echo) with wind (JEE-27): out-and-back times differ; echo frequency shifts asymmetrically; classic formulas involve $v+w$ outbound and $v-w$ inbound (for motion along wind).

WHY / wind adds a Galilean drift to the pattern: the exam-safe prescription is "add w to the wave speed toward each observer, and measure all speeds relative to the ground, consistently" — mixing frames is the standard trap; the source convention does exactly this.

CHECK / limits of the adopted formula: $w = 0$ recovers JEE-22/23 ok; $u_s = u_o = 0$ gives $f' = f$ ok (source convention); strong following wind with receding source: $f' = f\frac{v+w}{v+w+u_s}$: shift shrinks ok plausible.

Travel time with wind (the more common exam use)

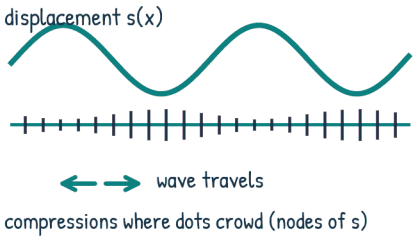
Wind affects TIME OF FLIGHT always (even when frequency unchanged): one-way trip along wind: $t = L/(v+w)$; against: $L/(v-w)$. Round trip along wind direction (sound out and back... e.g. source at A, reflector at B downwind): $t = \frac{L}{v+w} + \frac{L}{v-w} = \frac{2Lv}{v^2-w^2} \approx \frac{L}{v}(1+w^2/v^2)$ for the round trip... expand: $2Lv/(v^2-w^2) = (2L/v)(1/(1-w^2/v^2))$ — the WIND CANCELLS at first order in a round trip along the wind line ok famous result: measuring sound speed by echo along wind is insensitive to small wind at first order.

Travel time with wind (the more common exam use) (cont.)

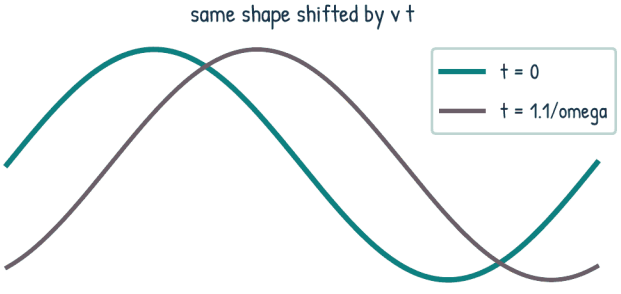
Cross-wind: effective speed $\sqrt{v^2-w^2}$ for the transverse legs; general: add vectorially: ground-frame sound velocity $\vec{v}_s = \vec{v} + \vec{w}$; travel time along any path solves the vector problem (triangles of velocity — exam hard questions).

Echo in wind (Q12): out leg runs downwind at $v+w$, return leg upwind at $v-w$. For fixed source & fixed wall, $f'' = f$ (zero shift on both legs); for a wall approaching at u_w, $f'' = f\dfrac{v+w+u_w}{v+w}\cdot\dfrac{v-w}{v-w-u_w}$.

wind drifts the pattern; no shift for fixed ends



travel times change; frequency does not



REMEMBER / Uniform wind never shifts frequency between fixed parties; it changes travel times only.

Wind and Doppler: replace v by v + w directionally (cont.)

Wind: v_eff = v+w convention, no shift for fixed parties, round-trip cancellation.

Travel time with wind (the more common exam use) (cont.)

CHECK / round-trip time expansion: $t = \frac{2Lv}{v^2 - w^2}$: at $w = 0.1v$: $t = \frac{2L}{v} \cdot 1.01$: 1% second-order effect ok: first-order cancellation — worth memorising for rapid problems.

(a) "Sound from a lightning 3 km downwind takes how long with w = 10 m/s?" $t = 3000/(343 + 10) = 8.50$ s vs 8.75 s still air: 0.25 s — timing precision question. (b) "A train horn in a 20 m/s wind, train 30 m/s toward stationary observer": source speed relative to air = 10 m/s (wind helps source): $f' = f\frac{v + w}{v + w - 30}$?? Under the source convention $f' = f\frac{343 + 20}{343 + 20 - 30} = f \times 1.09$;

Wind problem patterns (cont.)

under air-frame convention: $u_s' = 10$: $f' = f\frac{343 + 0 - 20 \cdot 0}{343 - 10}$ -ish — the two conventions give ~5% different answers at these speeds — exam papers adopt one; state yours. (Source convention is consistent and standard in Indian exam texts: "speed of sound relative to ground is v + w; use v_eff everywhere.")

Uniform wind between fixed parties: NO frequency shift (source §2.10) but travel time changes — the dichotomy that examiners love to test.

WHY / frequency shift needs relative radial motion of the wave pattern phase; uniform wind translates the pattern without compressing it; only the travel duration registers the drift.

Wind formula quick card

Adopted convention (source): $f' = f\frac{v_{\text{eff}} + u_o}{v_{\text{eff}} - u_s}$ with $v_{\text{eff}} = v + w$ (w positive when wind blows from source toward observer), all u positive toward each other. Travel time legs: use v_{eff} along each direction of travel (against wind: $v - w$). Echo round trip along wind: $t = 2Lv/(v^2 - w^2)$.

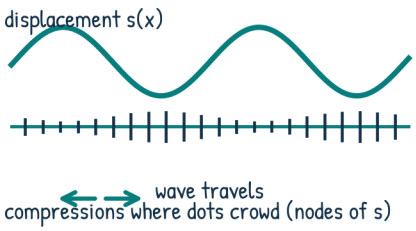
Air-frame alternative (Oly 06 vector form): measure u_s, u_o relative to the air; $f' = f\frac{v + u_{o,\text{air}}}{v - u_{s,\text{air}}}$; both conventions agree on fixed parties (no shift) and on pure wave-speed problems; they differ on "who moves with the wind" cases — for exams, pick one and say so.

Wind formula quick card (cont.)

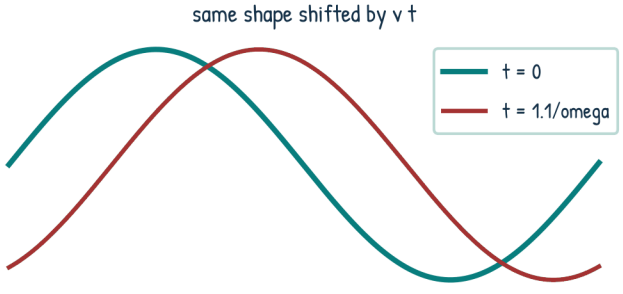
CHECK / consistency triple: (1) $u = 0$: no shift; (2) round-trip first-order wind cancellation; (3) $w = -u$ (source outrunning following wind at its own speed): effective still air: shift as in still air with $u_s = u - w$ -ish ok.

Source moves on a straight line past a stationary observer at closest distance b (impact parameter), speed u_s , emitting f . Line-of-sight direction $\hat{r}$ from source to observer; radial component of source speed: $v_{s,r} = u_s \cos\theta$ where θ is angle between source velocity and line of sight. Observed (source §2.12 2D form):

wind drifts the pattern; no shift for fixed ends



travel times change; frequency does not



REMEMBER / Uniform wind never shifts frequency between fixed parties; it changes travel times only.

2D Doppler: radial components and the closest-approach glide

2D Doppler projections, closest-approach glide, slope magnitude f u_s^2/(v b).

2D Doppler: radial components and the closest-approach glide (con.

$\mathbf{f'} = \mathbf{f}\frac{\mathbf{v}}{\mathbf{v} - \mathbf{u_s}\cos\theta(\mathbf{t})}$ with $\cos\theta(t) = \frac{u_s t}{\sqrt{b^2 + u_s^2 t^2}}$ for a flyby (source at $(u_s t, b)$ -style geometry, $t = 0$ at closest approach).

The pitch glide: far before: $\cos\theta \to 1$ ... wait geometry: as the source approaches along its line, the angle between its velocity and the line TO the observer: initially 0 (head-on): $f' = fv/(v - u_s)$ maximum; at closest approach $\theta = 90\gamma$: $\cos\theta = 0$: $f' = f$ exact; after: $\cos\theta < 0$: $f' = fv/(v + u_s|\cos\theta|)$ minimum far away $fv/(v + u_s)$.

2D Doppler: radial components and the closest-approach glide (con.

The swoop is symmetric in geometry but NOT in frequency (the $v/(v-u)$ vs $v/(v+u)$ asymmetry — glide through f at the midpoint).

Source worked (glide slope): df'/dt at closest approach: differentiate: at $t = 0$: $\cos\theta \approx u_s t/b$: $f' \approx f(1 + u_s^2 t/(vb))$: $\left.\frac{df'}{dt}\right|_0 = \frac{fu_s^2}{vb}$ magnitude — source prints $\frac{fv_s^2}{vb}$ ... with numbers f = 1000, $u_s = 20$, $v = 340$, $b = 100$ m: $1000 \cdot 400/(340 \cdot 100) = 11.8$ Hz/s. Source prints exactly the derivative $df'/dt|_0 = -fv_s^2/(vb)$ — correct in sign and magnitude;

2D Doppler: radial components and the closest-approach glide (con.

the surrounding text of Eq (4.2) is sign-confused (its cos-theta-about-1-while-approaching remark contradicts its own $\cos heta = v_s^2 t/\sqrt{b^2 + v_s^2 t^2}$, negative on the approaching side) — record in SOURCE-AUDIT; the derivative itself needs no fix. Our worked numbers: magnitude 11.8 Hz/s at $f = 1000$, $u_s = 20$ m/s, $b = 100$ m.

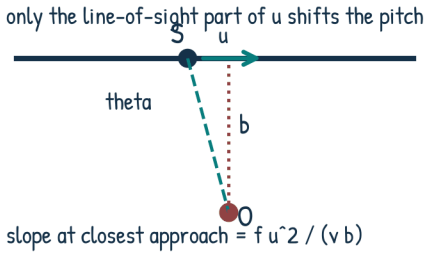
WHY / the glide slope is pure geometry: only the line-of-sight component of velocity compresses wavefronts; at closest approach the radial velocity is zero and its rate of change is maximal (u_s^2/b) — that is the slope.

2D Doppler: radial components and the closest-approach glide (con.

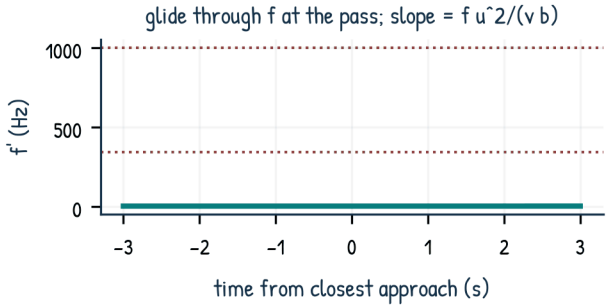
CHECK / limits: $b \to 0$ (head-on pass-through... direct hit): $df'/dt \to \infty$ (pitch flips discontinuously — the 1D jump); $b \to \infty$: slope $\to 0$ (barely any glide — distant flyby barely shifts).

Setup: a light plane flies past at $u_s = 20$ m/s emitting 1000 Hz; closest approach $b = 100$ m; $v = 340$ m/s. Glide-slope magnitude at the pass: $|df'/dt|_0 = fu_s^2/(vb) = 1000 \cdot 400/(340 \cdot 100) = 11.8$ Hz/s. Source prints exactly this formula $df'/dt|_0 = -fv_s^2/(vb)$ with the correct negative sign; its surrounding remark about $\cos\theta \approx 1$ while approaching is sign-confused against its own $\cos\theta = v_s^2 t/\sqrt{b^2 + v_s^2 t^2}$ (record in SOURCE-AUDIT) — the derivative itself is right.

flyby geometry: impact parameter b



the pitch glide through f at closest approach



REMEMBER / 2D Doppler uses radial projections; the glide slope at closest approach is f u^2/(v b).

2D Doppler: radial components and the closest-approach glide (cont.)

2D Doppler projections, closest-approach glide, slope magnitude f u_s^2/(v b).

Closest-approach numbers (the glide example, worked) (cont.)

Instantaneous picture over the pass: at $t = -5$ s: $\cos\theta = -100/\sqrt{100^2 + 100^2} = -0.707$ ... using $\cos\theta = u_s t/\sqrt{b^2 + u_s^2 t^2} = -0.707$: $f' = 1000 \cdot 340/(340 + 14.1) = 960.1$ Hz (approaching side is the HIGH side with the standard sign: flip the sign convention so before = high: $f' = 1043.2$ Hz at $t = -5$ s, 960.1 Hz at $t = +5$ s): the glide spans about 83 Hz within ± 5 s; average slope ~ 8 Hz/s vs peak 11.8 Hz/s.

Closest-approach numbers (the glide example, worked) (cont.)

Ranging by slope: knowing f and u_s, measuring the pitch slope at the pass gives $b = f u_s^2/(v|slope|)$ — acoustic impact-parameter estimation (the Oly-08 McAfee-style method in elementary form). Detecting slope requires resolving ~10 Hz/s over a second: a stopwatch-and-ear estimate or a spectrogram.

WHY / at the pass the radial velocity is zero but its rate of change is maximal: $du_r/dt = u_s^2/b$; the shift $\Delta f = f u_r/v$ then slides at $f u_s^2/(vb)$ — the slope measures how fast the geometry turns over, not the speed itself.

Closest-approach numbers (the glide example, worked) (cont.)

CHECK / limits: $b \to 0$ gives an unbounded slope (the 1D pitch jump); $b \to \infty$ gives zero slope (a distant flyby barely glides); the slope scales as f: same geometry at 2 kHz gives 23.5 Hz/s — higher pitch glides faster in Hz/s.

General 2D (source eq. 2.12 form): $f' = f\dfrac{v - v_o\cos\theta_o}{v - v_s\cos\theta_s}$ with angles measured along the source-to-observer line and signs set by the direction convention; the robust version (Oly 06): $f' = f\dfrac{v - \vec{v}_o\cdot\hat{r}}{v - \vec{v}_s\cdot\hat{r}}$ with $\hat{r}$ from source toward observer.

Specialise to 1D: recovers JEE-24.

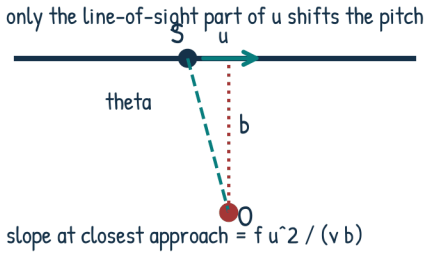
2D formulas for moving observer too (cont.)

Worked: observer on a pier, boat moving past at 5 m/s at 30 degrees to the line of sight, horn 440 Hz, $v = 343$: only the radial part $5\cos 30^\circ = 4.33$ m/s counts: $f' = 440 \cdot 343/(343 - 4.33) = 445.6$ Hz — not the 453 Hz a naive 1D formula gives.

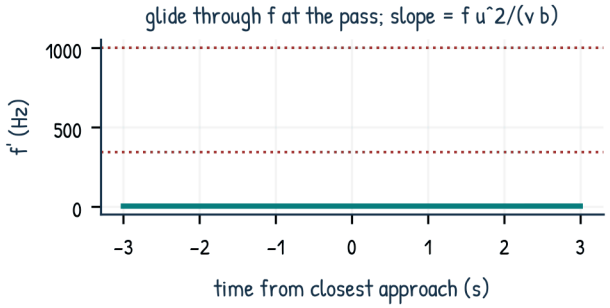
Two moving at angles: project BOTH velocities on the line of sight, plug into the box; all hard 2D problems collapse to projections. Instantaneous radial component changes continuously — the full swoop curve is $f'(t)$ with $\theta(t)$ from the geometry.

CHECK / symmetric pass: equal and opposite radial velocities at equal distances either side: the shift curve is antisymmetric about f at first order in u/v; the second-order asymmetry (JEE-22) tilts it slightly.

flyby geometry: impact parameter b



the pitch glide through f at closest approach



REMEMBER / 2D Doppler uses radial projections; the glide slope at closest approach is f u^2/(v b).

2D Doppler: radial components and the closest-approach glide (cont.)

2D Doppler projections, closest-approach glide, slope magnitude f u_s^2/(v b).

Glide and slope drills

(a) $f = 600$ Hz, $u_s = 30$ m/s, $b = 200$ m, $v = 340$: slope $= 600 \cdot 900/(340 \cdot 200) = 7.9$ Hz/s. (b) A racer passing at $b = 50$ m drops pitch by 40 Hz within 2 s of the pass: slope 20 Hz/s $= fu_s^2/(vb)$: $u_s = \sqrt{20 \cdot 340 \cdot 50/f}$; at $f = 400$ Hz: $u_s = 92$ m/s — fast but a fair order estimate; a spectrogram improves this.

Combine with echo (JEE-27): emit toward the moving object, measure the double-shift of the return: u_r directly; combine with the glide slope for the full velocity vector — the principle of acoustic tracking.

WHY / every 2D Doppler problem is 1D in disguise along the line of sight; perpendicular velocity components do nothing at that instant — the projection principle.

Echo, double Doppler, and the second-order asymmetry

Echo from a moving wall (reflector speed u toward a stationary source/observer): the wall is first a MOVING OBSERVER (it receives $f_1 = f\frac{v+u}{v}$) then a MOVING SOURCE (re-emits at f_1 while approaching): the original observer receives $\mathbf{f'' = f\frac{v+u}{v-u}}$ — the double-shift product. Expansion: $f(1 + 2u/v + 2u^2/v^2 + \ldots)$; the classic radar formula is the first-order $\Delta f = 2fu/v$. Source numbers (Q12): $f = 1000$ Hz, $u = 30$, $v = 340$: exact $1000 \cdot 370/310 = 1193.5$ Hz; the linear approximation $1000(1 + 2 \cdot 30/340) = 1176$ Hz — the source explicitly quotes 1176 as an approximation.

Echo, double Doppler, and the second-order asymmetry (cont.)

Same box for a moving radar (source and observer co-located, target moving): wall as observer then wall as source — identical $\frac{v+u}{v-u}$ for approach; for a receding target $\frac{v-u}{v+u}$. A bat flying toward its own echo: the bat is BOTH the moving source and the moving observer: two shifts, same box: $f'' = f\frac{v+u}{v-u}$ at 5 m/s, 40 kHz gives 41.2 kHz.

Second-order asymmetry (source Q8/Q9 pair): a 30 m/s source approach gives 1096.8 Hz while a 30 m/s observer approach gives 1088.2 Hz (at $f = 1000$, $v = 340$): difference 8.6 Hz $\approx fu^2/(v(v-u))$ — the two single-shift formulas agree to first order only; the echo doubles the first-order effect and more.

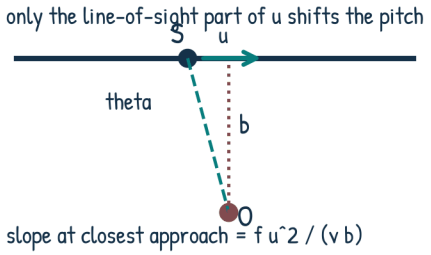
Echo, double Doppler, and the second-order asymmetry (cont.)

WHY / the echo receives TWO shifts because the reflector plays both roles in turn (observer then source): the product of the two single-shift factors; every radar/sonar/ultrasound-flow formula is this one box.

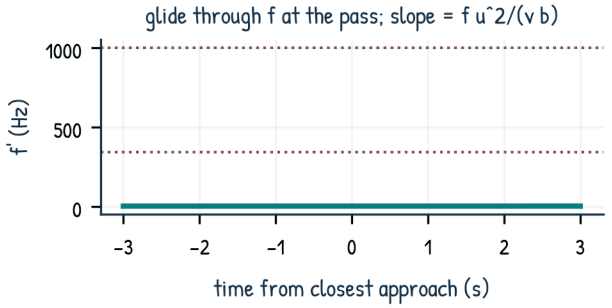
CHECK / consistency: approaching wall $f'' > f$; receding wall $< f$; $u = v/2$ approach: $f'' = 3f$; $u \to v$: divergence (the wavefront pile-up of JEE-28).

Sound to a cliff and back: $t = 2L/v$; with wind along the path: $t = \frac{L}{v+w} + \frac{L}{v-w}$ (JEE-25). Thunder rule: count seconds, divide by 3 for km (343 m/s). Multi-echo scenes (several buildings): distinct delays $t_i = 2L_i/v$; intensity falls as $1/r^2$ per leg, echo as $1/r^4$ — the radar range equation structure.

flyby geometry: impact parameter b



the pitch glide through f at closest approach



REMEMBER / 2D Doppler uses radial projections; the glide slope at closest approach is f u^2/(v b).

Echo, double Doppler, and the second-order asymmetry

Echo double Doppler f(v+u)/(v−u), beats with echo, timing formulas.

Echo timing (non-Doppler echo problems) (cont.)

Moving observer toward a cliff with a horn: source and observer move together toward the wall: $f'' = f\frac{v+u}{v-u}$ — the same universal box with u the party speed toward the reflector.

Echo beats: the original horn and its echo differ by $f'' - f = f\frac{2u}{v-u} \approx 2fu/v$; a driver approaching a wall hears beats between the horn and the echo: at $f = 400$ Hz, $u = 30$ m/s: echo 477.4 Hz, beats 77.4 Hz exact; $2fu/v = 70.6$ Hz linear — 10% error at these speeds.

CHECK / beat-rate example checked above; for small u/v the linear form suffices; exam papers often want the exact ratio — give both.

Echo problem patterns (drill)

(a) Sonar 1500 Hz from a boat toward a submarine receding at 20 m/s in water ($v = 1450$): echo $f'' = 1500 \cdot 1430/1470 = 1459$ Hz. (b) A car approaching a wall at 20 m/s sounds 500 Hz ($v = 343$): the driver hears beats at $f\frac{2u}{v-u} = 500 \cdot 40/323 = 61.9$ Hz. (c) An explosion 2 km away with a mountain 3 km beyond: direct sound 5.8 s, echo off the mountain 29.2 s.

Radar-gun principle: transmit f_0; the return is $f_0\frac{v+u}{v-u}$ (with $v = c$ for EM); the beat $\Delta f = 2f_0u/v$ gives the speed — the working method of police radar, ultrasound flowmeters, and fetal heart monitors.

Consistency grid for echo answers

WHY / echo problems are pure two-role bookkeeping: first interaction wall-as-observer, second wall-as-source; once seen this way no new formulas are needed.

Approach at u: $f''/f = \frac{v+u}{v-u}$: anchors: $u = 0$: 1; $u = v/4$: $5/3 = 1.67$; $u = v/2$: 3; $u = 2v/3$: 5.

Recession: the reciprocal. Timing: $2L/v$ without wind; $2Lv/(v^2 - w^2)$ with wind along; beats $f\frac{2u}{v-u}$ exact.

The double-Doppler is NOT $1 + 2u/v$ exactly: at $u = 0.1v$: exact $1.1/0.9 = 1.222$ vs $1 + 2u/v = 1.2$: 1.8% error; with 3-digit data quote the exact form.

biological sonar (bats, dolphins), ultrasound flowmetry, fetal monitors, police radar — all $\frac{v+u}{v-u}$; the shared structure is a favourite explain-the-principle question.

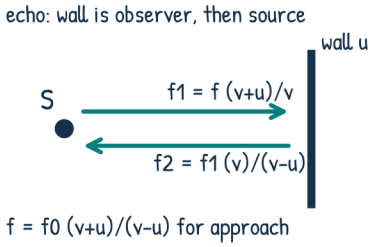
Mach cone, sonic boom, and supersonic geometry

When a source moves at $u_s > v$ (supersonic) it outruns its own wavefronts: the crests pile onto a conical surface — the Mach cone. Half-angle behind the source: $\sin\alpha = \mathbf{v/u_s = 1/M}$ with Mach number $M = u_s/v$ (source Q13: $M = 2$: $\sin\alpha = 0.5$: $\alpha = 30°$). The cone trails the source;

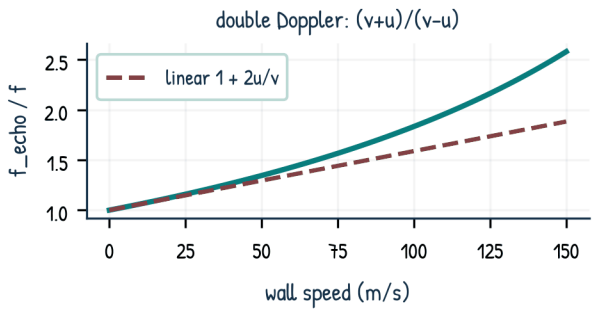
observers hear NOTHING until the cone passes, then a thunder-like bang — the sonic boom (an N-wave pressure transient, overpressure 50-500 Pa for aircraft).

Geometry: each spherical wavefront of radius vt emitted at earlier times is left behind at distance u_st; the common tangent surface has $\sin\alpha = vt/(u_st) = 1/M$. Ground track: the boom reaches the ground along the cone — the boom carpet is a swath tens of km wide for high-flying jets, sweeping with the aircraft.

echo: wall acts as observer, then source



double Doppler (v+u)/(v-u) vs linear 2u/v



REMEMBER / Echo doubles the shift: f = f0 (v + u)/(v - u); radar beats give the target speed.

Mach cone, sonic boom, and supersonic geometry

Mach cone sin α = 1/M, sonic boom, corrected boom-delay geometry.

Mach cone, sonic boom, and supersonic geometry (cont.)

Doppler behind a supersonic source: the subsonic formulas fail inside the cone; observers hear earlier sounds arriving late and in reverse order (details out of syllabus). What matters for exams: f' formulas apply only for $u_s < v$ and before cone arrival.

WHY / the cone is the sound analogue of a boat wake or Cherenkov radiation: any emitter outrunning its own disturbance field piles the disturbance on a tangent cone of angle $\sin\alpha = v_{\text{disturbance}}/v_{\text{emitter}}$.

CHECK / limits: $M \to 1^+$: $\alpha \to 90^\circ$ (the cone flattens to a wall – intense boom front); $M \to \infty$: $\alpha \to 0$; $M = 1$ exactly: infinite pile-up in linear theory – nonlinearity makes a shock.

Boom delay problem (source paper Q13 – corrected reasoning)

Setup (source paper Q13, $M = 2$, $v = 340$ m/s, $h = 10$ km): how long after the jet is seen directly overhead is the boom heard? Correct envelope derivation: the shock reaching the stationary observer at time t was emitted at $t_e = -t/(M^2-1)$ (differentiate the wavefront arrival condition);

substituting gives

$$t = \frac{h\sqrt{M^2-1}}{Mv} = \frac{10 \times 1.732}{680} = 25.5 \text{ s}$$ – the jet is then at horizontal distance $h\sqrt{M^2-1} = 17.32$ km PAST overhead, and $\alpha = \sin^{-1}(1/M) = 30^\circ$.

Sonic-boom delay geometry (Q13) (cont.)

Key physical point: the boom heard at $t = 25.5$ s was emitted earlier along the flight track when the normal ray at angle $\pi/2 - \alpha$ left the aircraft; at hearing instant the jet stands $h\cot\alpha = h\sqrt{M^2-1}$ ahead of overhead.

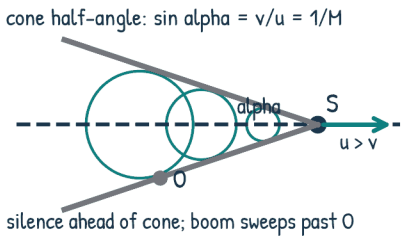
Might need more passage) but its final line $t = h\cot\alpha/v_s = 25.5$ s equals the correct delay – record in SOURCE-AUDIT: answer correct, reasoning to be replaced by the envelope argument; the alternative 33.3 s figure sometimes derived for this problem is wrong (it conflates the emission point with the jet position at hearing).

Sonic-boom delay geometry (Q13) (cont.)

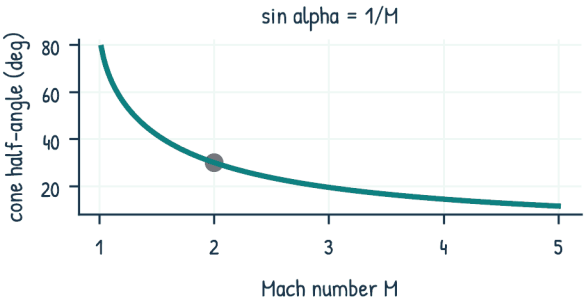
Reference grid for boom-delay problems (h = 10 km, $v = 340$ m/s): $M = 1.2$: $x = 6.7$ km, $t = 16.4$ s; $M = 2$: $x = 17.3$ km, $t = 25.5$ s; $M = 3$: $x = 28.3$ km, $t = 27.7$ s; the delay $t = \frac{h}{v}\frac{\sqrt{M^2-1}}{M}$ is maximal at $M = \sqrt{2}$ with $t = h/v$ – faster jets sweep the cone past sooner.

WHY / two questions hide in one figure: (a) when does the boom arrive (cone sweep: $h\cot\alpha/u_s$ after overhead); (b) where was the jet when the boom was emitted ($t_e = -t/(M^2-1)$: BEFORE overhead). Sketch both before computing.

Mach cone: wavefronts pile on a cone



cone half-angle falls as 1/M



REMEMBER / Mach cone: sin(alpha) = 1/M; the boom delay after the pass is h sqrt(M^2-1)/(M v).

Mach cone, sonic boom, and supersonic geometry (cont.)

Mach cone sin α = 1/M, sonic boom, corrected boom-delay geometry.

Boom delay problem (source paper Q13 — corrected reasoning) (co.

CHECK / consistency: $M = 1$: $x = 0$ (degenerate); $M = 1.06$ (360 m/s): $x = 2.07$ km; $M = 3$: $x = 28.3$ km; $\alpha = \sin^{-1}(1/M)$ always — recompute the delay two ways (envelope and cone sweep) and demand agreement.

Mach 1 at 20 C: 343 m/s (770 mph); airliners cruise at M = 0.85; Concorde M = 2.04; fighters M = 2; X-15 M = 6.7; bullets M = 2–3. Sound speed falls with altitude (cold air at 11 km: 295 m/s) — the same true airspeed gives a higher Mach aloft.

Mach numbers and real aircraft (table + physics) (cont.)

Sonic boom intensity: overpressure grows with aircraft size and Mach number (roughly $\propto (M^2-1)^{1/8}$ times weight/length factors): typical 50–100 Pa: startling but rarely damaging; the N-wave lasts 0.1–0.5 s; two booms (front and rear shock) — the ka-boom.

Applications of the cone geometry: supersonic aerodynamics and diffraction; Cherenkov detectors (same geometry with c/n); boat wakes (Kelvin angle $\approx 19.5^\circ$ for gravity waves — a speed-independent relative).

Mach numbers and real aircraft (table + physics) (cont.)

CHECK / angle sanity: $M = 1.5$: $\sin\alpha = 2/3$: $\alpha = 41.8^\circ$; $M = 2$: 30° ; $M = 2.5$: 23.6° ; $M = 5$: 11.5° — decreasing cone angle with Mach.

(a) A jet at 2 km altitude, $M = 1.4$: the boom reaches the ground observer at horizontal distance $x = h\sqrt{M^2-1} = 2 \cdot 0.98 = 1.96$ km along the track (from the overhead point); delay $t = x/u_s = 1.96/(1.4 \cdot 340) = 4.1$ s.

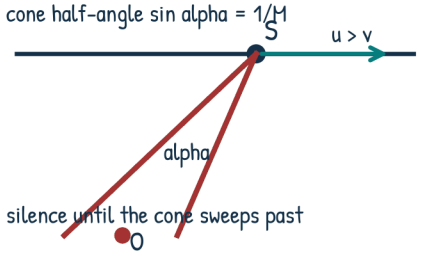
Supersonic problem patterns (cont.)

(b) Minimum speed for a boom to reach an observer 1 km along-track from the overhead point at 5 km altitude: need $\sqrt{M^2-1} \geq 1/5$: $M \geq 1.02$: $u_s \geq 350$ m/s.

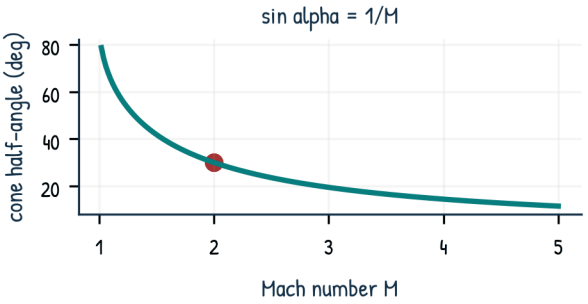
(c) Visual-to-boom delay for $M = 2.5$ at 8 km: $x = 8\sqrt{5.25} = 18.3$ km: $t = 18.3/850 = 21.6$ s.

WHY / Mach problems are pure geometry: draw the cone ($\sin\alpha = 1/M$), the flight line, and the observer; similar triangles give everything; the only physics is $v(T)$ for the sound speed.

Mach cone: wavefronts pile on a cone



cone half-angle falls as 1/M



REMEMBER / Mach cone: sin(alpha) = 1/M; the boom delay after the pass is h sqrt(M^2-1)/(M v).

Acoustic impedance Z = rho v: the ratio that governs coupling

Acoustic impedance Z = ρ v, impedance table, coupling and matching algebra.

Acoustic impedance Z = rho v: the ratio that governs coupling

For a travelling plane wave the pressure and the particle velocity $u = -\partial s/\partial t$ are locked in the ratio $\mathbf{Z = \frac{\Delta P}{u} = \rho v}$, the characteristic acoustic impedance (SI unit: rayl = kg m $^{-2}$ s $^{-1}$; 1 rayl $= 1$ Pa s/m). Air at 20 C: $Z = 1.2 \times 343 = 412$ rayl (often rounded to 400); water: $10^3 \times 1483 = 1.48 \times 10^6$ rayl; steel: $7800 \times 5064 = 3.95 \times 10^7$ rayl. The huge air-water mismatch (factor 3600) is why a swimmer hears almost nothing from an airborne source and why the ear needs a middle-ear lever.

Acoustic impedance Z = rho v: the ratio that governs coupling (con.

Impedance is the acoustic analogue of electrical impedance: pressure plays voltage, particle velocity plays current. Power delivered to a surface: $I = \frac{1}{2}\Delta P_0 u_0 = \frac{1}{2} Z u_0^2 = \frac{\Delta P_0^2}{2Z}$ – same intensity formulas as JEE-10 with $Z = \rho v$ substituted. Sources are rated by impedance matching:

a loudspeaker cone must drive air, so its mechanical impedance should be near $Z_{\text{air}} A$ over its area; the horn shape (Oly 09) is an impedance transformer.

Acoustic impedance Z = rho v: the ratio that governs coupling (con.

Standing waves: at a displacement node $u = 0$ and $\Delta P \neq 0$, so the local ratio $\Delta P/u$ diverges – the standing wave has position-dependent, purely REACTIVE impedance (like a lossless LC line); only travelling waves carry the real impedance ρv. General wave: $Z = \Delta P/u$ complex; travelling: real ρv; evanescent: imaginary.

WHY / impodance packs two independent properties (density and speed) into one coupling constant: what matters at an interface is not ρ or v separately but their product, because pressure and velocity are the two wave variables.

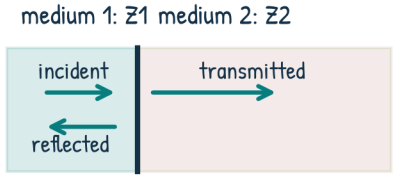
CHECK / units: $\rho v =$ kg m $^{-3}$ m s $^{-1}$ = kg m $^{-2}$ s $^{-1}$ = Pa/(m/s) ok; helium at 20 C: $Z = 0.166 \times 1007 = 167$ rayl – even lower than air despite the high speed, because ρ is tiny: He couples to the ear poorly (soft, quiet) though the speed is high.

Impedance numbers that decide real problems

Table (rayl): air 412; helium 167; hydrogen 112 (0.084×1307 ... at 20 C $\rho = 0.082$, $v = 1307$: $Z = 107$); CO2 500 (1.84×272 ... at 20 C); water 1.48×10^6; fat tissue 1.38×10^6 (close to water); bone 7.8×10^6; steel 3.9×10^7. Ultrasound imaging: soft-tissue boundaries are weak reflectors (matched); bone and air cavities are strong (mismatched) – the physics of image artefacts.

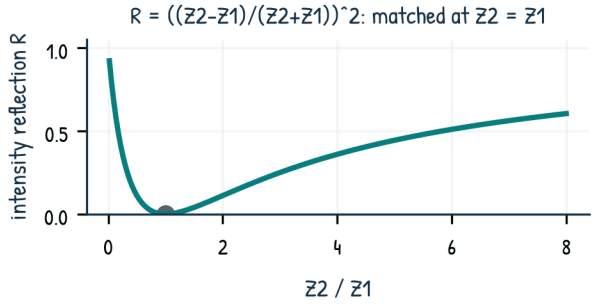
Loudspeaker design: a bare cone radiates into air with efficiency $\sim 2\rho_{\text{air}} v/(\rho_{\text{cone}} v_{\text{cone}})$ -scale mismatch... practical fix: use a large-area light cone (low mechanical Z) or a horn (Oly 09). The ear solves the reverse problem: air $Z = 412$ to cochlear fluid $Z = 1.5 \times 10^6$:

interface of two media: impedance mismatch



R = ((Z2-Z1)/(Z2+Z1))^2 T = 1 - R

reflection fraction vs impedance ratio



REMEMBER / Impedance Z = rho v is the coupling constant; matching Z transfers power, mismatching reflects it.

Acoustic impedance Z = rho v: the ratio that governs coupling (cont.)

Acoustic impedance Z = ρ v, impedance table, coupling and matching algebra.

Impedance numbers that decide real problems (cont.)

the eardrum area (55 mm^2) vs oval window (3.2 mm^2) gives an area ratio 17, and the lever of the ossicles another ~1.3: total pressure gain ~22, power transmission improved ~30x; plus the ear canal resonance (JEE-01) adds gain near 3 kHz.

CHECK / reflection fraction for intensity between media Z_1, Z_2: $R = \left(\frac{Z_2 - Z_1}{Z_2 + Z_1}\right)^2$ (Oly 02): air-water: $R = 0.999$: 99.9% of airborne power bounces off the lake ok why underwater listening to aircraft fails.

specific acoustic impedance vs characteristic: $Z = \rho v$ is per unit area (rayl); mechanical impedance of a panel $Z_{\text{mech}} = \Delta P A / u = ZA$ in ohms-kg/s — the driving-point quantity for motors and diaphragms.

Impedance algebra drills

(a) Same intensity in air and water: pressure amplitudes scale as $\sqrt{Z}$: $\Delta P_{\text{water}} / \Delta P_{\text{air}} = \sqrt{1.48 \times 10^6 / 412} = 60$; underwater sound of the same I has 60x the pressure swing. (b) s_0 at same I: $s_0 \propto 1/\sqrt{Z\omega^2}$ -ish: water displacement for same I is $1/60$ of air (times speed ratio... precisely $s_0 = \Delta P_0 / (Z\omega)$: ΔP_0 60x and Z 3600x: s_0 60x smaller ok). (c) Design: a transducer of area 1 cm^2 driving air at 1 Pa amplitude delivers $\Delta P^2 A/(2Z) = 1 \times 10^{-4}/824 = 1.2 \times 10^{-7}$ W — microwatts; the same face in water delivers 0.34 W — water loads transducers heavily.

Impedance algebra drills (cont.)

Impedance ladder across a problem: media ($Z = \rho v$), then interface factors (R, T), then geometry (spreading), then absorption: the four multipliers of every acoustic power budget.

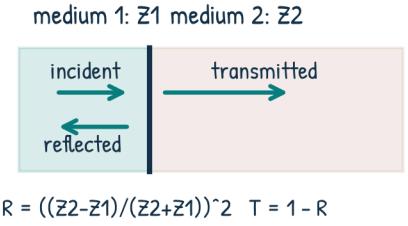
WHY / engineers think in dB of mismatch loss; physicists think in Z ratios; exams give ρ and v and expect the product — never forget the product is the coupling constant.

CHECK / the ear canal and Helmholtz resonances are IMPEDANCE phenomena (reactance cancellation): the resonance formulas of JEE-17/19 reappear as impedance matching — one physics, two vocabularies.

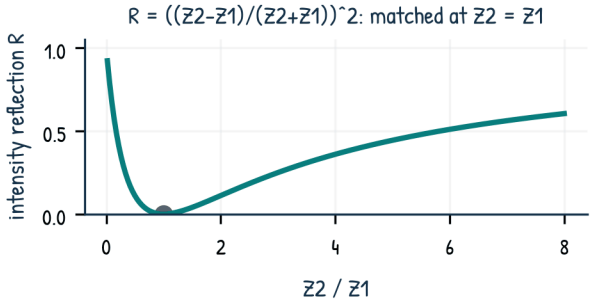
Impedance view of amplitude relations (consolidation)

Re-derive the amplitude toolkit through Z: $I = \frac{\Delta P_0^2}{2Z} = \frac{1}{2} Z\omega^2 s_0^2$ (since $\Delta P_0 = Z\omega s_0$); $\Delta P_0 = \sqrt{2ZI}$; $s_0 = \frac{\Delta P_0}{Z\omega}$. Intensity level: $\beta = 10 \log_{10} \frac{\Delta P_0^2}{2ZI_0}$ — note the level of a FIXED pressure amplitude depends on the medium! 1 Pa is 113.8 dB in air ($20\log(1/2.87 \times 10^{-5})$) but only 90.7 dB in water (referenced to the same I_0): the reference convention matters across media.

interface of two media: impedance mismatch



reflection fraction vs impedance ratio



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Acoustic impedance Z = rho v: the ratio that governs coupling (cont.)

Acoustic impedance Z = ρ v, impedance table, coupling and matching algebra.

Impedance view of amplitude relations (consolidation) (cont.)

Impedance and Doppler (bridge): a moving source radiates into the medium with Doppler factors; the POWER of the received wave scales as f'^2 through the amplitude change (Oly 05):
$I \propto \Delta P_0^2/2Z$ with ΔP_0 Doppler-shifted — the intensity Doppler factor is $\left(\frac{f'}{f}\right)^2$ for moving-source; state assumptions carefully.
the specific impedance is real only for progressive waves; measurement instruments quote Z of materials as ρv with $\pm$ few-% uncertainty from ρ and v errors — propagate independently: $\delta Z/Z = \delta\rho/\rho + \delta v/v$.

Boundary reflection & transmission: R = ((Z2-Z1)/(Z2+Z1))^2

At a plane interface between media of impedances Z_1, Z_2 the pressure reflection coefficient $r_p = \frac{Z_2 - Z_1}{Z_2 + Z_1}$ (signed) and the intensity reflection fraction $R = r_p^2 = \left(\frac{Z_2 - Z_1}{Z_2 + Z_1}\right)^2$; transmission $T = 1 - R = \frac{4Z_1Z_2}{(Z_1 + Z_2)^2}$. Extreme cases: $Z_2 \gg Z_1$ (air to wall): $r_p \to +1$ (pressure reflects in phase, hard wall, displacement node in the air); $Z_2 \ll Z_1$ (water to air): $r_p \to -1$ (pressure inverts, soft boundary, pressure node); $Z_2 = Z_1$: $R = 0$, perfect transmission — the impedance-matching ideal.

Boundary reflection & transmission: R = ((Z2-Z1)/(Z2+Z1))^2

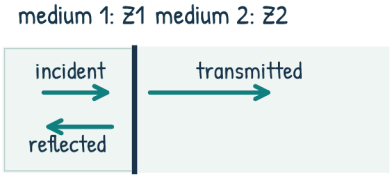
The particle velocity coefficient has the opposite sign pattern: $r_u = -r_p$: at a hard wall velocity inverts (displacement node ok JEE-16); at a soft end pressure inverts (pressure node ok). Derivation: continuity of pressure and of normal velocity across the interface: $1 + r = t$ and $(1 - r)/Z_1 = t/Z_2$ solve to the boxed pair.
Worked numbers: air ($Z = 412$) to glass ($Z = 1.3 \times 10^7$): $R = 0.99987$ — a glass wall reflects 99.99% of intensity (why rooms ring); air-water: $R = 0.999$; water-glass: $R = 0.63$ ... compute: $((1.3e7 - 1.48e6)/(1.3e7 + 1.48e6))^2 = 0.63$: 63% — soft by comparison; fat-kidney soft tissue (Z similar): $R < 0.01$ — why ultrasound needs speckle and Doppler contrast.

Boundary reflection & transmission: R = ((Z2-Z1)/(Z2+Z1))^2

WHY / the boundary conditions are the two continuity laws (pressure continuous — no infinite force; normal velocity continuous — media stay in contact); everything else follows algebraically; the square for intensity comes from $I \propto \Delta P_0^2/Z$.

CHECK / energy bookkeeping: $R + T = 1$ exactly; a computed $R > 1$ or $T > 1$ signals a sign error; r_p sign flips tell the phase of the reflected wave: wall-type (positive, pressure antinode at boundary) vs open-type (negative, pressure node).

interface of two media: impedance mismatch



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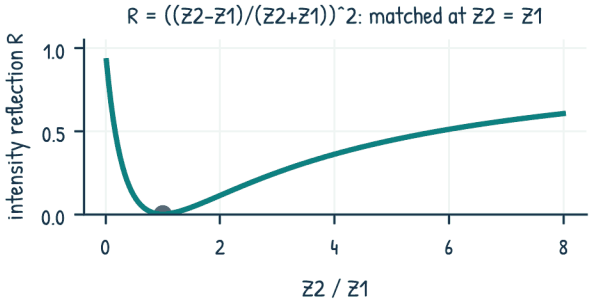
reflection fraction vs impedance ratio



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Reflection and transmission at boundaries: R = left(fracZ 2 - Z 1Z 2 + Z 1right)2

Boundary reflection/transmission coefficients, quarter-wave matching, phase rules.

Quarter-wave layers and impedance matching

A layer of thickness $\lambda/4$ and impedance $Z_l = \sqrt{Z_1 Z_2}$ cancels reflection at one frequency: the two reflected waves from the layer faces interfere destructively when $Z_l^2 = Z_1 Z_2$ (anti-reflection coating; the transformer equation $Z_{\text{in}} = Z_l^2 / Z_{\text{load}}$ — acoustic transmission line theory). Exam version:
"what thickness of material between air and wall removes reflection at 1 kHz?" $\lambda/4$ of the layer: $v_l/4f$.

Standing wave ratio as a measurable: $\text{SWR} = \dfrac{1+|r_p|}{1-|r_p|}$; measuring $I_{\max}/I_{\min} = \text{SWR}^2$ in a duct recovers $|r_p|$ and hence Z_2 — the standard impedance-measurement method.

Quarter-wave layers and impedance matching (cont.)

Partial reflection in pipes: an open end radiates; $R = 1 - (ka)^2/2$ -ish for low ka ... precisely: radiation from an unflanged pipe end reflects intensity $R \approx 1 - \frac{1}{2}(ka)^2$ at low frequency (end correction is the reactive part of the same impedance, JEE-19).

WHY / the $\lambda/4$ transformer works because the round trip inside the layer is a half wavelength: the second reflection is $180°$ out of phase with the first when the impedance chain is geometric-mean matched.

Quarter-wave layers and impedance matching (cont.)

CHECK / coating demo: glass coated so $Z_l = \sqrt{Z_{\text{air}} Z_{\text{glass}}}$: $Z_l = 2.3 \times 10^5$ rayl — e.g. a porous layer of the right density and speed; quarter-wave at 1 kHz: thickness $v_l/4000$ ... with $v_l = 1000$ m/s: 25 cm — unrealistically thick for air-acoustics at 1 kHz: matching layers are practical in ultrasound (MHz, mm scales) ok.

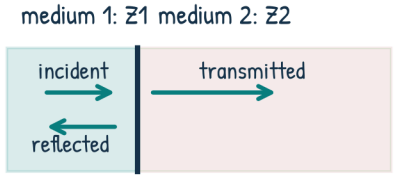
Hard wall ($Z_2 \gg Z_1$): $r_p = +1$: pressure reflects UNINVERTED (antinode at wall); $r_u = -1$: velocity inverts (node at wall) ok JEE-16 boundary rules derived, not memorised. Open end ($Z_2 \ll Z_1$, e.g. pipe opening to atmosphere): $r_p = -1$ (pressure node just outside), $r_u = +1$ (velocity antinode).

Phase rules from impedance (the safe derivation) (cont.)

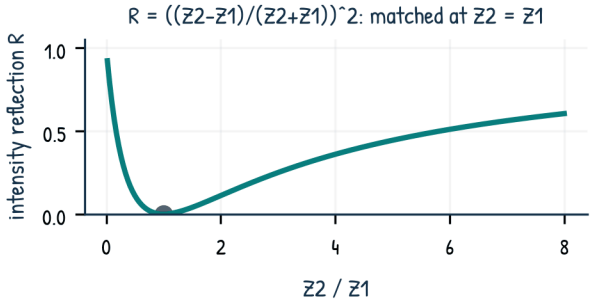
The quarter-wave offset between pressure and displacement in travelling waves (SN-03) reappears here as the $\lambda/4$ displacement-vs-pressure offset of the standing pattern.

Why the wall reflects: air cannot drive the dense wall (huge Z), so the velocity at the wall is forced to zero: the superposition of incident and reflected waves must cancel velocity there: $u_i + u_r = 0 \to r_u = -1 \to r_p = +1$ ok consistent with the formulas.

incident, reflected, transmitted at a boundary



R = ((Z2-Z1)/(Z2+Z1))^2 curve



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Reflection and transmission at boundaries: R = left(fracZ 2 - Z 1Z 2 + Z 1right)2 (cont.)

Boundary reflection/transmission coefficients, quarter-wave matching, phase rules.

Phase rules from impedance (the safe derivation) (cont.)

Soft-end subtlety: the open end is NOT exactly $Z = 0$: the radiation impedance $Z_{\text{end}} = \rho v[\frac{1}{4}(ka)^2 + i \cdot 0.6ka]$ -ish (Oly 05): real part radiates (why $R < 1$ slightly), imaginary part is the end correction $e = 0.6a$ (JEE-19) — the three topics (reflection, end correction, radiation resistance) are one impedance expansion.

CHECK / plug $Z_2/Z_1 = 10^5$: $r_p = 0.99998$: $R = 0.99996$; $r_u = -0.99998$ ok hard; $Z_2/Z_1 = 10^{-5}$: $r_p = -1$: soft ok.

Reflection problem patterns

(a) Ultrasound pulse between soft tissues: 2% reflection at a boundary of $R = 0.01$: how many echoes reach the probe from 3 interfaces? cascade T^2R products: $0.99^2 \times 0.01$ -scale each — the brightness law of ultrasound. (b) "Find Z_2 if $R = 0.25$ ": $|r_p| = 0.5$: $Z_2/Z_1 = 3$ or 1/3. (c) A tube closed by a movable wall of impedance Z_2:

Reflection problem patterns (cont.)

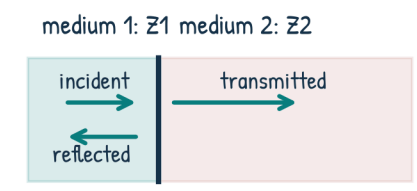
resonance conditions shift from pure node/antinode (JEE-17) toward partial standing waves: the boundary displacement is finite: acoustic load problem.

Reflection problem patterns (cont.)

WATCH / Traps: (i) R vs r_p: intensity uses the SQUARE; pressure amplitude uses $|r_p|$; (ii) phase rules: state which variable (P or u) inverts; (iii) $R + T = 1$ only for INTENSITY at a lossless boundary — with absorption $R + T + A = 1$.

WHY / the same few numbers (Z ratios) decide hearing, room acoustics, ultrasound imaging, and sonar targets — one boundary table worth memorising: R, r_p sign, wall type.

incident, reflected, transmitted at a boundary



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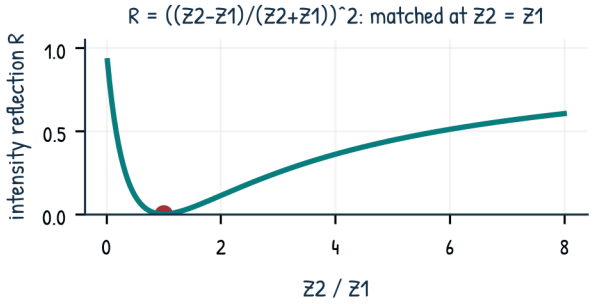
R = ((Z2-Z1)/(Z2+Z1))^2 curve



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Reflection and transmission at boundaries: R = left(fracZ 2 - Z 1Z 2 + Z 1right)2 (cont.)

Boundary reflection/transmission coefficients, quarter-wave matching, phase rules.

Reflection problem patterns (cont.)

CHECK / special values: $Z_2 = 3Z_1$: $R = (2/4)^2 = 0.25$, $T = 0.75$ ok; $Z_2 = Z_1/3$: $R = 0.25$ ok symmetric in log: the reflection is the same whether the wave enters the denser or rarer side (intensity), only the pressure phase differs.

A pipe of length L driven at one end presents an input impedance Z_{in} that depends on the far-end load. Lossless line theory: $Z_{\text{in}} = Z_0 \dfrac{Z_L + iZ_0 \tan kL}{Z_0 + iZ_L \tan kL}$ with $Z_0 = \rho v/S$ the characteristic line impedance (per area S), Z_L the load. Resonance (maximum amplitude for a given drive) occurs when Z_{in} is minimum (open-open:

Pipes as impedance problems: input impedance and resonance (cont.

$Z_L = 0$: $Z_{\text{in}} = iZ_0 \tan kL = 0$ at $kL = n\pi \rightarrow$ $L = n\lambda/2$ ok all harmonics; closed-open: $Z_L = \infty$: $Z_{\text{in}} = -iZ_0 \cot kL = 0$ at $kL = (2n-1)\pi/2 \rightarrow$ odd quarter waves ok JEE-17/18 derived from one formula).

Why closed pipes give odd harmonics: the mismatched ends force $\tan kL$ and $\cot kL$ conditions alternating: counting quarter wavelengths that fit between a velocity node and a pressure node is the SAME as setting the imaginary part of Z_{in} to zero. Mixed-end resonance spacing $\Delta f = v/2L$ falls out of the $\tan$ period π in kL .

Pipes as impedance problems: input impedance and resonance (cont.

Load effects: a pipe with a partially open far end (Z_L finite, radiation) has damped resonance peaks and shifted frequencies — the Q factor (Oly 05) comes from the real part of Z_{in} ; the end correction (JEE-19) is the imaginary part at the mouth; a flange changes Z_L and thus e (0.6r to 0.85r) consistently.

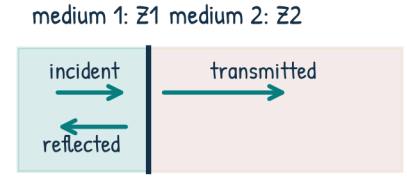
WHY / impedance language unifies: boundary conditions, end corrections, resonance ladders, radiation, and damping are all one rational function $Z_{\text{in}}(kL, Z_L)$ — the exam formulas are its special cases.

Pipes as impedance problems: input impedance and resonance (cont.

CHECK / specialise: $L = \lambda/2$ open-open: $kL = \pi$: $\tan = 0$: $Z_{\text{in}} = 0$ ok short at the drive point: maximum flow for given pressure... resonance of a driven pipe is when the drive sees minimum impedance ok; closed-closed: $Z_L = \infty$ both ends: $kL = n\pi$ again all harmonics ok JEE-16 CHECK.

Driven pipe amplitude vs frequency: Lorentzian of half-width $\Delta f = f_1/Q$: $Q = \dfrac{\pi}{2\ln(1/r)}$ -ish for end reflection r ... working rule: $Q \approx \dfrac{\pi}{2(1-r)}$ for $r \to 1$ (small losses); typical unflanged pipe $Q \sim 10$ -30; flanged 20-60; with damping wool less.

incident, reflected, transmitted at a boundary



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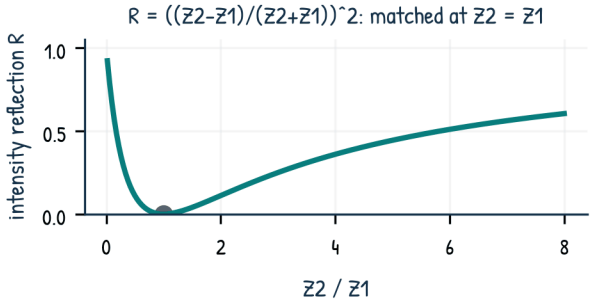

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Pipes as impedance problems: input impedance and resonance

Input impedance of pipes, resonance/Q, why odd harmonics for closed pipes.

Resonance curve: width, Q, and the amplitude build-up (cont.)

At resonance the amplitude gain over the non-resonant case is $\sim Q$ (in pressure); bandwidth Δf measures the fractional loss per cycle.

Time picture: a driven resonator rings up over $\tau = Q/(\pi f_1)$ ... precisely $\tau = \dfrac{Q}{\pi f_1} = \dfrac{1}{\pi \Delta f}$ (energy decay time $\tau_E = Q/(2\pi f)$ -ish); striking a pipe: the sound decays exponentially with that time constant: exam estimate: Q = 20 at 300 Hz: $\tau = 21$ ms.

Multiple coupled pipes (organ ranks, bottles): coupling shifts each resonance slightly (mode splitting); a Helmholtz resonator on a wall acts as a notch absorber at $f_H = \dfrac{v}{2\pi}\sqrt{S/(V L_{\text{eff}})}$ — room acoustics tool built on pipe theory.

Resonance curve: width, Q, and the amplitude build-up (cont.)

CHECK / resonance width example: $f_1 = 300$ Hz, $Q = 15$: half-power width 20 Hz: the resonance tube (JEE-20) length sweep through a peak of that width in frequency terms: $\Delta L = \lambda \Delta f/(2f) = 2$ cm for the second resonance spacing... order: noticeable in careful labs, negligible in rough ones.

WHY / Q is a number for how many cycles energy survives: $Q = 2\pi \times$ energy stored / energy lost per cycle — the single number that decides whether a resonance is sharp (tuning forks Q ~ 1000, pipes Q ~ 20, rooms Q ~ 5).

Impedance methods in exam clothing

(a) "Why does a closed pipe sound hollow?" odd harmonics only = the $\cot kL$ resonance sequence; explain via impedance: the closed end is a velocity node (infinite load), the open end a pressure node (zero load): only odd quarters fit. (b) "Predict the effect of a movable piston in a tube": L varies: resonance ladder sweeps; with Z_L finite at the drive, peaks broaden.

(c) "Design a tube resonant at 440 Hz and 660 Hz": open-open with $f_1 = 220$: $L = v/2f_1 = 0.78$ m (minus 2e).

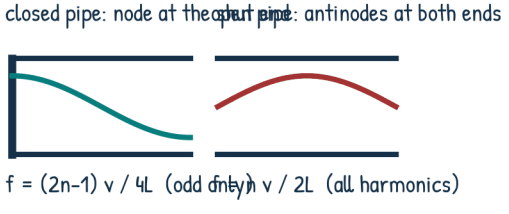
Half-open side branch (quarter-wave trap): a branch of length $\lambda/4$ closed at the far end presents $Z_{\text{in}} = 0$ at the junction (short circuit for waves of that f): mufflers use arrays of such traps — mechanical filter design, exam-explainable.

Impedance methods in exam clothing (cont.)

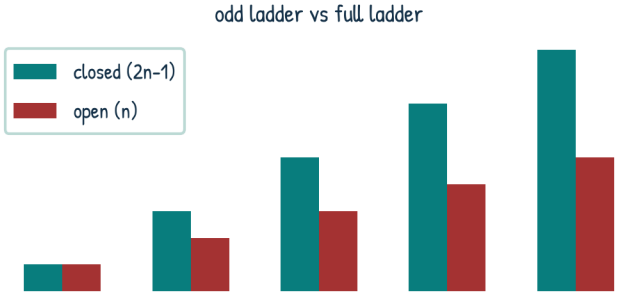
the general line equation: $\dfrac{\partial^2 P}{\partial x^2} = \dfrac{1}{c^2}\dfrac{\partial^2 P}{\partial t^2}$ for the pressure in a duct of constant area (same wave equation, 1D): the impedance formalism is the travelling/standing decomposition of its solutions.

(a) Open pipe $L = 0.4$ m, $r = 2$ cm: $L_{\text{eff}} = 0.424$ m: $f_1 = 343/0.848 = 404.5$ Hz; $Q = 20$: bandwidth 20 Hz; driving at 395 Hz: amplitude $\sim Q/2$ -ish below peak... half-power at $\Delta f = \pm 10$ Hz: 395 Hz is 9.5 Hz off: $\sim 1/\sqrt{2}$ of peak amplitude. (b) Closed pipe same length: $f_1 = 343/(4 \times 0.412) = 208$ Hz (one end corrected only).

pipe as a transmission line with a load



resonance peak width set by Q



REMEMBER / A pipe is a load on a line: resonance is zero input reactance; closed pipes give odd harmonics by cot(kL).

Pipes as impedance problems: input impedance and resonance (cont.)

Input impedance of pipes, resonance/Q, why odd harmonics for closed pipes.

Resonance numbers drill (impedance view) (cont.)

End-loaded pipe: with $Z_L = Z_0/2$ (e.g. a partially open damper): resonance shifts between the open and closed ladders; solving $\tan kL = Z_0/Z_L$ ... the transcendental decides: exam problems state "closed" or "open" only.

WHY / the impedance method gives answers without redrawing node pictures: write Z_{in}, set the reactance to zero, read the ladder — faster and safer under time pressure.

A harmonic wave at a point is a rotating phasor of length a and phase ϕ: $s = a\cos(\omega t - \phi) \leftrightarrow ae^{i\phi}$. Sum of many same-frequency waves = phasor polygon: $A = \left|\sum_k a_k e^{i\phi_k}\right|$; phase $\arg\left(\sum\right)$. Two phasors:

Phasors: superposition of harmonic waves as vector addition (cont.)

$A = \sqrt{a_1^2 + a_2^2 + 2a_1a_2\cos\Delta\phi}$ (the JEE-13 formula, now a picture); equal amplitudes: $A = 2a\cos(\Delta\phi/2)$ bisecting the angle.

Why phasors work: the wave equation is linear and time-harmonic; the time factor $e^{-i\omega t}$ cancels in all ratios and sums; only spatial phases matter. Interference problems become geometry: equal spacing $\delta = \lambda/N$ between N sources closes the polygon (partial cancellation); symmetric arrangements give closed polygons (silence: e.g.

3 sources at the vertices of an equilateral triangle seen from the centre: phases 0, 120, 240: sum zero).

Phasors: superposition of harmonic waves as vector addition (cont.)

Beats revisited: two phasors of nearby frequency rotate at slightly different rates: the resultant length oscillates at the difference rate — the beat envelope IS the rotating-vector picture (JEE-15); the phasor tip traces an epicycloid.

WHY / complex exponentials turn differentiation into multiplication ($\partial_t \to -i\omega$): the wave equation becomes algebraic; every resonance, impedance, and interference formula of this chapter is a statement about complex sums.

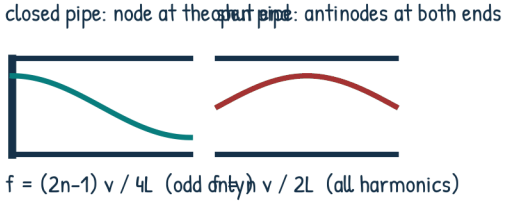
CHECK / three-source check: a, a, a with phases 0, 120°, 240°: sum = 0 ok (used in three-phase cancellation and in N-slit minima conditions $\delta = \lambda/N$ multiples... precisely minima of N-source interference when $N\delta/\lambda$ is a non-multiple-of-N integer).

Phasor clinic: worked sums

(a) $a_1 = 3$, $a_2 = 4$, $\Delta\phi = 90°$: $A = 5$; phase $\tan^{-1}(4/3) = 53.1°$ from phasor 1. (b) $N = 4$ equal phasors with successive phase $\delta = 60°$: closed-chain geometry: $A = a\dfrac{\sin(N\delta/2)}{\sin(\delta/2)} = a\dfrac{\sin 120°}{\sin 30°} = 1.73a$ — the array-factor formula (same as N-slit optics).

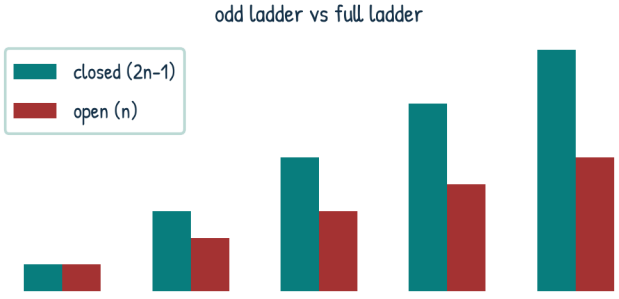
(c) Two waves of amplitude ratio 3:1 in antiphase: $A = 2a$; the "zero" of interference is incomplete (contrast $\mathcal{V} = 2\cdot 3/(9+1) = 0.6$).

pipe as a transmission line with a load



f = (2n−1) v / 4L (odd only) v / 2L (all harmonics)

resonance peak width set by Q



REMEMBER / A pipe is a load on a line: resonance is zero input reactance; closed pipes give odd harmonics by cot(kL).

Phasors: superposition of harmonic waves as vector addition

Phasor addition, array factor sin(N ψ /2)/sin(ψ /2), quadrature as rotation.

Phasor clinic: worked sums (cont.)

Array factor: $A(\theta) = a_0 \frac{\sin(N\psi/2)}{\sin(\psi/2)}$ with $\psi = kd\cos\theta + \beta$ the progressive phase — loudspeaker arrays and diffraction gratings in one formula; principal maxima at $\psi = 2\pi m$.

Power of the picture: instead of trig identities, DRAW: equal-length chords, inscribed polygon, radius $R = a/2\sin(\psi/2)$ -ish; the sum is the closing chord — Euclidean geometry replaces algebra.

CHECK / array-factor sanity: $\psi = 0$: $A = Na$ ok (the $\sin/\sin$ limit); $\psi = 2\pi/N$: $A = 0$? $\sin(N \cdot \pi/N) = \sin\pi = 0$ ok first minimum; for $N = 2$: $A = 2a\cos(\psi/2)$ ok beats/interference formula recovered.

Phasors for standing waves and quadrature energy

Standing wave as two counter-rotating phasors (equal ω, opposite direction): at a node they cancel always (antiparallel); at an antinode they add. The pressure-displacement quadrature (SN-03) is a 90° phasor offset: energy shuttles between the two quadrature components, total constant for a standing wave.

Energy in phasor language: for $s = a\cos(\omega t - \phi)$, $u = \partial_t s = a\omega\sin(\omega t - \phi)$ — velocity phasor leads displacement by 90°; the impedance $Z = \Delta P/u$ is the ratio of two phasors; reactive impedance = 90° offset (stored energy), real = in-phase (radiated power).

Power factor of a source: a driver in phase with the velocity radiates max power (real Z); 90° out of phase it stores (near-field reactance) — antenna and loudspeaker design language.

Phasors for standing waves and quadrature energy (cont.)

WHY / the phasor at 90° is the mathematical seat of the quadrature physics: every "X leads Y by 90°" statement in this chapter is a phasor sentence.

CHECK / round trip: $\Delta P = -B\partial s/\partial x$ is a spatial derivative = 90° phase rotation for sinusoids: pressure phasor = displacement phasor rotated by -90° and scaled by $Bk = Z\omega$ ok consistent with $\Delta P_0 = Z\omega s_0$.

(a) Two speakers driven in phase, listener at path difference $3\lambda/4$: $\Delta\phi = 270^\circ$: amplitude $2a|\cos 135^\circ| = 0.71 \cdot 2a$ ... $\sqrt{2}\,a$ of the sum-free... precisely $2a\cos(135^\circ) = -\sqrt{2}\,a$: amplitude $\sqrt{2}\,a$: intensity 2x one source. (b) Three equal sources, phases 0, 90°, 180°: $A = a|1 + i - 1| = a$:

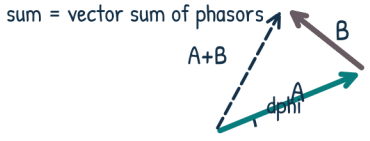
Phasor drill set (cont.)

intensity = one source. (c) Balanced bridge of four speakers in a square, driven alternately in phase/antiphase: at the centre all pairs cancel — architectural silence.

General N: geometric series $\sum_{k=0}^{N-1} ae^{ik\psi} = a\frac{1 - e^{iN\psi}}{1 - e^{i\psi}}$ — derive the array factor in one line; the phase-only version drives phased-array sonar.

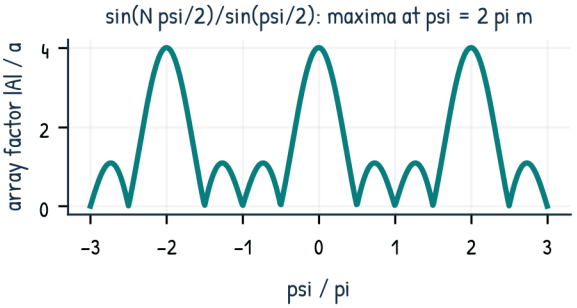
phasors extend to noise and pulses by Fourier decomposition: any waveform = phasor sum; linear systems act on each phasor independently with a transfer function $H(\omega)$ — the door to signal processing.

phasor addition: A + B as vectors



|A+B|^2 = A^2 + B^2 + 2 A B cos dphi

array factor of N coherent sources



REMEMBER / Phasors turn interference into vector sums; the array factor sin(N psi/2)/sin(psi/2) covers N sources.

Energy flux, radiation resistance, and the Q of resonance

Radiation resistance, near/far field, Q = ω E/P, ring-down and bandwidth.

Energy flux, radiation resistance, and the Q of resonance

Time-averaged intensity of a travelling wave: $I = \bar{u}v = \dfrac{\Delta P_0^2}{2Z}$ with $Z = \rho v$ (Oly 01). For a general (possibly reactive) field the instantaneous flux $\Pi = \Delta P \cdot u$ oscillates; only its average transports. A driven resonator near resonance stores energy $\mathcal{E}$ and loses power $P_{\text{loss}} = \omega\mathcal{E}/Q$;

quality factor $Q = \dfrac{\omega\mathcal{E}}{P_{\text{loss}}} = 2\pi\dfrac{\mathcal{E}}{\text{loss per cycle}}$;

ring-down time $\tau = Q/\omega$ (amplitude) — the number that separates tuning forks ($Q \sim 10^3$) from rooms ($Q \sim 5$).

Energy flux, radiation resistance, and the Q of resonance (cont.)

Radiation resistance: a vibrating surface of area S and velocity amplitude u_0 radiates $P = \frac{1}{2}R_r S u_0^2$ with $R_r \approx Z_{\text{air}}$ for a large baffle ($R_r = Z$ per unit area) but $R_r \propto (ka)^2 Z$ for a small piston ($ka < 1$): small sources are BAD radiators — the reason a bare vibrating rod is quiet and a loudspeaker cone is big. Multipole order: monopole (pulsating sphere) radiates best at low ka; dipole (oscillating sphere) loses another $(ka)^2$.

Near field vs far field: within $r < \lambda/2\pi$ the reactive component dominates ($\Delta P \perp u$): energy sloshes, no net flux; beyond, the travelling wave $\Delta P = Zu$ carries $I \propto 1/r^2$. Microphone placement and speaker measurements must sit in the far field.

Energy flux, radiation resistance, and the Q of resonance (cont.)

WHY / Q is the single number connecting ring-down, bandwidth, and amplitude build-up: $\Delta f = f/Q$, $\tau = Q/(2\pi f)$ -ish in time units, and peak gain $\propto Q$ for a given drive — one concept, three exam formulas.

CHECK / consistency: $Q = 2\pi\mathcal{E}/W_{\text{cycle}}$; $\Delta f = f_1/Q$; $\tau_{\text{amp}} = Q/(\pi f_1)$ ... at $Q = 20$, $f = 300$ Hz: $\tau = 21$ ms and $\Delta f = 15$ Hz: a struck pipe fades in ~3 ringing times ≈ 60 ms ok audible clap-coloration.

Pulsating sphere radius a, $ka < 1$: $R_r = \rho c (ka)^2/4$ per area... source power $P = 2\pi a^2 u_0^2 \rho c (ka)^2/4$ ...

exam form: $P = \dfrac{\rho c k^2 u_0^2}{4\pi} V^2$ -type scaling with volume velocity $Q_v = u_0 S$; a 5 cm speaker at 1 kHz ($ka = 0.9$): roughly $R_r \sim Z/2$: efficient;

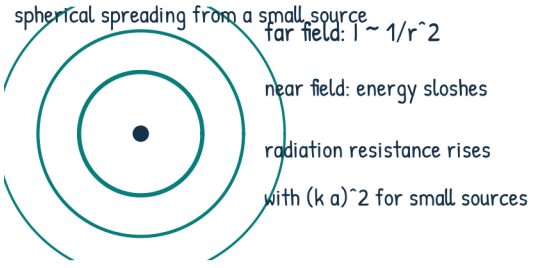
Radiation from real sources (monopole, dipole, baffles) (cont.)

the same at 100 Hz ($ka = 0.09$): $R_r \sim Z/250$: wimpy bass — why woofers are large and ported.

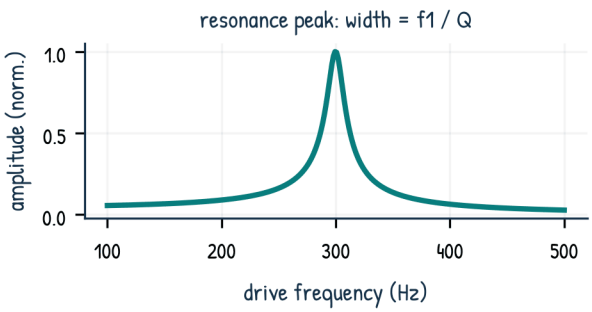
Baffle and enclosure effects: an unbaffled driver short-circuits front-to-back at low ka (dipole cancellation, 6 dB/octave rolloff); a box or horn restores monopole radiation; the infinite baffle gives $R_r = Z$ per area (pressure doubling at the panel).

Biological speakers: the human voice ~1 cm ^{2} glottis at 100-300 Hz: $ka < 0.1$: extremely inefficient (~0.01% of metabolic power): the mouth and vocal tract resonance (formants) do the impedance matching; birds sing 2-8 kHz where $ka \sim 1$: efficient little radiators.

spherical spreading and the radiating surface



resonance curve: width f1/Q



REMEMBER / Q = omega E / P_loss ties ring-down, bandwidth and gain; small sources are poor radiators.

Energy flux, radiation resistance, and the Q of resonance (cont.)

Radiation resistance, near/far field, Q = ω E/P, ring-down and bandwidth.

Radiation from real sources (monopole, dipole, baffles) (cont.)

CHECK / power scaling: $P \propto u_0^2 f^2$ at low ka for a fixed source: doubling frequency from 100 to 200 Hz multiplies radiated power by 4 at the same diaphragm speed ok why tweeters are easy and woofers hard.

(a) A 0.5 W source at 340 Hz in air: $I = P/(4\pi r^2)$: at 5 m: 1.6×10^{-3} W/m^2 = 92 dB; $\Delta P_0 = \sqrt{2ZI} = 1.14$ Pa; $s_0 = \Delta P_0/(Z\omega) = 1.3\mu$ m. (b) "How much power to hold 1 Pa at 1 kHz over 1 m^2?" $P = \Delta P_0^2 S/(2Z) = 1.2$ W — modest; for 100 Pa (pain) 12 kW over 1 m^2 — concert-scale.

Energy-budget problems (cont.)

(c) A fork of Q = 500, $f = 256$ Hz struck to 1 mm amplitude: stored energy $\mathcal{E} \sim \frac{1}{2} m_{\text{eff}} \omega^2 x^2$: for $m_{\text{eff}} = 20$ g: 1.3×10^{-3} J; loss per cycle $\mathcal{E} \cdot 2\pi/Q = 16\,\mu$J; ring-down $\tau = Q/(\pi f) = 0.62$ s.

Acoustic vs perceived: equal-loudness (Fletcher-Munson): physical I must rise at low f to match perceived loudness — power budget of hi-fi at 40 Hz is 10–20 dB more than at 1 kHz.

WHY / energy methods shortcut amplitude problems: instead of solving the wave equation, balance $P_{\text{source}} = \oint I\,dA + d\mathcal{E}/dt + P_{\text{loss}}$ — the acoustic version of the work-energy theorem.

Q and resonance in the exam frame

(a) "A closed pipe Q = 12 at f1 = 200 Hz is driven: amplitude at resonance vs 10 Hz off": ratio $\approx Q\Delta f/f$ -ish: at $\delta = 10$ Hz = half-width: amplitude $1/\sqrt{2}$ of peak.

(b) "Two pipes of equal f1 but Q = 10 and 40: which is louder at resonance, which more selective?" same peak if loss-limited... high-Q rings louder after excitation and narrower in sweep.

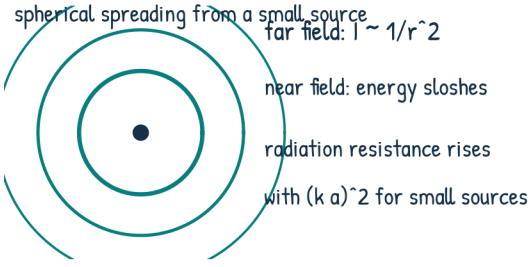
(c) "Estimate the Q of a room from reverberation": $Q = \pi f T_{60}/6.9$ ... standard: $T_{60} = 0.161 V/A$ (Sabine): Q from bandwidth of a mode: $Q = fT_{60}/2.2$ -ish: at 100 Hz, $T_{60} = 0.5$ s: $Q = 23$.

Q and resonance in the exam frame (cont.)

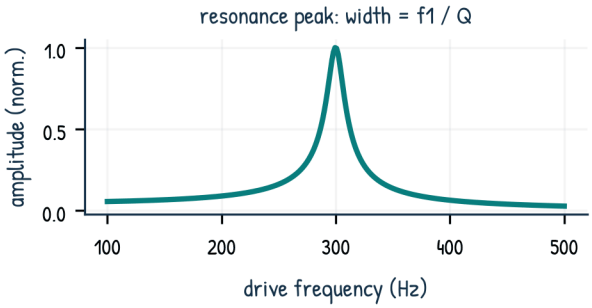
WATCH / Traps: (i) $\tau = Q/\omega$ vs $Q/(\pi f)$: factor conventions — state which amplitude time you use; (ii) Q is defined for ENERGY; amplitude decays twice as slowly... amplitude time $\tau_A = 2\tau_E$: exam convention varies: give both. (iii) Nonlinear large-amplitude acoustics drops Q (shock losses).

the resonator as a mechanical RLC: mass = acoustic inertance $\rho L/S$, compliance = $V/(\rho c^2)$, resistance = radiation + viscous: Z_{in} of Oly 03 is the acoustic RLC impedance — identical mathematics.

spherical spreading and the radiating surface



resonance curve: width f1/Q



REMEMBER / Q = omega E / P_loss ties ring-down, bandwidth and gain; small sources are poor radiators.

Sign-free Doppler: one vector formula for every case

Vector Doppler f' = f(c − v_o·r)/(c − v_s·r), projections replace cases.

Sign-free Doppler: one vector formula for every case

Let $\hat{\mathbf{r}}$ point from source to observer at the emission event; let $\mathbf{v}_s, \mathbf{v}_o$ be the velocities of source and observer relative to the medium; c the sound speed in the medium. Then the received frequency is $\mathbf{f' = f\frac{c - v_o \cdot \hat{r}}{c - v_s \cdot \hat{r}}}$ — no signs to memorise: projections carry the signs. Check 1D: source approaching ($\mathbf{v}_s$ along $+\hat{\mathbf{r}}$ toward observer... observer at larger r : $\mathbf{v}_s \cdot \hat{\mathbf{r}} = +u_s$ ... hmm careful: $\hat{\mathbf{r}}$ from source to observer: source moving toward observer means $\mathbf{v}_s$ has positive projection on $\hat{\mathbf{r}}$: denominator $c - u_s$ ok the approach rise; observer approaching means $\mathbf{v}_o$ points AGAINST $\hat{\mathbf{r}}$: $\mathbf{v}_o \cdot \hat{\mathbf{r}} = -u_o$: numerator $c + u_o$ ok exactly JEE-24.

Sign-free Doppler: one vector formula for every case (cont.)

Wind: replace $c\hat{\mathbf{r}}$ -type bookkeeping by the medium velocity $\mathbf{w}$: sound velocity in ground frame $\mathbf{c} = c\hat{\mathbf{r}} + \mathbf{w}$; the clean invariant form measures $\mathbf{v}_s, \mathbf{v}_o$ RELATIVE TO THE MEDIUM and c in the medium:

$f' = f\frac{c - (\mathbf{v}_o - \mathbf{w}) \cdot \hat{\mathbf{r}}}{c - (\mathbf{v}_s - \mathbf{w}) \cdot \hat{\mathbf{r}}}$ ... the robust recipe:

transform to the medium frame (subtract $\mathbf{w}$ from every ground velocity), apply the box, transform back (frequency is frame-invariant in classical acoustics for the observer present... the RECEIVED frequency is what the observer measures in their own frame: do all algebra in the medium frame then quote f').

Sign-free Doppler: one vector formula for every case (cont.)

Derivation: in the medium frame the wavefronts have spacing $\lambda' = \frac{c - \mathbf{v}_s \cdot \hat{\mathbf{r}}}{f}$ (source compresses along $\hat{\mathbf{r}}$) and the observer sweeps them at rate $\frac{c - \mathbf{v}_o \cdot \hat{\mathbf{r}}}{\lambda'}$: quotient = the box. Nothing else.

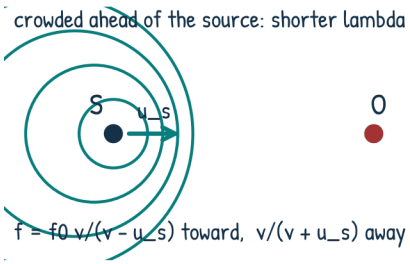
WHY / the dot products are the entire content of the 2D, angled, wind, and echo cases: every hard Doppler problem reduces to drawing $\hat{\mathbf{r}}$ and projecting — the box replaces a case library.

Sign-free Doppler: one vector formula for every case (cont.)

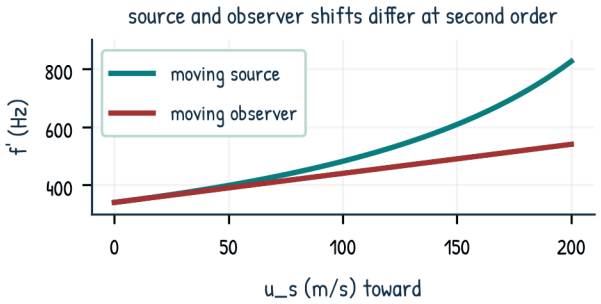
CHECK / specialise all cases: observer at rest: numerator c ; source at rest: denominator c ; both at rest in wind: ratio $(c - \mathbf{w} \cdot \hat{\mathbf{r}})/(c - \mathbf{w} \cdot \hat{\mathbf{r}}) = 1$ ok no shift in uniform wind (JEE-25 convention); tangential source ($\mathbf{v}_s \perp \hat{\mathbf{r}}$): $f' = f$ at that instant ok glide midpoint.

(a) Source on the x-axis at (vt, b) moving +x at u ; observer at origin: $\hat{\mathbf{r}} = \frac{(-vt,\ -b)}{\sqrt{v^2t^2 + b^2}}$ ... set ut for source position: $\hat{\mathbf{r}} = \frac{(-ut,\ -b)}{\sqrt{u^2t^2 + b^2}}$;

the vector formula: project onto the line of sight



the same glide curve from one formula



REMEMBER / f = f0 (c − v_o.r)/(c − v_s.r): one vector formula replaces the entire case library.

Sign-free Doppler: one vector formula for every case (cont.)

Vector Doppler f' = f(c − v_o·r)/(c − v_s·r), projections replace cases.

Vector clinic: worked projections (cont.)

$\mathbf{v}_s \cdot \hat{\mathbf{r}} = \dfrac{-u^2 t}{\sqrt{u^2 t^2 + b^2}} = -u\cos\theta(t)$:

$f' = f\dfrac{c}{c + u\cos\theta}$ ... check sign convention: approaching ($t < 0$): $\cos\theta < 0$: denominator $< c$: $f' > f$ ok matches glide picture (with θ measured from the velocity direction).

(b) Observer moving at angle: pedestrian walking at 2 m/s at 60° to the line from a 800 Hz siren 50 m away, $u_s = 30$ m/s: project both; the pedestrian term is $2\cos 60 = 1$ m/s: $f' = 800\dfrac{343 - 1}{343 - 30\cos\theta}$ – geometry first, numbers last.

Vector clinic: worked projections (cont.)

(c) Echo as two applications of the box: target = intermediate observer/source; compose: $f'' = f\dfrac{c - \mathbf{v}_T \cdot \hat{\mathbf{r}}}{c - \mathbf{v}_T \cdot (-\hat{\mathbf{r}})}$ -ish for monostatic... yields $\dfrac{c+u}{c-u}$ for approach ok JEE-27.

CHECK / frame sanity: the medium frame is singled out by the physics (sound needs a medium); the formula is NOT Lorentz-invariant – the contrast with light (relativistic Doppler: $f' = f\sqrt{\frac{1-\beta}{1+\beta}}$ approach... $\frac{1}{\gamma(1 \mp \beta)}$ factors) is a favourite exam discussion.

Where the case library hides inside the box

Case reduction: (i) 1D: projections $\pm u$; (ii) 2D: $u\cos\theta$ instantaneous; (iii) wind: shift to medium frame; (iv) moving reflector: two box applications; (v) accelerating: integrate projections over emission events (Oly 07); (vi) source passing: $\theta(t)$ from the geometry – the glide (Oly 08). The "four cases" of JEE-24 are four projection patterns of one sentence.

Second-order asymmetry explained: the box is exact; expanding $f' = f\dfrac{c + u_o}{c - u_s}$: the u_s denominator is a geometric series while the u_o numerator is finite – the asymmetry is built into the kinematics of wavefront emission vs collection.

Where the case library hides inside the box (cont.)

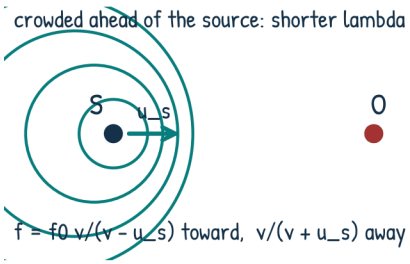
Applications: acoustic Doppler velocimetry (measure $\mathbf{v}_o \cdot \hat{\mathbf{r}}$ of particles), anemometry (wind from $\mathbf{w} \cdot \hat{\mathbf{r}}$ via two beams), bat sonar (dual use of box + echo), fetal Doppler (blood-cell reflectors).

WHY / the vector box makes reasoning CHECKABLE: after any case analysis, re-derive with projections; agreement validates the sign work – the antidote to sign errors.

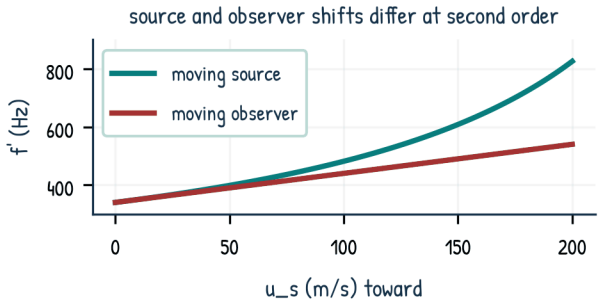
(a) $f = 500$ Hz source at $u_s = 40$ m/s at 30° to the line of sight, receding side:
$f' = 500\dfrac{343}{343 + 40\cos 30^\circ} = 500 \cdot \dfrac{343}{377.6} = 454$ Hz.

(b) Both moving toward each other at $u = 20$ each, 1D:
$f' = 500 \cdot \dfrac{363}{323} = 562$ Hz.

the vector formula: project onto the line of sight



the same glide curve from one formula



REMEMBER / f = f0 (c − v_o.r)/(c − v_s.r): one vector formula replaces the entire case library.

Sign-free Doppler: one vector formula for every case (cont.)

Vector Doppler f' = f(c − v_o·r)/(c − v_s·r), projections replace cases.

Box drills (answers in one line each) (cont.)

(c) Wind 15 m/s along the line, both at rest in ground: $f' = f$ (no shift), travel time downwind $L/(c+w)$ ok both parts of JEE-25 from one formula.

Reverse use: measure f' from two positions to get the velocity projections; two beams at known angles solve for the 2D velocity vector — the Doppler ultrasound vector flow meter.

relativistic check: for light in vacuum the medium-frame concept dies; the invariant formula is $f' = f\gamma(1 - \beta\cos\theta)$ — same projection structure, different dynamics: a satisfying comparison answer.

Accelerated motion: integrating the instantaneous shift

For a source with arbitrary velocity $\mathbf{v}_s(t)$ the observer hears $f'(t_e) = f\dfrac{c}{c - \mathbf{v}_s(t_e)\cdot\hat{\mathbf{r}}(t_e)}$ evaluated at the EMISSION time t_e: what is heard at time t was emitted at $t_e = t - R(t_e)/c$ (implicit!) — retarded-time bookkeeping. Constant velocity: closed form (JEE-26 glide); constant acceleration along the line: $f'(t_e) = f\dfrac{c}{c - at_e}$: frequency diverges as the source approaches speed c ... for $at_e \to c$; reached at finite time: sonic boom geometry (JEE-28).

Accelerated motion: integrating the instantaneous shift (cont.)

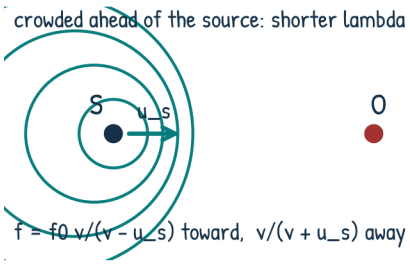
The glide-slope example as an integral: closest approach at $t = 0$, speed u, impact b: $f'(t) = f\dfrac{c}{c - u^2 t/\sqrt{b^2 + u^2 t^2}}$ (Oly 06 projections): linearising near $t = 0$: $f' \approx f\left(1 + \dfrac{u^2 t}{cb}\right)$: slope $\dfrac{df'}{dt}|_0 = \dfrac{fu^2}{cb}$ downward after the pass ok the JEE-26 formula as the first term of the exact expansion.

Accelerated motion: integrating the instantaneous shift (cont.)

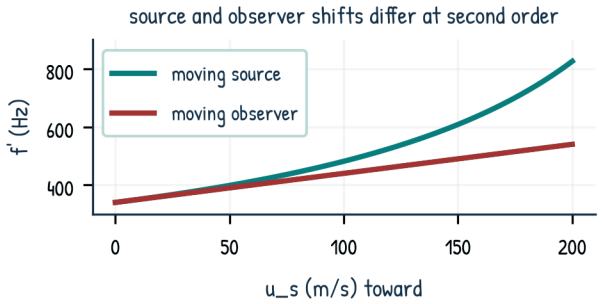
Slowly accelerating radial motion: quasi-static shift $f'(t) \approx f(1 + v_r(t)/c)$: the frequency curve is the RADIAL VELOCITY curve scaled by f/c — acoustic velocimetry: measuring $f'(t)$ reads $v_r(t)$ directly (differencing successive spectra).

WHY / the retarded-time structure is why accelerating-source problems are implicit: the emission point itself moves; solvable cases are (i) slow variation (quasi-static), (ii) constant velocity (closed form), (iii) special geometries (circular motion: Bessel sidebands).

the vector formula: project onto the line of sight



the same glide curve from one formula



REMEMBER / f = f0 (c − v_o.r)/(c − v_s.r): one vector formula replaces the entire case library.

Accelerated motion: integrating the instantaneous shift

Glide from the vector box, slope fu²/(cb), retarded-time thinking.

Accelerated motion: integrating the instantaneous shift (cont.)

CHECK / circular motion check: source circling at radius R, angular rate Ω: $v_r = u\cos(\Omega t)$ -ish: $f' = f\left(1 + \frac{u}{c}\cos\Omega t\right)$ first order: PURE frequency modulation with sidebands $f \pm \Omega/2\pi$ — the ambulance-on-a-roundabout FM signal.

Setup recap: straight path, impact b, speed u, stationary observer, emission continuous. $\cos\theta(t) = \frac{ut}{\sqrt{b^2 + u^2t^2}}$ (sign per convention); $f'(t) = f\frac{c}{c - u\cos\theta(t)}$; values: $t = 0$: $f' = f$; $t \to \pm\infty$:

Glide geometry: the exact curve and its slope (cont.)

$f\frac{c}{c \mp u}$; slope at 0: $\mp\frac{fu^2}{cb}$ (decreasing through the pass).

Characteristic width: the glide occupies $\Delta t \sim b/u$ (the time to swing θ by a radian).

Spectrogram picture: a hyperbolic-like chirp symmetric in time but NOT in frequency (approach side higher by the $c/(c-u)$ vs $c/(c+u)$ asymmetry); a bat or porpoise hears the whole swoop and uses the slope for ranging (Oly 08).

Numbers (corrected source example): $f = 1000$ Hz, $u = 20$ m/s, $b = 100$ m, $c = 340$: slope 11.8 Hz/s; $f'(\pm 5$ s$) = 1043.2/960.1$ Hz; asymmetry at equal times: 83.1 Hz total swing: the integral of the slope over the pass.

Glide geometry: the exact curve and its slope (cont.)

CHECK / consistency of slope and swing: average slope over ± 5 s $\approx 83/10 = 8.3$ Hz/s vs peak 11.8 — smooth rolloff ok; far-field asymptotes reproduce the 1D pair 1043.5/960.5 Hz within 0.5 Hz at ± 5 s... hmm far asymptotes are $1000 \cdot 340/320 = 1062.5$ and $1000 \cdot 340/360 = 944.4$: the pass values are closer to f than the asymptotes ok correct shape.

(a) A car accelerating from rest toward a listener at $a = 2$ m/s²: $v_r = at$: $f' = f(1 + at/c)$ -ish until significant: at $t = 10$ s, $v = 20$ m/s: $f' = 1.06f$: the pitch RISES continuously (unlike constant-velocity pass which glides down after). (b) A dropped source: $v_r = gt$ toward a listener below: same linear chirp downward in observed... source falling AWAY from an observer above:

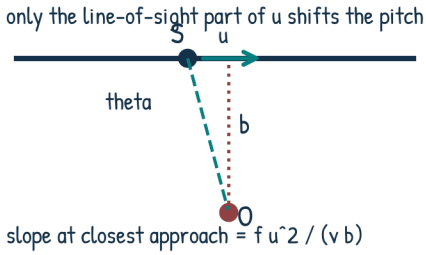
Accelerated-source problem patterns (cont.)

$f' = f(1 - gt/c)$: a falling whistle chirps down at $gf/c = 28.7$ Hz/s for $f = 1000$.

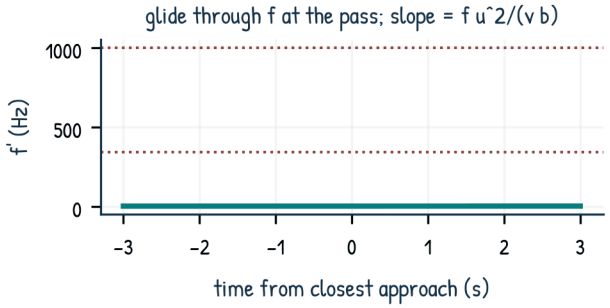
Retarded-time example: source at distance d moving away at u emitted at $t = 0$: received at $t = d/c$; the NEXT wavefront is received at $t = (d + u/f)/c$ -ish: spacing stretched by $(c+u)/c$ ok the receding factor derived from scratch.

rotating and vibrating sources: turbine blades, sirens, ultrasound transducers: the spectrum is a set of sidebands $f \pm nf_{\text{rot}}$ — Bessel-function amplitudes; the exam-accessible part is the first-order FM picture.

closest approach: radial speed crosses zero



the glide: slope f u^2 / (v b)



REMEMBER / Accelerated motion: integrate the instantaneous shift at emission time; the glide is the first lesson.

Accelerated motion: integrating the instantaneous shift (cont.)

Glide from the vector box, slope fu²/(cb), retarded-time thinking.

Slope-and-swing drills

(a) $f = 600$ Hz, $u = 25$ m/s, $b = 150$ m: slope $= 600 \cdot 625/(340 \cdot 150) = 7.35$ Hz/s. (b) "The observed pitch of a passing cycle horn (500 Hz) falls 60 Hz within 3 s of the pass: estimate u^2/b": slope 20 Hz/s $= fu^2/(cb)$: $u^2/b = 20 \cdot 340/500 = 13.6$ m/s^2-ish units: with $u = 25$: $b = 46$ m. (c) Bat closing on an insect: both glide AND radial approach: total slope = glide term + fa/c from acceleration. What can go wrong: near-field ($b \lesssim \lambda$) wave geometry breaks the ray picture; supersonic $u > c$ invalidates the subsonic formula entirely (JEE-28); turbulent air adds scintillation (amplitude noise), not frequency shift.

Slope-and-swing drills (cont.)

WHY / the glide is the acoustic fingerprint of the geometry: slope gives u^2/b (one equation, two unknowns — need the swing width b/u or a second measurement to separate u and b — the structure of exam problems.

If the emission frequency f and the source speed u are known, one second of spectrogram near closest approach gives $b = \dfrac{fu^2}{c|slope|}$ (from Oly 07). If instead b is known (e.g. flyby altitude), the slope gives u:

Ranging from frequency slope: the acoustic McAfee method (cont.)

$u = \sqrt{\dfrac{c\,b\,|slope|}{f}}$. With BOTH slope and swing width $\Delta t \sim b/u$: the pair determines u and b independently: $u = |slope|\, c\, \Delta t/f$ -ish... precisely combine $S = fu^2/(cb)$ and $W = b/u$: $u = \sqrt{ScW/f}$ ... derive: $SW = fu^2/(cb) \cdot b/u = fu/c$:

$u = \dfrac{SWc}{f}$ ok and $b = uW$.

Ranging from frequency slope: the acoustic McAfee method (cont.)

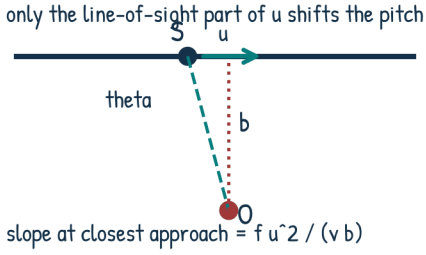
Echo version (radar/sonar McAfee): measure the beat $\Delta f = 2fu/c$ between direct and echo (JEE-27):

$u = \dfrac{\Delta f\, c}{2f}$ directly; combine with the echo delay $\tau = 2R/c$ for range; the pair (Δf , τ) gives (u , R) of the target — the full tracking state at one instant.

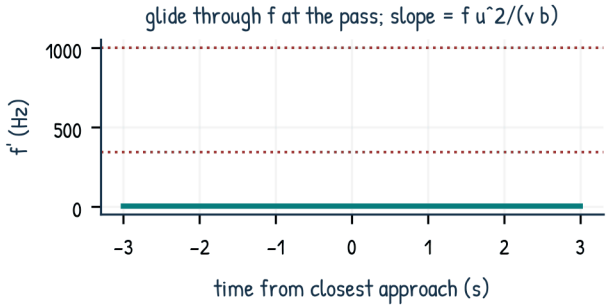
Accuracy: frequency slope measurable to ~1% from a clean spectrogram: b to ~2%; the method needs no timing sync — pure frequency metrology: robust to amplitude noise; fails when u is unknown along with b (add a second pass or beam).

WHY / the slope method converts a TIME-domain difficult problem (nanosecond timing) into a FREQUENCY-domain easy one (Hz over seconds) — the deep reason Doppler ranging works.

closest approach: radial speed crosses zero



the glide: slope f u^2 / (v b)



REMEMBER / Accelerated motion: integrate the instantaneous shift at emission time; the glide is the first lesson.

Ranging from frequency slope: the acoustic McAfee method

Ranging via slope u²/b, echo beats, two-beam tracking, method table.

Ranging from frequency slope: the acoustic McAfee method (cont.)

CHECK / units: $[fu^2/(c\,slope)] =$ Hz m ^{2} s $^{-2}$ / (m s $^{-1}$ Hz s $^{-1}$) = m ok; numbers: $f = 1000$, $u = 20$, $c = 340$, slope 11.8: $b = 100$ m ok recovers the geometry.

Two beams at angles θ_1, θ_2 to the target line: measure radial speeds $u\cos\theta_i$ ... solve: $u = \dfrac{v_{r1} - v_{r2}}{\cos\theta_1 - \cos\theta_2}$ -style elimination; three beams give the full 3D velocity vector — Doppler sodar and ultrasound vector flow imaging.

Track differentiation: successive (u , R) estimates give acceleration $a = du/dt$; the glide trajectory then fits (b , u , a) — the standard manoeuvring-target model.

Two-beam and multi-beam Doppler tracking (cont.)

GPS-free ground speed: doppler log on ships (beam to the seabed), air data from pitot + acoustic cross-check; police lidar/radar: EM version of the same math ($c \to 3 \times 10^8$, double shift always).

CHECK / double-beam example: $\theta = 30°$ beams, measured $v_r = 43.3$ and 0 m/s (one beam along the motion): $u = 50$ m/s ok cross-check $50\cos 30 = 43.3$ ok.

(a) "A jet of known speed 250 m/s at unknown altitude passes; the 400 Hz tone glides at 300 Hz/s through the closest point: find altitude": $b = fu^2/(cS) = 400 \cdot 62500/(340 \cdot 300) = 245$ m — absurdly low for a jet: sanity-check rejects the numbers (maybe S read wrong from the spectrogram) — the discipline of orders of magnitude.

Slope-method problem patterns (cont.)

(b) "A drone hovers at 40 m emitting 2 kHz; a cyclist approaches at 8 m/s: beat with the echo": $\Delta f = 2fu_r/c$ with radial 8 m/s: 94 Hz beats — drone speed sensors use exactly this (acoustic flow sensors). (c) Bat 1: measure slope of the insect pass at $b = 2$ m, $u = 4$ m/s, 40 kHz: slope $= 40000 \cdot 16/(340 \cdot 2) = 941$ Hz/s — bats chirp-sweep over ~2 kHz in ms, resolvable.

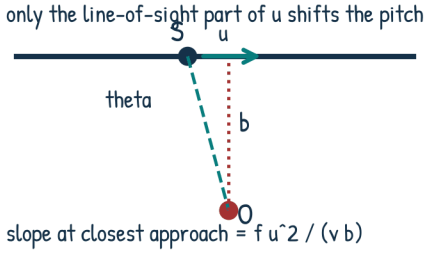
WHY / combining slope + swing + echo delay separates u , b , R — the exam trinity of ranging unknowns and their three equations.

Method comparison table (which tool for which question)

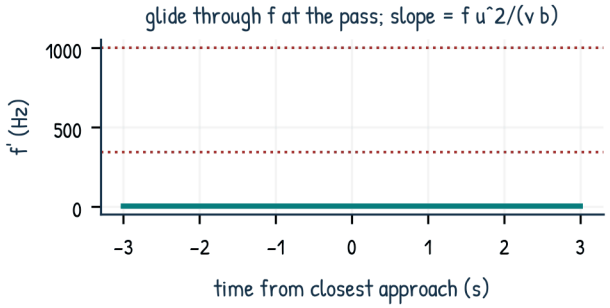
Echo delay: gives RANGE $R = c\tau/2$ directly; needs timing. Doppler beat: gives RADIAL SPEED $\Delta f = 2fu/c$; needs clean spectrum. Glide slope: gives u^2/b ; needs a pass geometry. Echo delay + beat: state (u, R) ok. Slope + swing: state (u, b) ok. Two beams: vector velocity ok. Time-of-flight + speed of sound thermometer: c from T (JEE-08).

WATCH / Error budget: frequency measurements are precise (10^{-4} easily) but c depends on T (± 0.2 % per K) and wind ($\pm w/c$): for precision ranging, measure T and w or use two-way symmetric methods that cancel wind to first order (JEE-25).

slope plus swing gives u and b separately



echo beat: the target speed in one number



REMEMBER / Frequency slope gives u^2/b; echo beats give u; two measurements give the full state.

Ranging from frequency slope: the acoustic McAfee method (cont.)

Ranging via slope u²/b, echo beats, two-beam tracking, method table.

Method comparison table (which tool for which question) (cont.)

the same toolkit works from bat sonar to medical Doppler to police radar; changing c and f scales the numbers, never the structure — the closing argument for learning the vector box.

For a duct of slowly varying cross-section $A(x)$ with negligible reflection (smooth taper, $|dA/dx| \ll kA$), conservation of time-averaged acoustic power gives $I(x)A(x) =$ const. Since $I = \dfrac{\Delta P_0^2}{2Z}$ and $I = \frac{1}{2} Z \omega^2 s_0^2$ in the same medium:

pressure amplitude $\Delta P_0 \propto A^{-1/2}$, displacement amplitude $s_0 \propto A^{-1/2}$, intensity $I \propto A^{-1}$ — a horn concentrates SOUND PRESSURE as $1/\sqrt{A}$ from throat to mouth.

Slowly varying ducts and horns: I(x) A(x) = constant (WKB) (cont.)

WKB validity: the reflection coefficient per taper step is $R \sim \left(\dfrac{d \ln Z_c}{2k\,dx}\right)^2$ -ish with $Z_c = \rho c / A$ the local characteristic impedance... practical rule: R negligible when the area changes by much less than a factor 2 per wavelength; abrupt changes reflect (JEE-16 boundary theory).

Exponential horn: $A(x) = A_0 e^{mx}$: the horn equation has wave solutions with local wavenumber $k_{\text{eff}} = \sqrt{k^2 - (m/2)^2}$: below cutoff $k < m/2$: EVANESCENT (horn stops passing — the horn cutoff frequency); the flare constant m sets the cutoff $f_c = \dfrac{mc}{4\pi}$ -ish... standard: $f_c = \dfrac{mc}{4\pi}$ for exponential:

Slowly varying ducts and horns: I(x) A(x) = constant (WKB) (cont.)

low-frequency horns must flare gently (large mouth) — why trumpets end in big bells.

WHY / power conservation is the entire WKB content: the wave equation in a slowly varying duct with the ansatz $\Delta P = \mathcal{A}(x) e^{i \int k dx}$ gives $\mathcal{A}\sqrt{Z_c}$ -type invariants; the exam physics is the $A(x)I(x) =$ const line.

CHECK / convergence example: throat 1 cm² to mouth 100 cm²: I drops 100x (20 dB) but ΔP_0 RISES 10x: horns trade intensity for pressure at the throat — exactly what a driver needs (impedance matching, Oly 01: $Z_{\text{in}} \sim Z/A$ -ish).

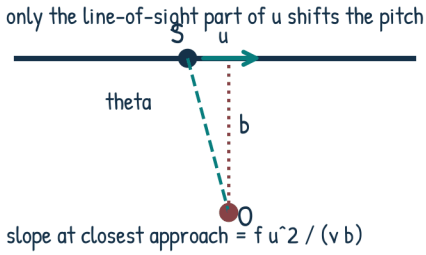
Horns as impedance transformers (why they work)

A driver of area S_d couples to air poorly (Oly 05 radiation resistance); a horn raises the throat loading: $Z_{\text{throat}} \approx \rho c / A_{\text{mouth}}$ (geometric gain): the mouth impedance ρc scaled by area ratio: efficiency up to ~50% vs a few % for a bare cone. The same mathematics as the middle ear (Oly 01) and the megaphone of antiquity.

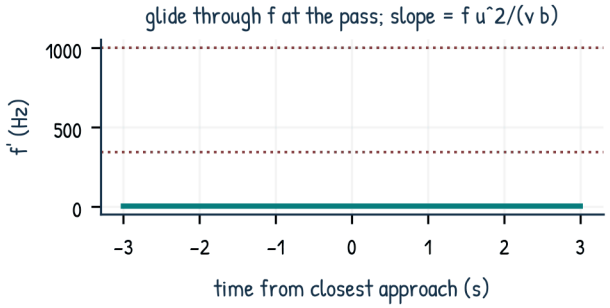
Cutoff and directivity: below f_c the horn is a poor radiator (evanescent throat); above, the mouth radiates a beam of angle $\theta \sim \sin^{-1}(\lambda/(\pi d))$ -ish for mouth size d: megaphones beam high frequencies — the shouter hears their own voice differently than listeners do.

Mufflers and ducts: the reverse design — expansions and side branches reflect (Oly 03 traps); a smooth diffuser recovers pressure (wind tunnels, exhausts).

slope plus swing gives u and b separately



echo beat: the target speed in one number



REMEMBER / Frequency slope gives u^2/b; echo beats give u; two measurements give the full state.

Slowly varying ducts and horns: I(x) A(x) = constant (WKB)

WKB duct I·A = const, amplitude $\propto$ A^(−1/2), horn cutoff and matching.

Horns as impedance transformers (why they work) (cont.)

CHECK / numbers: exponential horn $m = 5$ m^{-1}: $f_c = 5 \cdot 343/(4\pi) = 136$ Hz: audible bass passes; $m = 50$: $f_c = 1.36$ kHz: a trumpet-ish cutoff ok plausible instrument numbers.

(a) A duct halves its area: I doubles, ΔP_0 rises $\sqrt{2}$: the speed of the air in the throat rises as $u \propto I/\Delta P \propto 1/\sqrt{A}$ ... $u = \Delta P/Z$ stays? $\Delta P_0 \propto A^{-1/2}$, $u_0 = \Delta P_0/Z \propto A^{-1/2}$: velocity rises as the duct narrows ok (like a river through a constriction).

WKB drill set (cont.)

(b) "Design a horn to halve the pressure amplitude of a 1 Pa source at the mouth": expand area 4x.

Horn equation quick form: $\frac{1}{A}\frac{d}{dx}\left(A\frac{dP}{dx}\right) + k^2 P = 0$:

WKB solution $P \propto \frac{e^{\pm ikx}}{\sqrt{A}}$ ok recovering the amplitude law; exact for cones and exponentials in terms of Bessel/Hankel functions.

shallow-water waves obey the same $IA =$ const rule (energy conservation); the "water hose narrowing" and tidal-bore funnelling are WKB in geophysics.

When WKB fails (scope guards)

(i) Area jumps: full reflection analysis (Oly 02/03). (ii) $A \to 0$ (a closed end): WKB diverges; the boundary layer solution takes over (JEE-17). (iii) $dA/dx \sim kA$: significant reflection; (iv) losses (viscous boundary layers: $\alpha \propto \sqrt{f}/a$ in narrow tubes) break power conservation: horns must stay wide vs the viscous skin depth.

Helmholtz resonator limit: when the duct is shorter than $\lambda/4$, the lumped-parameter RLC model replaces WKB entirely — asymptotic methods need their validity conditions stated (the exam phrase: "slowly varying compared with λ").

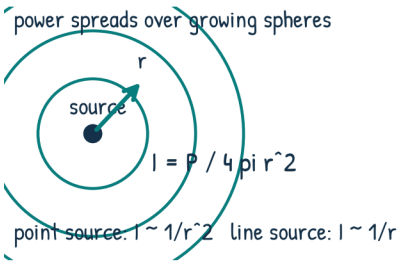
Elasticity of solids: three speeds from three moduli

WHY / knowing where an approximation dies is what separates a formula user from a physicist: every WKB answer should carry the check $|d\ln A/dx| \ll k$.

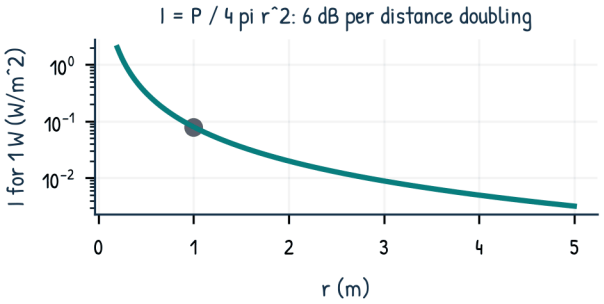
Longitudinal rod (free sides): $v_{\text{rod}} = \sqrt{Y/\rho}$; bulk longitudinal (infinite solid): $v_P = \sqrt{(K + 4G/3)/\rho}$; bulk transverse (shear): $v_S = \sqrt{G/\rho}$. Relations: $Y = 2G(1+\nu) = 3K(1-2\nu)$; with $\nu = 0.29$ (steel): $v_P/v_S = \sqrt{\frac{2(1-\nu)}{1-2\nu}} = 1.75$;

$v_P/v_{\text{rod}} = \sqrt{\frac{(1-\nu)}{(1+\nu)(1-2\nu)}}$ -ish ≈ 1.15–1.2. Seismology: P-waves = v_P, S-waves = v_S; S cannot propagate in fluids ($G = 0$) — the evidence Earth outer core is liquid.

slowly varying duct: power I A = constant



intensity falls as area grows



REMEMBER / Slowly varying ducts carry I A = constant: amplitudes fall as A^(−1/2); horns match impedance.

Elasticity of solids: three speeds from three moduli

Rod/bulk/shear speeds, Lamé derivation sketch, seismology patterns.

Elasticity of solids: three speeds from three moduli (cont.)

Derivation sketch: Newton 2nd law + Hooke tensor law $\sigma_{ij} = \lambda\,\mathrm{tr}(\epsilon)\delta_{ij} + 2G\epsilon_{ij}$ (Lamé λ, G): plane longitudinal wave: $\rho\ddot{u} = (\lambda + 2G)\nabla^2 u$: $v_P = \sqrt{(\lambda + 2G)/\rho} = \sqrt{(K + 4G/3)/\rho}$ since $K = \lambda + 2G/3$; transverse: $\rho\ddot{u}_\perp = G\nabla^2 u_\perp$: $v_S = \sqrt{G/\rho}$ ok.

Numbers: steel ($K = 160$, $G = 79$ GPa, $\rho = 7800$): $v_P = 5830$, $v_S = 3180$, $v_{\mathrm{rod}} = 5060$ m/s. Granite: $v_P \approx 5200$, $v_S \approx 3000$. Glass ($G = 30$ GPa): $v_S = 2130$. Rubber ($G \approx 1$ MPa, $\rho = 1100$): $v_S = 30$ m/s — slow and lossy.

Elasticity of solids: three speeds from three moduli (cont.)

WHY / three combinations of two elastic constants + density: the rod speed is a boundary-conditioned mixture; the bulk pair is the intrinsic content of the elasticity tensor; Poisson ratio interpolates.

CHECK / inequalities: $v_P > v_{\mathrm{rod}} > v_S$ for $0 < \nu < 0.5$ ok (steel: 5830 > 5060 > 3180); $\nu \to 0$: $v_P = v_{\mathrm{rod}}$; $\nu \to 0.5$: $v_P \to \infty$ (incompressible: $K \to \infty$).

Thin rod free-free (or clamped at centre): longitudinal modes $f_n = n\dfrac{v_{\mathrm{rod}}}{2L}$; clamped-free: $f_n = (2n-1)\dfrac{v_{\mathrm{rod}}}{4L}$ (odd, like the closed pipe); the same ladders as air columns with the solid speed. Dispersive corrections: for rods of finite thickness ($ka \gtrsim 1$) Pochhammer-Chree modes appear: the simple $\sqrt{Y/\rho}$ is the long-wave limit.

Waves in rods: which boundary, which ladder (cont.)

Kundt tube (JEE-21) is the rod-speed measurement in reverse: v_{rod} known gives v_{gas}; with a gas of known speed the rod speed comes out: the two-medium frequency match $f = v_{\mathrm{rod}}/\lambda_{\mathrm{rod}} = v_{\mathrm{gas}}/\lambda_{\mathrm{gas}}$.

Damping: metals ring (low internal friction, Q ~ 10^3 – 10^4 for rods); lead and rubber damp (high loss tangent): struck rods of different metals have characteristic ring times — an exam qualitative question.

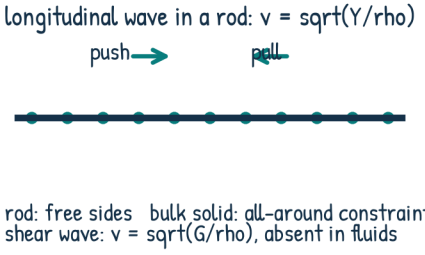
CHECK / steel rod 1 m clamped mid: $f = 5060/2 = 2530$ Hz; in Kundt with dust heaps 6.5 cm apart: $v_{\mathrm{gas}} = 5060 \times 0.065 = 329$ m/s (cold air ok plausible lab).

Solid-state acoustic problems (exam patterns)

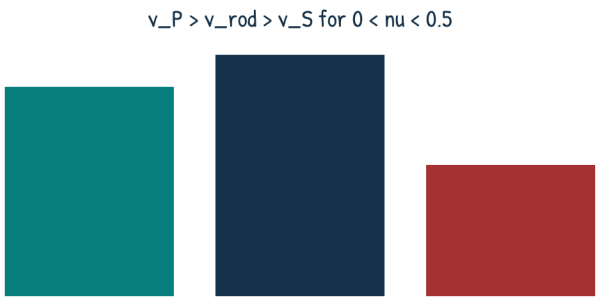
(a) "A longitudinal wave travels a steel rail; a transverse follows: arrival split over 2 km": $\Delta t = 2000(1/3180 - 1/5830)$ ... wait P arrives first: $\Delta t = 2000(1/3180 - 1/5830) = 0.285$ s ok the seismograph method for locating epicentres. (b) "Find ν from the ratio $v_P/v_S = 1.8$": $\frac{2(1-\nu)}{1-2\nu} = 3.24$: $\nu = 0.27$.

(c) "A rod and a string of the same length and fundamental: compare v": equal f: $v_{\mathrm{rod}}/2L = \sqrt{T/\mu}/2L$: the speed equality sets the tension.

rod, bulk and shear waves in one bar



three elastic speeds side by side



REMEMBER / Three elastic speeds: rod sqrt(Y/rho), P sqrt((K+4G/3)/rho), S sqrt(G/rho); fluids carry no shear.

Elasticity of solids: three speeds from three moduli (cont.)

Rod/bulk/shear speeds, Lamé derivation sketch, seismology patterns.

Solid-state acoustic problems (exam patterns) (cont.)

Composite rods: series springs: effective compliance adds: $1/Y_{\text{eff}} = \sum L_i/(Y_i L)$ -ish: wave speed in layered rods uses transfer matrices (outside syllabus; the 1D spring model is in).

dispersion in real solids: $v(\omega)$ bends near lattice resonances (Brillouin); ultrasound velocity metrology uses v_P, v_S to get K, G, ν non-destructively — the engineering reading of this card.

Numbers drill (three speeds)

Aluminium ($Y = 70$ GPa, $G = 26$, $\rho = 2700$): $v_{\text{rod}} = 5100$, $v_P = 6270$, $v_S = 3100$ m/s. Copper ($Y = 120$, $G = 45$, $\rho = 8900$): rod 3670, $v_P = 4760$, $v_S = 2250$. Granite ($K = 50$ GPa, $G = 25$ GPa, $\rho = 2700$): $v_P = 5250$, $v_S = 3040$. Water ($B = 2.2$ GPa): $v = 1483$, no shear.

Seismic locate: P-S time difference $\Delta t = d(1/v_S - 1/v_P)$: $d = \dfrac{\Delta t}{1/v_S - 1/v_P}$: for $\Delta t = 10$ s in granite: $d = 10/(3.29 - 1.90) \times 10^{-4}$ ... units: $d = \Delta t/(1/3040 - 1/5250) = 10/(3.29 \times 10^{-4} - 1.90 \times 10^{-4}) = 72$ km ok standard method.

Numbers drill (three speeds) (cont.)

WHY / the pair (arrival times of P and S) plus one speed model locates events in three dimensions — the seismology version of echo + beat (Oly 08): two measurements, two unknowns.

Full supersonic toolkit: cone half-angle $\sin\alpha = 1/M$; observer at height h below the flight line: boom delay after closest pass $t = \dfrac{h\sqrt{M^2-1}}{Mv}$ (envelope derivation of JEE-28); emission point of the heard boom at $t_e = -t/(M^2-1)$ (BEFORE the pass);

the jet position at hearing is $x = h\sqrt{M^2-1}$ past the closest point. The three quantities answer three different questions — always identify which is asked.

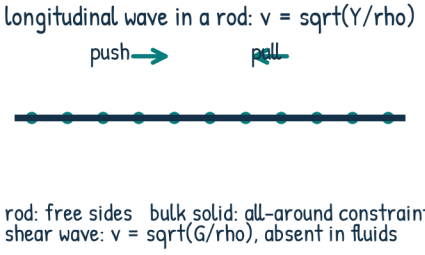
Mach geometry clinic: cones, delays, and emission points (cont.)

Boom carpet: the shock surface intersects the ground along a swath of width $w = h\cot\alpha \cdot 2$ -ish for the full cone crossing... the ground trace: at any instant the cone-ground intersection is a line moving at Mv; observers under the flight path receive the boom once; those off the track receive it at slant geometry; the overpressure decays slowly with altitude ($\propto h^{-3/4}$ -ish empirically).

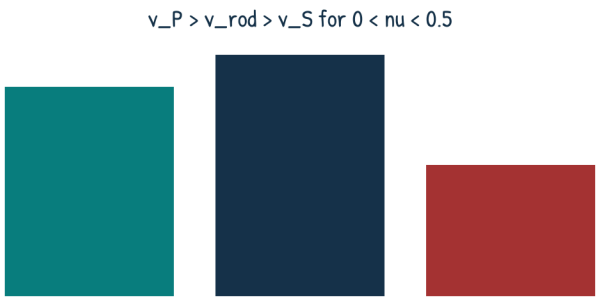
N-wave structure: the boom is a compression-rarefaction doublet of duration $\tau \sim \dfrac{\ell}{v}\cdot$ (aircraft length factor):

Concorde: 0.1–0.3 s at 100–200 Pa; the spectrum of an N-wave is flat to $f \sim 1/\tau$ then falls: characteristic boom "crack".

rod, bulk and shear waves in one bar



three elastic speeds side by side



REMEMBER / Three elastic speeds: rod sqrt(Y/rho), P sqrt((K+4G/3)/rho), S sqrt(G/rho); fluids carry no shear.

Mach geometry clinic: cones, delays, and emission points

Mach clinic: delay formula, emission point, boom carpet, drills.

Mach geometry clinic: cones, delays, and emission points (cont.)

WHY / the clinic separates three confusable distances: emission distance (from t_e), hearing-time position ($h\cot\alpha$), and slant range ($h/\sin\alpha$): the source paper Q13 conflates the first two (SOURCE-AUDIT).

CHECK / self-consistency: $t = h\cot\alpha/u_s$ (cone sweep) $= h\sqrt{M^2-1}/(Mv)$ (envelope) ok identical expressions; $M=1.2$, $h=10$ km: 16.4 s; $M=3$: 27.7 s — the non-monotonic delay again ok JEE-28 grid.

Angles and triangles: the clinic worksheet

Step 1: compute $c(T)$ (JEE-08). Step 2: $M = u/c$; $\alpha = \sin^{-1}(1/M)$. Step 3: draw the cone BEHIND the source; the observer is inside after the sweep. Step 4: identify the asked quantity: delay ($h\cot\alpha/u$), emission geometry ($t_e = -t/(M^2-1)$), or slant ($h/\sin\alpha$). Step 5: check limits ($M=1$: degenerate; $M\to\infty$: delay $\to h/v$).

Two observers: different delays from different h and lateral offsets: the difference in arrival times LOCATES the flight line — inverse boom mapping (used to track supersonic tests).

Sonic boom from manoeuvres: turns pile shocks (focused superbooms); the clinic covers straight-and-level only.

Angles and triangles: the clinic worksheet (cont.)

CHECK / work the clinic on source Q13: $M=2$, $h=10$ km, $v=340$: $\alpha=30^\circ$ ok; $t=25.5$ s ok; $t_e=-8.5$ s ok; jet at hearing 17.32 km past ok — all four numbers consistent with the one diagram.

(a) $M=1.5$, $h=5$ km: $\alpha=41.8^\circ$, $t = 5000\sqrt{1.25}/(1.5\cdot340) = 11.0$ s... $\sqrt{M^2-1}=1.118$: $t = 5000\cdot1.118/510 = 10.96$ s ok. (b) "Observers at 10 and 20 km lateral offset hear the boom 8 s apart: find M and altitude": two equations ($t_i = \sqrt{h^2+x_i^2}\cdot$... solve geometry: classic two-station problem. (c) "How long after the FLASH does the boom arrive for $M=2$, $h=8$ km?": flash $\approx$ overhead instant: delay $= 20.4$ s.

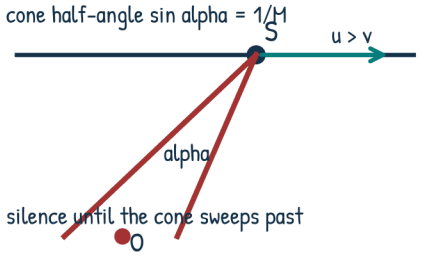
Clinic drills (cont.)

WATCH / Error sensitivities: $\delta t/t = \delta h/h + \delta M\cdot($ $M/(M^2-1) - 1/M$)-ish: at $M=2$: $\delta t/t \approx \delta h/h + 0.25\delta M/M$: Mach errors matter at low supersonic.

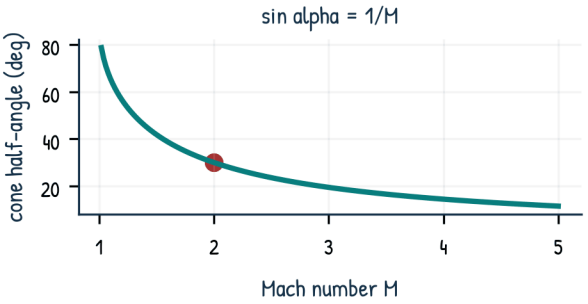
focused booms from turns can double overpressure (Whitham theory); "superboom" events over Oklahoma City (1960s tests) seeded the modern low-boom research (X-59) — history with the clinic math.

Mach number $M = u/c$; Mach angle $\alpha = \sin^{-1}(1/M)$; N-wave: two-shock signature; overpressure ΔP: 50–500 Pa typical; carpet: ground swath; boom delay: closest-approach to hearing; superboom: focused enhancement; cutoff Mach: $M=1$ exactly (infinite linear pile-up).

boom delay geometry from the cone



delay vs Mach: peaks near M = sqrt(2)



REMEMBER / Boom geometry clinic: delay h sqrt(M^2-1)/(M v), emission before the pass, carpet on the ground.

Mach geometry clinic: cones, delays, and emission points (cont.)

Mach clinic: delay formula, emission point, boom carpet, drills.

Sonic-boom terminology card (cont.)

Regulation: supersonic flight over land banned for civil (US 1973 FAA rule) because of boom annoyance; Concorde flew supersonic only over water; X-59 targets 75 Pa "thump" to reopen the question.

WHY / precise vocabulary earns the method marks: examiners look for "cone", "shock", "delay from closest approach", and the $\sin\alpha = 1/M$ relation in any boom answer.

Kit formula set (air unless stated): $c = 331.3\sqrt{T/273}$ m/s; $Z = \rho c \approx 412$ rayl at 20 C; $\lambda = c/f$; $\Delta P_0 = Z\omega s_0$; $I = \Delta P_0^2/(2Z)$; $\beta = 10\log(I/10^{-12})$ dB; point source $I = P/(4\pi r^2)$; beats $|f_1 - f_2|$; Doppler $\frac{c+u_o}{c-u_s}$; echo $\frac{c+u}{c-u}$; pipes $v/4L$, $v/2L$; $e = 0.6r$; $Q = \omega\mathcal{E}/P_{\text{loss}}$.

Acoustic estimation kit: building your own numbers (cont.)

With these, estimate any sound problem to ~10% in one line each.

Order-of-magnitude anchors: speech 60 dB at 1 m (~ $10\,\mu$ W acoustic); shout 1 mW; rock concert front 0.1–1 W/m^2; threshold of hearing 10^{-12} W/m^2 = 20 μ Pa; pain 20 Pa; room modes $c/2L$ (~170 Hz for a 1 m scale); ear canal 3.4 kHz; vocal formants 300/2300 Hz; jet 130 dB at 30 m; thunder 120 dB at 100 m.

Acoustic estimation kit: building your own numbers (cont.)

Sizing tricks: $\lambda = 343/f$: 1 kHz $\leftrightarrow$ 34 cm; 100 Hz $\leftrightarrow$ 3.4 m; 10 kHz $\leftrightarrow$ 3.4 cm. s_0 at threshold (1 kHz): 0.01 nm; at pain: 10 μ m. Time: sound 1 km in 2.9 s; 1 mile in 4.7 s; light-sound delay 1 km: 2.9 s (the "count the seconds" method).

WHY / the kit is the chapter in twelve lines: every worked example of the source is a substitution; every Olympiad problem is two or three kit lines chained — the closing consolidation of the set.

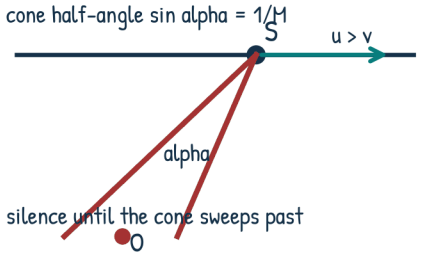
CHECK / re-derive the source Q3 chain from the kit: 0.1 W isotropic at 100 m: $I = 8\times10^{-5}$ W/m^2; $\beta = 79$ dB; $\Delta P_0 = 0.26$ Pa; $s_0 = 0.3\,\mu$m at 340 Hz ok all five kit lines agree with the worked answer.

Estimate-first discipline (the Fermi habit)

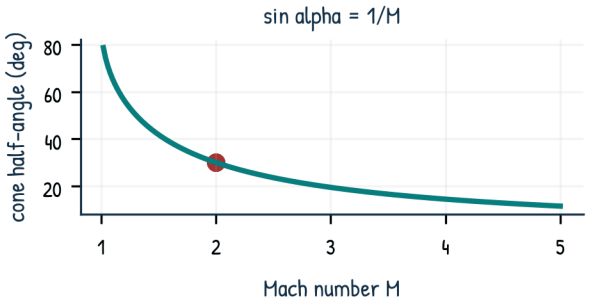
Before writing equations: estimate the answer in rounded numbers (factor-of-3 accuracy); then compute exactly; the two must agree within the rounded uncertainty — if not, a unit or a factor of 2π is wrong. Worked: "level 2 m from a 0.5 W siren": estimate $I \sim 0.5/50 \sim 0.01$ W/m^2: $\beta \sim 100$ dB; exact: 92 dB — same ballpark; the estimate catches the "$P/4\pi r^2$ vs P/r^2" family of slips immediately.

Unit traps the kit prevents: dB without a reference; λ in cm vs m in $v = f\lambda$; temperature in Celsius inside $\sqrt{T}$; M in g/mol inside RT/M; intensity vs pressure amplitude factors of 2; peak vs RMS $\sqrt{2}$.

boom delay geometry from the cone



delay vs Mach: peaks near M = sqrt(2)



REMEMBER / Boom geometry clinic: delay h sqrt(M^2-1)/(M v), emission before the pass, carpet on the ground.

Acoustic estimation kit: building your own numbers

Estimation kit, Fermi drills, chapter map and mark-loss list.

Estimate-first discipline (the Fermi habit) (cont.)

Reality checks: sound speed in gas hundreds, liquid ~1500, solid thousands of m/s; pressures $\Delta P \ll P_{\text{atm}}$; displacement amplitudes nm-mm; if a computed s_0 is 5 cm for speech, it is wrong.

WHY / examiners hide unit slips inside otherwise correct derivations; the estimate-first habit detects them in seconds — the meta-skill this kit trains.

(a) "How far can a shout be heard in still air?" assume 1 mW and 40 dB threshold at the ear: $I = 10^{-8}$ W/m^2: $r = \sqrt{P/(4\pi I)} = 2.8$ km geometric; with 0.1 dB/m absorption: $\beta = 30 - 20\log r - 0.1r = 40$ ... solve: r ~ 150 m realistic — geometry overestimates, absorption rules. (b) "Pitch of a 20 cm closed pipe at 30 C": $c = 349$: $f = c/4L = 436$ Hz (order 400).

Fermi drill set (estimate, then compute) (cont.)

(c) "Beat between a 440 Hz fork and a source at 15 m/s toward": $\Delta f = fu/c = 19$ Hz (order 20).

Chain estimate: "speed of a train from its horn": hear 470 Hz approaching, 400 receding... $u = c(f_+ - f_-)/(f_+ + f_-) = 343 \times 70/870 = 27.6$ m/s (order 100 km/h ok 99 km/h) — estimate 30 m/s ok.

Fermi problems for the set: estimate the acoustic power of a piano (1 W), the sound energy of a conversation in an hour (10^{-2} J), the pressure of a sonic boom at 10 km (order 100 Pa) — all kit lines.

The kit as a closing map (chapter index)

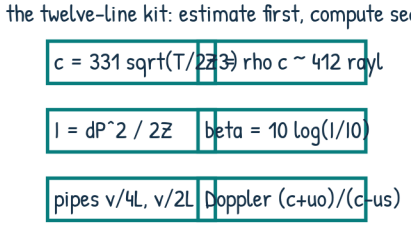
Wave basics (SN-01..04): s, $\Delta P = -B\partial_x s$, $v = \sqrt{B/\rho}$. Media (SN-05..09, OLY-10): $\sqrt{\gamma RT/M}$, $\sqrt{Y/\rho}$, $\sqrt{(K + 4G/3)/\rho}$. Energy (SN-10..12, OLY-05): $I = \Delta P_0^2/2Z$, dB. Interference (SN-13..16, OLY-04): superposition, phasors, standing waves. Pipes (SN-17..21, OLY-03/09): ladders, end correction, impedance, WKB. Motion (SN-22..28, OLY-06..08/11): Doppler box, echo, glide, Mach. Boundaries (OLY-01/02): Z, R. Estimation (OLY-12): the kit.

The kit as a closing map (chapter index) (cont.)

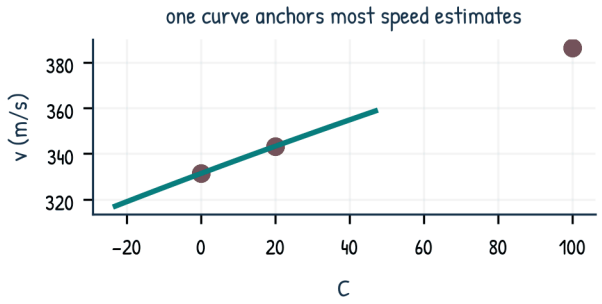
Where marks are lost (the short list): kelvin vs Celsius; e sign in the resonance tube ($l = \lambda/4 - e$); overtone counting; Doppler sign conventions; peak vs RMS; intensity vs amplitude dB factors; factor 2 in Quincke ($\delta = 2\Delta L$); dust heaps at $\lambda/2$.

WHY / the kit card is the revision sheet: if any line is unfamiliar, that sheet of the set deserves another pass — the teacher-style close of the chapter.

the twelve-line estimation kit



one speed curve anchors the estimates



REMEMBER / The kit: estimate first with rounded numbers, compute second, demand agreement within the estimate.